Soil Hydraulic Memory and the Recoverability of Evaporation Savings and the Closure of the Water Balance in a Simulation Study with Daisy

Naziru Halilu[1,2,*]

[1] Agricultural University of Athens, Athens, Greece

[2] Public University of Navarre, Pamplona, Spain

ORCID: 0009-0006-2324-9599

[*] Correspondence: halilunaziru73@gmail.com

Abstract

Water kept from evaporating is routinely counted as water gained, and process-model outputs that support this count are treated as closed accounts. Both premises are tested on one designed ensemble of Daisy simulations covering ten soils and many experiments. Hydraulic memory, the share of released antecedent storage that drains, differs strongly between soils, follows the pore-size parameter and the saturated conductivity, transfers to independent evaporation and mulch experiments, and is predicted for held-out soils by a rate-competition law with one parameter. Warming raises and wetter climates lower the recoverable share, while nitrogen, a negative control, has no effect. A blind search for linear invariants recovers the water balance and reveals an unattributed evaporative term that is a large part of evapotranspiration. A compact law predicts evapotranspiration in withheld experiments and soils as well as gradient boosting, whereas drainage resists every law tried. Judged against reported soil evaporation alone, the drainage share of an evaporation change is misjudged in most soils, so a saving must be judged against the closed non-transpiration term. The paper gives a soil- and climate-specific discount for evaporation savings, a template for deciding which model outputs can be trusted, and testable predictions for lysimeters. All findings concern a one-dimensional matrix-flow model and need confirmation with measured drainage.

Keywords. soil hydraulic memory; evaporation savings; drainage partitioning; water-balance closure; equation discovery; process-model auditing

List of symbols

Table 1: Symbols, abbreviations and units of hydraulic memory and of the accounting audit.

Symbol	Meaning	Unit
Soil water and hydraulic properties		
h, h_0	suction head; initial suction in E9	cm
θ, S_e	volumetric water content; effective saturation	–
α, n, m	van Genuchten scale and shape parameters, $m = 1 - 1/n$	cm^{-1}, –, –
K_s, K, K_r	saturated, unsaturated and relative conductivity	$\mathrm{cm\,d}^{-1}$, –
λ	pore-connectivity exponent (0.5)	–
TAW	total available water to 1 m	mm
Water balance		
P_r, I, W	rainfall, irrigation, water input $P_r + I$	$\mathrm{mm\,yr}^{-1}$
ET, T	evapotranspiration, transpiration	$\mathrm{mm\,yr}^{-1}$
E_s, E_c, ET_{nt}	soil evaporation, canopy evaporation, non-transpiration term $ET - T$	$\mathrm{mm\,yr}^{-1}$
R	unattributed evaporative term, Eq. (11)	$\mathrm{mm\,yr}^{-1}$
D, ΔS, Q_r	drainage at 1 m, storage change, runoff	$\mathrm{mm\,yr}^{-1}$
ε	closure residual, Eq. (12)	–
Hydraulic memory		
$\bar{R}$	storage release, Eq. (4)	$\mathrm{mm\,yr}^{-1}$
ϕ_x	memory fate of flux x, Eq. (5)	–
ψ_x	loss fate of flux x, Eq. (6)	–
ρ_R	recoverable fraction, $1 - \psi_D$	–
q^*	effective withdrawal scale in Eq. (8)	$\mathrm{cm\,d}^{-1}$
κ	irrigation compensation, Eq. (16)	–
Accounting audit		
$\mathbf{F}$, σ_k, $\mathbf{v}_k$	flux matrix, singular values and right singular vectors	–
$\mathbf{c}$	invariant coefficients, Eq. (17)	–
R^2_{cv}	cross-validated coefficient of determination, Eq. (15)	–
DPF, NLF, IE	drainage fraction D/W; nitrogen-leaching fraction; irrigation efficiency	–
Experiments, validation and methods		
E1–E17	numbered experiments of the Daisy ensemble (E9, E16, E7, E11 and E14 carry hydraulic memory)	–
LOSO, LOEO	leave-one-soil-out and leave-one-experiment-out validation	–
RMSE	root-mean-square error	–
SVD, PCA	singular-value decomposition; principal component analysis	–
GBM	gradient-boosting machine used as the flexible benchmark	–

1. Introduction

Evaporation is the largest single outflow of water from non-saturated terrestrial surfaces, and its response to warming is central to projected changes in soil moisture, drought and irrigation de-

mand (Berg and Sheffield, 2018; Dai, 2013; Seneviratne et al., 2010; Sherwood and Fu, 2014). At the global scale most terrestrial evapotranspiration passes through plants (Jasechko et al., 2013; Schlesinger and Jasechko, 2014), but the partitioning is sensitive to hydrologic connectivity, that is, to how readily water is exchanged between storage and the fluxes that deplete it (Good et al., 2015). A warmer atmosphere raises the evaporative demand that the surface can, in principle, impose on the soil (Brutsaert and Parlange, 1998; Roderick and Farquhar, 2002), and in managed systems this extra demand is met by irrigation or by drawing down storage. Practical interventions, including residue and straw mulches (Balwinder-Singh et al., 2011), deficit and scheduled irrigation (Fereres and Soriano, 2007; Geerts and Raes, 2009) and soil-sensor-based scheduling (Datta and Taghvaeian, 2023), are justified by the premise that evaporation can be curtailed and the saved water kept for crops.

That premise is less secure than it looks. Ward and Pulido-Velazquez (2008) and Grafton et al. (2018) showed at basin scale that apparent savings frequently do not appear as increases in the water that is usable, because the water "saved" by a technical improvement was previously a return flow or was being consumed elsewhere. The same accounting problem exists inside a single soil profile. If a mulch or a lower evaporation factor suppresses soil evaporation, the water not evaporated has several possible fates: it may remain in the root zone and reduce the irrigation requirement, it may be taken up by roots, or it may percolate below the root zone and be lost to the crop. Only the first two of these constitute a recoverable saving. The partitioning among them is controlled by the soil's ability to retain water against gravity, which depends on the pore-size distribution, on its connectivity, and on the unsaturated hydraulic conductivity at the moisture states the crop and the irrigation scheduler actually visit.

The relevant physics is old. Capillary flow in porous media is described by the Richards equation (Richards, 1931), closed with a water-retention function and a relative-conductivity function. The van Genuchten–Mualem family (Mualem, 1976; van Genuchten, 1980) reduces these to a pore-size-distribution parameter n, a scaling parameter α and a saturated conductivity K_s, and its parameters can be estimated from texture by pedotransfer functions (Saxton and Rawls, 2006; Schaap et al., 2001; Weber et al., 2024). Similitude of capillary flow (Miller and Miller, 1956), tortuosity and connectivity of the pore space (Blunt, 2017; Ghanbarian et al., 2013), and the roles of macropores and structure (Beven and Germann, 1982, 2013; Fatichi et al., 2020; Jarvis, 2007; Strudley et al., 2008) are all recognised as controls on the drainage response. What is missing is a compact, dimensionless diagnostic that links these pore-scale and hydraulic controls to the

managerial question "if evaporation changes by one millimetre, how much of that millimetre is recoverable?"

The term *hydraulic memory* is used here for the dependence of the water-balance partitioning on the antecedent storage state of the profile. Soil moisture memory is well established in land–atmosphere studies (Entekhabi et al., 1996; Koster and Suarez, 2001; Koster et al., 2004), where it describes the persistence of soil-moisture anomalies. A narrower and operationally measurable definition is adopted here: the fraction of the antecedent storage that is released from the root-zone store which leaves as drainage, as opposed to supplying evaporation or transpiration. Because a profile that is wet at the start of the season releases more water than one that starts dry, this fate fraction measures how "leaky" the soil store is in the sense of water-balance partitioning. It is related to, but distinct from, the Budyko-type partitioning of annual precipitation (Budyko, 1974; Milly, 1994; Zhang et al., 2001), which concerns the long-term mean, and to the ecohydrological theory of soil-moisture-dependent loss functions (Rodriguez-Iturbe and Porporato, 2004).

A few studies have simulated how irrigation and evaporation interact with soil type, and the Daisy soil–plant–atmosphere model has been calibrated and validated widely (Hansen et al., 1991, 2012; Holbak et al., 2026), including for evaporation from structured soils (Leuther and Diamantopoulos, 2024), freeze–thaw effects on hydraulic properties (Leuther and Schlüter, 2021; Miranda-Vélez et al., 2023), residue-induced changes in hydraulic properties (Leuther et al., 2026) and mulching (Vuaille et al., 2024). None of these defines a transferable recoverability metric, and none tests whether the metric is organised by a connectivity proxy that can be known in advance for a soil. This study therefore (i) defines the memory and loss fates; (ii) computes them from a systematic virtual experiment spanning ten soils; (iii) tests whether pore-connectivity proxies predict them out of sample; (iv) tests how climate and management modify the recoverable share; and (v) registers the outcome of every hypothesis tested, including the formulations that identify what the memory fate is not.

Three developments make a transferable metric timely. First, irrigation water is increasingly allocated on the basis of expected on-farm savings, so that the premise that an evaporation reduction is a saving carries economic and regulatory weight (Pereira et al., 2012; Perry et al., 2017); an erroneous premise directs investment to measures whose benefit is mostly a reallocation of drainage. Second, sensor-based scheduling and digital agriculture tools are proliferating (Datta and Taghvaeian, 2023; García et al., 2020), and they are usually parameterised by soil class; a class-level statement of how much of an evaporation change is recoverable could directly improve

their advice. Third, process-based models are increasingly used as virtual laboratories, and the surrogate and emulator literature (Asher et al., 2015; Razavi et al., 2012) requires interpretable, low-dimensional summaries of what the process model does if it is to be trusted by users. The diagnostics proposed here are exactly such summaries: two dimensionless numbers per soil that can be computed from a small designed experiment.

(1) The memory fate ϕ_D, the loss fate ψ_D and the recoverable fraction $\rho_R = 1-\psi_D$ are defined with explicit estimators and uncertainty. (2) Within a mechanistic simulator and ten soils, these diagnostics are shown to be ordered by conductivity-related hydraulic proxies and to transfer across experiments. (3) A one-parameter rate-competition law is provided with out-of-sample validation, and the conditions that bound it are quantified. (4) The shifts in the recoverable share caused by warming and rainfall are quantified, including a continuous climate surface for one soil. (5) A hypothesis register records the test and outcome of every formulation examined, so that the definitions are tied to evidence.

The analysis addresses four linked problems. Soils differ in how much of the released antecedent storage drains, and it is tested whether this memory fate ϕ_D follows pore-connectivity proxies. Transferability is examined next: ϕ_D, measured from initial moisture, is used to predict ψ_D from evaporation-change experiments. A single competition law with one fitted parameter is then evaluated against purely structural predictors under leave-one-soil-out validation. Finally, the response of the recoverable share $1 - \psi_D$ to warming, rainfall, irrigation trigger, dose and nitrogen supply is quantified. Section 3.1 reports the alternative formulations that were tested and what each establishes about the definition of the memory fate.

The accounting question. Conservation of mass is the most reliable organising principle in hydrology, and it is the foundation of every water-balance model from the bucket to the Earth system model. Yet in practice the budget of a managed soil profile rarely closes exactly in the output of a process model: the fluxes that are reported separately (transpiration, soil evaporation, canopy interception) need not add to the evapotranspiration that enters the balance; change in storage is reported or implied but its size relative to the fluxes is seldom audited; and boundary fluxes at the base of a one-dimensional column are, by construction, definitions rather than measurements (Hillel, 1998; Refsgaard and Henriksen, 2004). At the same time, a strand of scientific machine learning has argued that conservation laws, governing equations and low-dimensional invariants can be *discovered* from data, using sparse regression over libraries of candidate terms (Brunton et al., 2016; Champion et al., 2019; de Silva et al., 2020), symbolic regression (Schmidt

and Lipson, 2009; Udrescu and Tegmark, 2020), neural networks that learn invariants (Iten et al., 2020; Liu and Tegmark, 2021), and physics-informed learning (Karniadakis et al., 2021; Raissi et al., 2019). In hydrology, deep learning has delivered impressive predictive skill (Kratzert et al., 2018, 2019; Shen, 2018), with an ongoing debate about what such models reveal about processes (Nearing et al., 2021; Reichstein et al., 2019) and about whether interpretable models should be preferred for consequential decisions (Rudin, 2019). Enforcing conservation analytically in learned emulators is also an active topic (Beucler et al., 2021).

Three gaps limit confidence in discovery claims in soil-water science. First, discovery methods are rarely validated against a case with a known answer before being applied to an unknown. Second, a discovered relation is typically validated within the data from which it was found, without leave-one-*experiment*-out tests that ask whether the relation survives a change of regime, which is the relevant test of a "law" (Roberts et al., 2017). Third, negative results (where sparse laws fail to beat black-box benchmarks, or where a presumed constant is not constant) are rarely documented, so the reader cannot estimate the base rate of success (Kirchner, 2006; Oreskes et al., 1994).

A large designed ensemble of process-model runs, in which the true balance is known to hold up to the model's own storage term, is taken as a laboratory for method evaluation. A four-part pipeline is built (Fig. 1): (1) a linear-invariant search by singular-value decomposition of the flux matrix with a permutation null, used first as a positive control; (2) a unattributed-term search by auditing partitions that should sum to a total; (3) a manifold analysis of water-fate composition by compositional PCA (Aitchison, 1986; Egozcue et al., 2003); and (4) interpretable sparse regression over a dimensionless library with LOSO and LOEO validation against a gradient-boosting benchmark (Friedman, 2001; Hastie et al., 2009; Tibshirani, 1996). One fixed benchmark, cross-validation grouping and stopping rule are used for all targets, and every law that failed is reported.

The paper asks what a blind method can and cannot recover from a flux matrix. A linear-invariant search must first return the water balance with calibrated uncertainty, which sets the credibility of everything that follows. Accounting terms that the model's own output columns leave open are then sought, and the number of independent directions governing the partition of water among fates is counted. Compact, dimensionally sensible laws are tested against a flexible machine-learning model in *unseen experiments and unseen soils*. The last aim is to separate relations that hold across regimes from those that only appear universal.

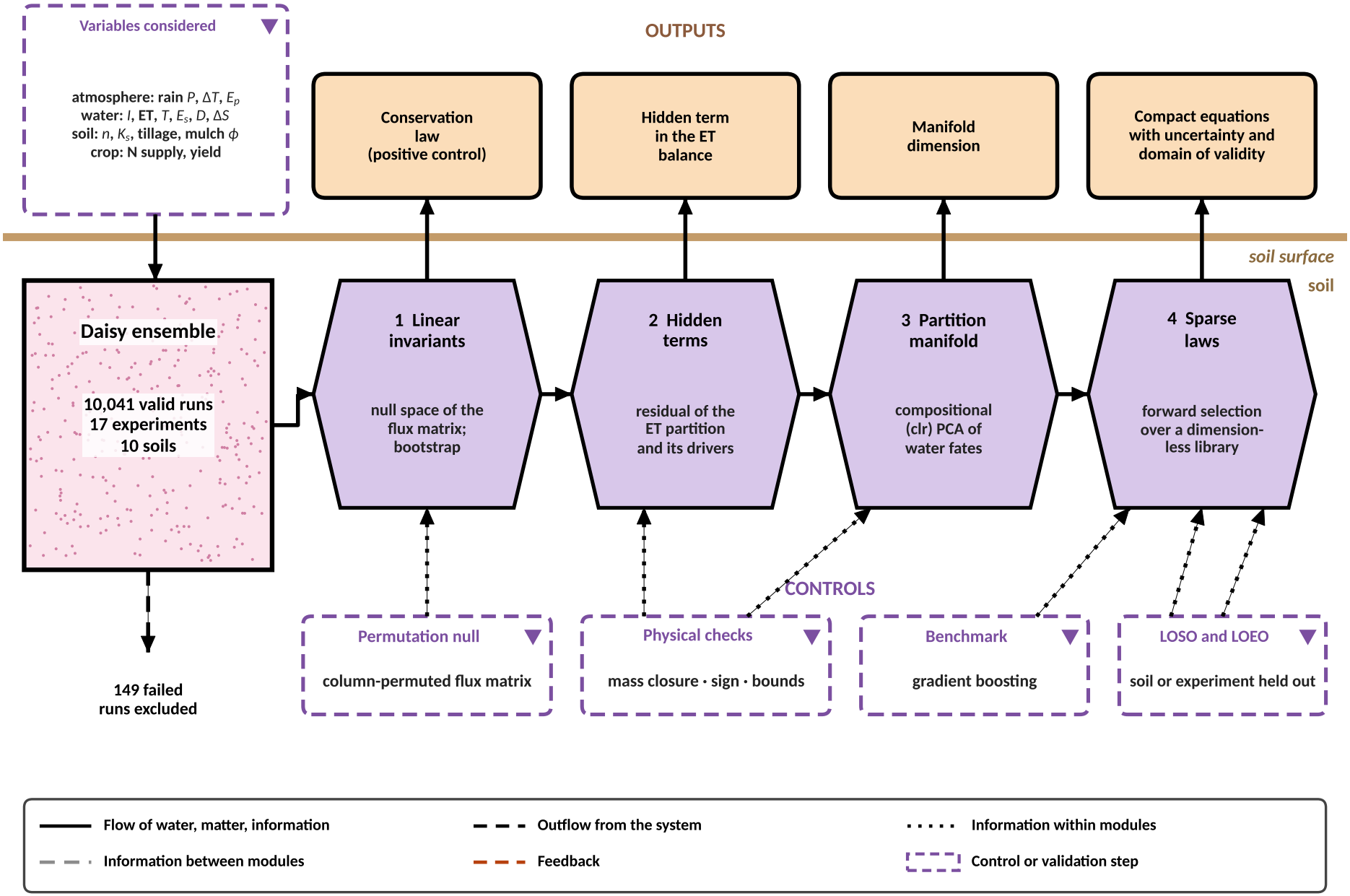


Figure 1: Pipeline for interpretable discovery. The pipeline recovers the known water balance as a positive control, audits partitions, measures the dimension of the partition manifold and fits sparse laws with LOSO/LOEO validation against gradient boosting. Controls (dashed purple boxes) are shown at the bottom and outputs at the top; purple hexagons are processing blocks, the pink stippled box is the data store, peach boxes are outputs and dotted lines are flows of information. The 149 runs excluded as numerical failures are analysed in Section 2.2.

Study design and levels of evidence. The unit of observation is the 20-year mean water balance of one simulated soil under one combination of experimental factors, and statements about soils are made at the level of the ten soils and not of the individual runs. Four levels of evidence are kept apart. Computational reproducibility, the regeneration of every reported number from archived code and data, is established. Predictive validity is assessed by leave-one-soil-out prediction and by transfer between independent experimental designs. Mechanistic validity is bounded by the process model, which represents matrix flow in one dimension. Field transferability is not tested, and the conclusions are therefore stated as predictions that lysimeter experiments can confirm or refute. For the accounting audit the runs of an experiment are correlated because they share a design, so validation withholds whole experiments and whole soils, and a flexible benchmark is reported even where it wins.

What is new. Evaporation savings are normally counted as gains, and the outputs of a process model as closed accounts. This paper tests both on one ensemble. (1) The recoverable share of a saving becomes a measured soil property: the memory fate ϕ_D, estimated from initial moisture alone, predicts the loss fate ψ_D of independent evaporation-change experiments across soils and climates, and a one-parameter rate-competition law explains the ordering with out-of-sample validation and stated bounds. (2) The accounting is audited before any partition is used as a target: a blind search must return the water balance first, with a permutation null, and then exposes an unattributed term of 30 % of evapotranspiration and a storage gap of 16 % in two experiments; laws are validated across experiments and soils and not only across random folds. (3) The two are joined: judged against reported soil evaporation instead of the closed non-transpiration term, the recoverable share changes by more than 0.1 in six of ten soils and reverses sign in two (Section 3.2). (4) Every formulation that was tried is kept in a hypothesis register with its outcome, including the failures, and the protocol is given as seven rules that other process models can adopt.

2. Methods

2.1 Memory analysis

Overview of the virtual-experiment design. The analysis uses a pre-existing ensemble of 10 190 simulations (149 failed and were excluded, leaving 10 041 valid runs) produced with Daisy (Hansen et al., 2012; Holbak et al., 2026), organised into 17 experiments (E0–E16) that vary irrigation dose, trigger pressure, sensor depth, method, soil hydraulic dynamics, mulch, initial

moisture, rainfall, temperature, evaporation factor, nitrogen rate, depth and amount (Table 2). Ten soils (S01–S10) span the texture triangle from sand to silty clay loam. All runs were driven by the same 20-year Taastrup (Denmark) weather series; unless stated otherwise every reported flux is a 20-year mean in mm yr^{-1}, which suppresses inter-annual noise but also means that results describe long-term mean partitioning, not event response. Daisy was selected because it resolves a one-dimensional Richards-equation soil column, coupled to a crop module, a mulch/residue layer, and nitrogen dynamics, which allows all components of the water balance (rainfall, irrigation, evapotranspiration split into transpiration, soil evaporation and canopy evaporation, drainage at 1 m, and change in storage) to be extracted for every run.

Table 2: Experiments of the ensemble used in this paper. "Valid" counts exclude runs that failed to complete. Experiments marked $*$ are the primary ones for the diagnostics defined here.

ID	Name	Valid runs	Factors varied (per soil unless noted)
E0	Baselines	10	One baseline per soil
E1	Doses	189	Irrigation dose
E2	TriggerPressure	197	Trigger suction
E3	SensorDepth	200	Sensor depth
E4	Methods	96	Surface, overhead, subsurface drip
E5	SHP_Dynamics	300	Time-varying soil hydraulic properties
E6	Mediterranean_Pollution	193	Climate/N scenarios
E7*	Mulching	200	Mulch vapour-flux factor (VFF) $\times$ mulch storage capacity (CAP), 20 cells
E8	Rain_Intensity_Moisture	1000	Rain intensity $\times$ initial moisture (S01)
E9*	Soil_Moisture_Rain	987	Initial suction h_0 (10 levels) $\times$ rain scale P (10 levels)
E10	Amount_Timing_Soil	954	Irrigation amount and timing
E11	Temp_Evap_Rain	1000	Temperature shift $\times$ evaporation factor $\times$ rain (S01)
E12	Sensor_Trigger_Rain	1000	Depth $\times$ trigger $\times$ rain (S01)
E13	N_Water_Soil	975	Nitrogen rate $\times$ water
E14	Mulch_Rain	1000	Mulch cover $\times$ capacity $\times$ rain (S01)
E15	Method_Depth_Amount	1000	Method $\times$ depth $\times$ amount (S01)
E16*	Edge_Cases	740	2^6 factorial per soil (P, T, h, D, N, E_p) and 13 further edge cases per soil

Daisy solves the one-dimensional Richards equation in a layered profile with the retention and conductivity functions above, couples it to a crop module (light interception, photosynthesis, root growth and water stress), a surface mulch/residue layer that intercepts rain and suppresses soil evaporation, an energy-balance-based potential evapotranspiration, and nitrogen transformations (Hansen et al., 1991, 2012; Holbak et al., 2026). The irrigation scheduler in the ensemble applies a fixed dose whenever the suction at a sensor depth exceeds a trigger. Drainage is the flux at 1 m

depth; no groundwater table is modelled so that net capillary rise is possible but is a boundary-condition artefact in a few soils (Table 27). Because the soil columns are one-dimensional, lateral flow, run-on, and spatial variability are absent, and the 20-year mean precludes interpretation at the event scale. Model structural limits are discussed in Refsgaard and Henriksen (2004) and Kirchner (2006): the emphasis here is on whether the model produces the *right answers for the right reasons*, which the present study can address only for hydraulic mechanisms internal to the model.

Soil hydraulic description. The retention and conductivity functions are the van Genuchten–Mualem pair (Mualem, 1976; van Genuchten, 1980):

$$S_e(h) = \left[1 + (\alpha h)^n\right]^{-m}, \qquad m = 1 - \tfrac{1}{n}, \tag{1}$$

$$K(S_e) = K_s\, S_e^{\lambda}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2, \tag{2}$$

where h is the suction head (cm), $S_e = (\theta - \theta_r)/(\theta_s - \theta_r)$ the effective saturation, α (cm^{-1}) and n ($-$) shape parameters, K_s the saturated conductivity (cm d^{-1}) and λ the pore-connectivity (tortuosity) exponent, conventionally $\lambda = 0.5$ (Mualem, 1976). In the diagnostics below, the relative conductivity $K_r = K/K_s$ is used with $\lambda = 0.5$. The shape parameter n is a pore-size-uniformity index: large n denotes a narrow pore-size distribution (sand, rapid desaturation), small n a broad distribution (fine-textured soils) with a long tail of fine pores that retain water at high suction. For the ten soils, effective (α, n) were taken from the profile-averaged values in the project documentation (Table 3). Saturated conductivities K_s were available from the class-mean values of Carsel and Parrish (1988) for eight of the ten soils; for the two soils (S01 coarse loam, S02 coarse sand) whose class does not map to a single USDA texture class, K_s is treated as unknown and implied by the fitted law (Section 3.1). *It is important that K_s here is a literature class value and not the Daisy input conductivity of each soil horizon.* This is a limitation that propagates to the rate-competition results.

Table 3: Soil set. n and α: effective van Genuchten parameters; *TAW*: total available water to 1 m used in the project documentation; $S_e(100)$: effective saturation at 100 cm suction from Eq. (1).

ID	Texture	α (cm^{-1})	n	*TAW* (mm)	$S_e(100)$	ϕ_D (E9)
S01	Coarse loam	0.0961	1.353	130	0.445	0.029
S02	Coarse sand	0.0536	1.600	105	0.356	0.265
S03	Sand	0.1450	2.680	60	0.011	0.846
S04	Loamy sand	0.1240	2.280	75	0.040	0.741
S05	Sandy loam	0.0750	1.890	95	0.166	0.643
S06	Loam	0.0360	1.560	150	0.468	0.289
S07	Silt loam	0.0200	1.410	220	0.688	0.147
S08	Sandy clay loam	0.0590	1.480	160	0.418	0.285
S09	Clay loam	0.0190	1.310	175	0.753	0.087
S10	Silty clay loam	0.0100	1.230	195	0.882	0.022

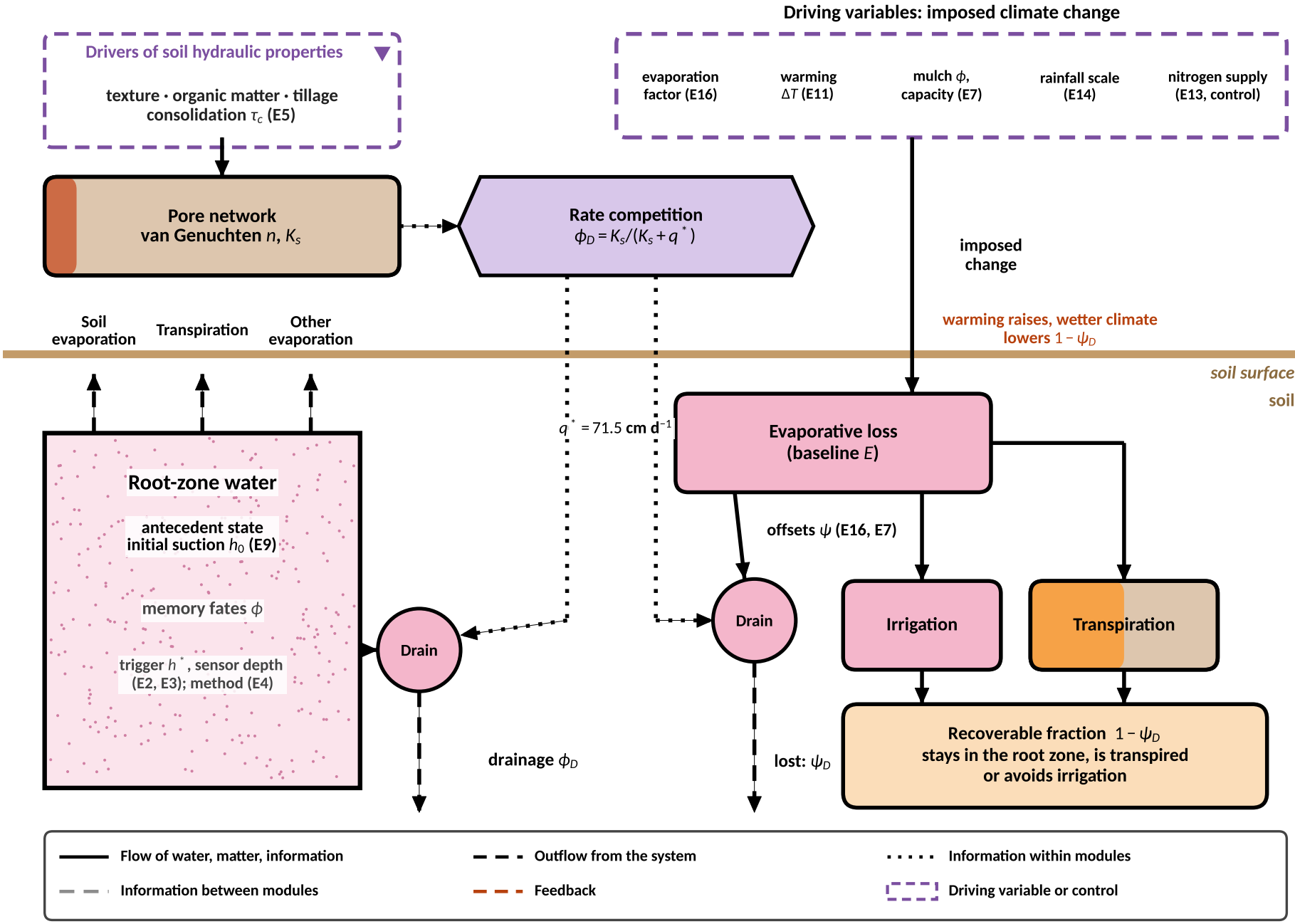


Figure 2: Conceptual and analytical framework. Antecedent storage released from the root zone can leave as drainage, soil evaporation, transpiration or other evaporation (memory fates ϕ, estimated from E9). An imposed change in evaporative loss is offset by drainage, irrigation or transpiration (loss fates ψ, estimated from E16 and E7). The recoverable fraction of an evaporation saving is $1 - \psi_D$. Boxes indicate storages and solid lines flows; dashed black lines are flows out of the system, dotted lines flows of information; dashed purple boxes are drivers of soil hydraulic properties and the imposed climate changes. Pink indicates water, orange the crop, light brown the soil and rust the pore network; the purple hexagon is the rate-competition process and peach boxes are results. The brown line is the soil surface.

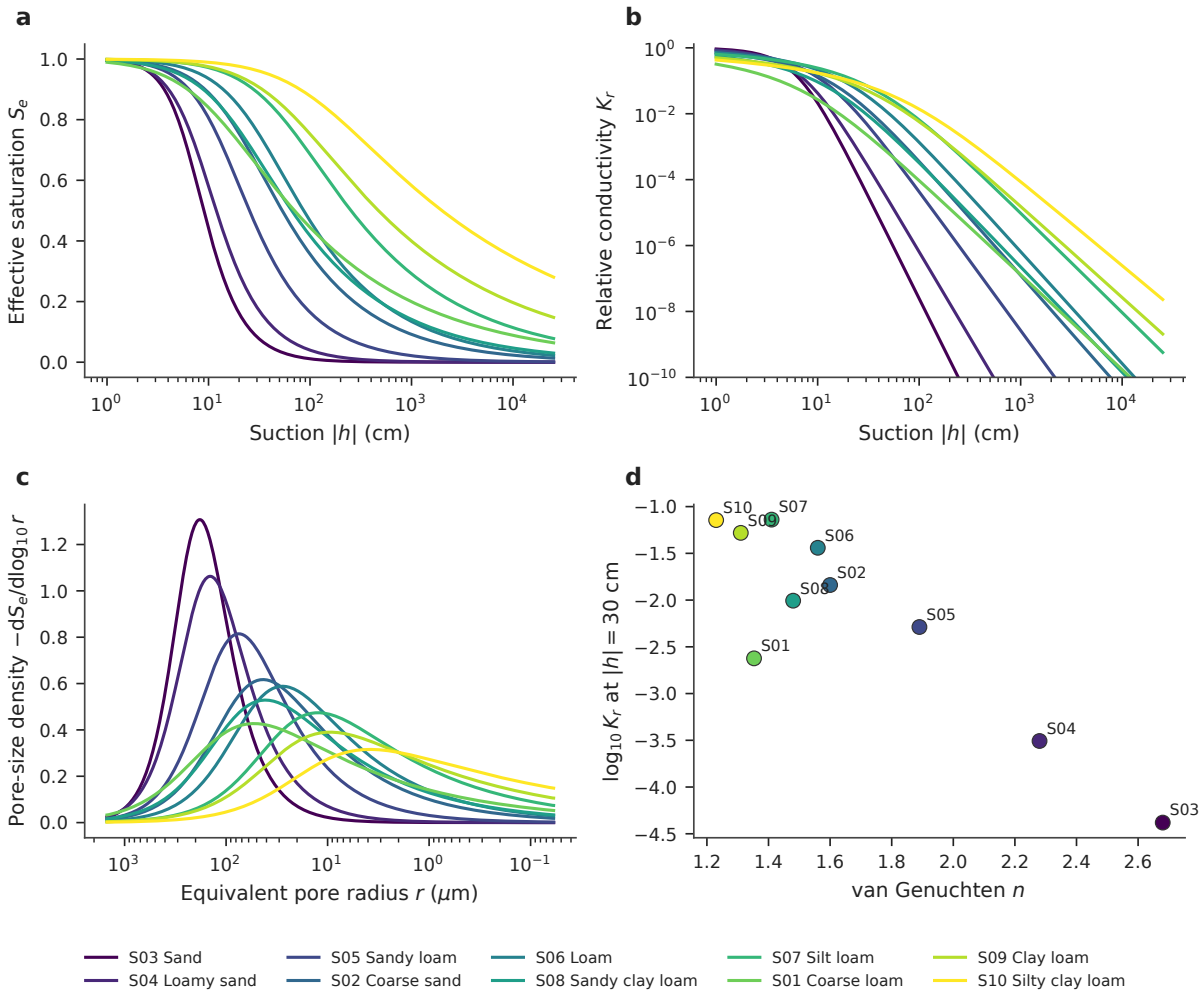


Figure 3: Hydraulic properties of the ten soils from Eqs. (1) and (2). The ordering by shape parameter n is the ordering on which the main results are organised.

For each run the 20-year mean annual water balance is

$$P_r + I = ET + D + \Delta S + Q_r, \tag{3}$$

where P_r is rainfall, I irrigation, $ET = T + E_s + E_c$ evapotranspiration (transpiration T, soil evaporation E_s, canopy evaporation E_c), D drainage across the 1 m boundary, ΔS the change in profile storage and Q_r runoff. Over the 20-year runs $\Delta S/W$ has a median of 0.9 % of the input $W = P_r + I$ (0.65 % excluding E8 and E9, whose initial moisture is deliberately perturbed; 95th percentile 2.3 %). The *storage release* is defined

$$\bar{R} = ET + D - P_r - I, \tag{4}$$

which is positive when the profile supplies water to the fluxes and negative when it accumulates

water; it equals $-(\Delta S + Q_r)$ by Eq. (3). The non-transpirational evapotranspiration is further defined as $ET_{\text{nt}} = ET - T$, which in the model is $E_s + E_c$ plus a small unattributed residual of the model's evapotranspiration bookkeeping.

Memory fate ϕ (experiment E9). Experiment E9 imposes, for each soil, an initial suction h_0 (ten values from 30 to 3000 cm) and a rainfall scale P (ten values from 0.5 to 1.4), a 10×10 lattice (100 runs per soil; 87 for S03 because of failed runs). Wetter initial states (small h_0) release more storage, so $\bar{R}$ varies along the h_0 axis. Writing $x \in \{E_s, T, D, \text{oth}\}$ for the soil-evaporation, transpiration, drainage and other-evaporation fluxes (with oth $= ET - T - E_s$), the *memory fate* of a flux is the within-P pooled regression slope of that flux on $\bar{R}$:

$$\phi_x \;=\; \frac{\sum_P \sum_{c\in P} (\bar{R}_c - \overline{\bar{R}}_P)(x_c - \bar{x}_P)}{\sum_P \sum_{c\in P} (\bar{R}_c - \overline{\bar{R}}_P)^2}, \qquad \phi_{E_s} + \phi_T + \phi_D + \phi_{\text{oth}} = 1, \tag{5}$$

where the sums run over lattice cells c of one soil and bars denote means within each rain level P. The unity constraint follows from Eq. (4) when the irrigation response to $\bar{R}$ is negligible within the lattice. Pooling within P removes the first-order effect of the rainfall change on $\bar{R}$ and isolates the response to antecedent state. ϕ_D is called the *hydraulic memory* of drainage: the fraction of an additional millimetre of antecedent storage released that is shed as drainage. Confidence intervals are obtained by bootstrapping the 100 cells (1 000 resamples). A field-capacity-referenced storage capacity is also computed, $\text{cap}_{FC} = \bar{R}(h_0{=}100\,\text{cm}) - \bar{R}(h_0{=}1500\,\text{cm})$ at $P = 1$.

Loss fate ψ and recoverable fraction (E16 and E7). E16 is a 2^6 factorial per soil with factors rainfall scale $P \in \{0.5, 1.5\}$, temperature shift $T \in \{2, 6\}$ °C, trigger suction $h \in \{50, 2000\}$ cm, dose $D \in \{1.2, 30\}$ mm, nitrogen rate $N \in \{25, 150\}$ kg ha^{-1} and soil-evaporation factor $E_p \in \{0.3, 1.2\}$; 13 further edge cases per soil (extremes of method, mulch, rooting depth and sensor depth) complete the experiment and are not used here. For each of the $2^5 = 32$ combinations of the other factors, the paired difference between $E_p = 1.2$ and $E_p = 0.3$ is formed. The loss fate of component x is the ratio of summed paired differences:

$$\psi_x \;=\; \frac{\sum_k \big[x_k(E_p{=}1.2) - x_k(E_p{=}0.3)\big]}{\sum_k \big[ET_{\text{nt}k}(E_p{=}1.2) - ET_{\text{nt}k}(E_p{=}0.3)\big]} \times s_x, \tag{6}$$

with $s_x = -1$ for drainage and transpiration (which fall when evaporation rises) and $s_x = +1$ for irrigation (which rises). Thus ψ_D is the drainage given up, per millimetre of extra non-transpirational evaporation; ψ_I the extra irrigation and ψ_T the transpiration lost. In a closed

balance $\psi_D + \psi_I + \psi_T + \psi_S \approx 1$, where ψ_S is the storage term. Because the shares sum to approximately one (Table 28), a value $\psi_D > 1$ is possible only when another term, here irrigation, changes sign; it occurs in sand (Section 3.1). *Recoverable fraction*: because drainage below the root zone is water the crop cannot use, whereas reduced irrigation or increased transpiration represents water retained in the system, the recoverable fraction is defined

$$\rho_R \;=\; 1 - \psi_D. \tag{7}$$

E7 varies the mulch vapour-flux factor (VFF, 1 = bare soil) and mulch storage capacity (CAP, mm) in a lattice of 20 cells per soil. Mulch reduces ET_{nt}, and each flux is regressed on ET_{nt} across the 20 cells; $\psi_D = -\mathrm{d}D/\mathrm{d}ET_{\mathrm{nt}}$ and $\psi_I = \mathrm{d}I/\mathrm{d}ET_{\mathrm{nt}}$. A bootstrap confidence interval over cells is computed. Because E7 acts through the physical mulch layer rather than a bulk evaporation factor, it is an independent realisation of the evaporation-saving mechanism.

Rate-competition law. To explain ϕ_D mechanistically two competing pathways are considered for water in the transmission zone after a wetting event. Gravity-driven percolation proceeds at a rate of order K evaluated at the high-moisture end of the profile, while evaporation and root uptake withdraw water at an effective rate q^*. If the two processes act in parallel on the same pool, the fraction lost to percolation is

$$\phi_D \;=\; \frac{K_s}{K_s + q^*}, \tag{8}$$

where q^* (cm d^{-1}) is a single effective withdrawal scale shared by all soils. This is a competing-rates (parallel-sink) closure, not a derivation from the Richards equation; it is the simplest functional form that (i) is bounded in $[0, 1]$, (ii) tends to 0 for impermeable and to 1 for freely draining soils, and (iii) has a single free parameter. q^* is estimated by nonlinear least squares on the eight soils with literature K_s and validated by leave-one-soil-out (LOSO) prediction. A variant uses the unsaturated conductivity at a fitted characteristic suction h^*, $\phi_D = K(h^*)/[K(h^*) + q]$ (Appendix A).

From Eqs. (3) and (4), any perturbation δ of the system that leaves rainfall unchanged satisfies $\delta\bar{R} = \delta ET + \delta D - \delta I$. Dividing by $\delta\bar{R}$ and splitting $ET = T + E_s + \mathrm{oth}$ gives $1 = \phi_T + \phi_{E_s} + \phi_{\mathrm{oth}} + \phi_D - \phi_I$ with $\phi_I = \delta I/\delta\bar{R}$. Within the E9 lattice irrigation responds weakly to initial moisture and the identity reduces to the unity constraint in Eq. (5). For an evaporation perturbation δE,

the same identity with $\delta ET = \delta ET_{\mathrm{nt}} + \delta T$ and a closed balance ($\delta \bar{R} \approx 0$ in steady state) gives

$$\delta ET_{\mathrm{nt}} = -\delta T - \delta D + \delta I \implies \psi_T + \psi_D + \psi_I \approx 1, \quad \psi_I = \frac{\delta I}{\delta ET_{\mathrm{nt}}},\ \psi_D = -\frac{\delta D}{\delta ET_{\mathrm{nt}}},\ \psi_T = -\frac{\delta T}{\delta ET_{\mathrm{nt}}}, \tag{9}$$

which is what Table 28 verifies numerically (sums 1.00–1.03; 1.13 for S10, where a storage term is evidently non-negligible). Equation (9) makes the logic of recoverability explicit: an evaporation change is paid for by three accounts, and drainage is the only one that is lost to the crop.

For $K_s \ll q^*$, $\phi_D \approx K_s/q^*$ is linear in conductivity; a tenfold lower K_s gives a tenfold lower memory. For $K_s \gg q^*$, $1 - \phi_D \approx q^*/K_s$ and the retained share declines inversely with conductivity. The logistic form in $\log K_s$ has its steepest response at $K_s = q^* = 71.5\,\mathrm{cm\,d^{-1}}$, which lies within the range of the sandy soils (Fig. 9a), and this is why the sandy soils differ most among themselves in ϕ_D while the fine soils are compressed near zero.

With ϕ_D dimensionless, and the governing variables K_s ($\mathrm{L\,T^{-1}}$), q^* ($\mathrm{L\,T^{-1}}$), n, and αh^*, the Buckingham Π theorem (Buckingham, 1914) implies $\phi_D = F(K_s/q^*,\, n,\, \alpha h^*)$. Equation (8) corresponds to dropping the dependence on n and αh^* beyond that carried by K_s; the variant in Appendix A retains a dependence on (n, α) through $K(h^*)$. That K_s alone suffices to explain 97% of the variance in eight soils indicates that, in this ensemble, n and α enter chiefly via their covariance with K_s in the texture classes, as expected from pedotransfer relationships (Saxton and Rawls, 2006; Schaap et al., 2001).

Four soil-level predictors of ϕ_D were compared: (i) *TAW*; (ii) $S_{\mathrm{e}}(100)$ from Eq. (1); (iii) n; and (iv) $\log_{10} K_s$. For (i)–(iii) all ten soils are used, for (iv) eight. Each predictor enters a one-variable linear regression clipped to $[0, 1]$. Performance is the RMSE of LOSO predictions, compared with the "mean-only" benchmark. Association among diagnostics uses Spearman ρ with (a) exact permutation p values and (b) percentile bootstrap intervals (10 000 resamples). Paired effects of climate factors on ψ_D in E16 use per-soil paired differences between factor levels with a two-sided exact sign test across soils. With $n = 10$ soils, power is limited; intervals are wide and are reported alongside point estimates (Efron and Tibshirani, 1993; Good, 2005).

Analyses used Python with NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), scikit-learn (Pedregosa et al., 2011) and Matplotlib (Hunter, 2007). Figure 2 uses a brown, orange, pink, rust and purple scheme; the data figures use the Okabe–Ito colour-blind-safe palette. The ensemble and the scripts are available in the project repository (Data, Code), consistent with the FAIR principles (Wilkinson et al., 2016) and with calls for reproducible hydrology (Hutton et al.,

2016).

2.2 Accounting audit

Data. The Daisy (Hansen et al., 2012; Holbak et al., 2026) virtual-experiment ensemble is used, as described in the methods of the memory analysis: 10 190 runs in 17 experiments (E0–E16), ten soils (S01–S10, van Genuchten $n = 1.23$–2.68, Table 6), 20-year Taastrup weather; 149 runs failed and are excluded (Section 2.2), and 10 041 valid runs are analysed. Each run contributes 20-year mean annual rainfall P_r, irrigation I, evapotranspiration *ET*, transpiration T, soil evaporation E_s, canopy evaporation E_c, drainage at 1 m D, and (derived) crop yield, nitrogen leaching and irrigation efficiency. Experiments differ in structure: single-soil (coarse loam) lattices (E8, E11, E12, E14, E15), all-soil factorials and lattices (E9, E16) and one-factor series (E1–E4). The distributions of the key variables are in Table 5.

The data in the repository. Every output of every run is open. The folder `Data/Comprehensive_Summary` of the repository holds, for each of the 17 experiments, one compressed table with every value of every Daisy log of every run, year by year (the annual water balance, the nitrogen balance, the harvest and the crop state), the one-row-per-run summary of all 10 190 runs with the settings read from each set-up file, the per-experiment summary, and the script that made them from the log files (Table 4 and Fig. 4); the 20-year means of these tables reproduce the run table used here within 2 %. The complete raw archive is on Zenodo (Halilu, 2026a).

Table 4: The data behind the papers: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E1	`Doses`	200	189	soil, rate, dose	5.13	198	98	564	79	195
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E5	`SHP Dynamics`	300	300	soil, method, tillage	4.13	152	43	579	84	189
E6	`Mediterranean Pollution`	210	193	soil, method, trigger, N rate	3.64	84	239	620	62	118
E7	`Mulching`	200	200	soil, mulch	5.29	183	96	555	74	161
E8	`Rain Intensity Moisture`	1000	1000	initial suction, rain scale	0.68	0	-1	282	-0	6
E9	`Soil Moisture Rain`	1000	987	soil, initial suction, rain scale	0.75	0	19	247	33	7
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E11	`Temp Evap Rain`	1000	1000	rain scale, warming, demand	3.81	324	18	754	66	151
E12	`Sensor Trigger Rain`	1000	1000	trigger, sensor depth, rain scale	5.88	260	35	670	32	155
E13	`N Water Soil`	1000	975	soil, method, rate, dose, trigger, N rate	4.00	162	83	547	79	207
E14	`Mulch Rain`	1000	1000	rain scale, mulch	6.15	235	43	634	25	173
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157
E16	`Edge Cases`	770	740	soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand	2.17	344	185	600	93	224

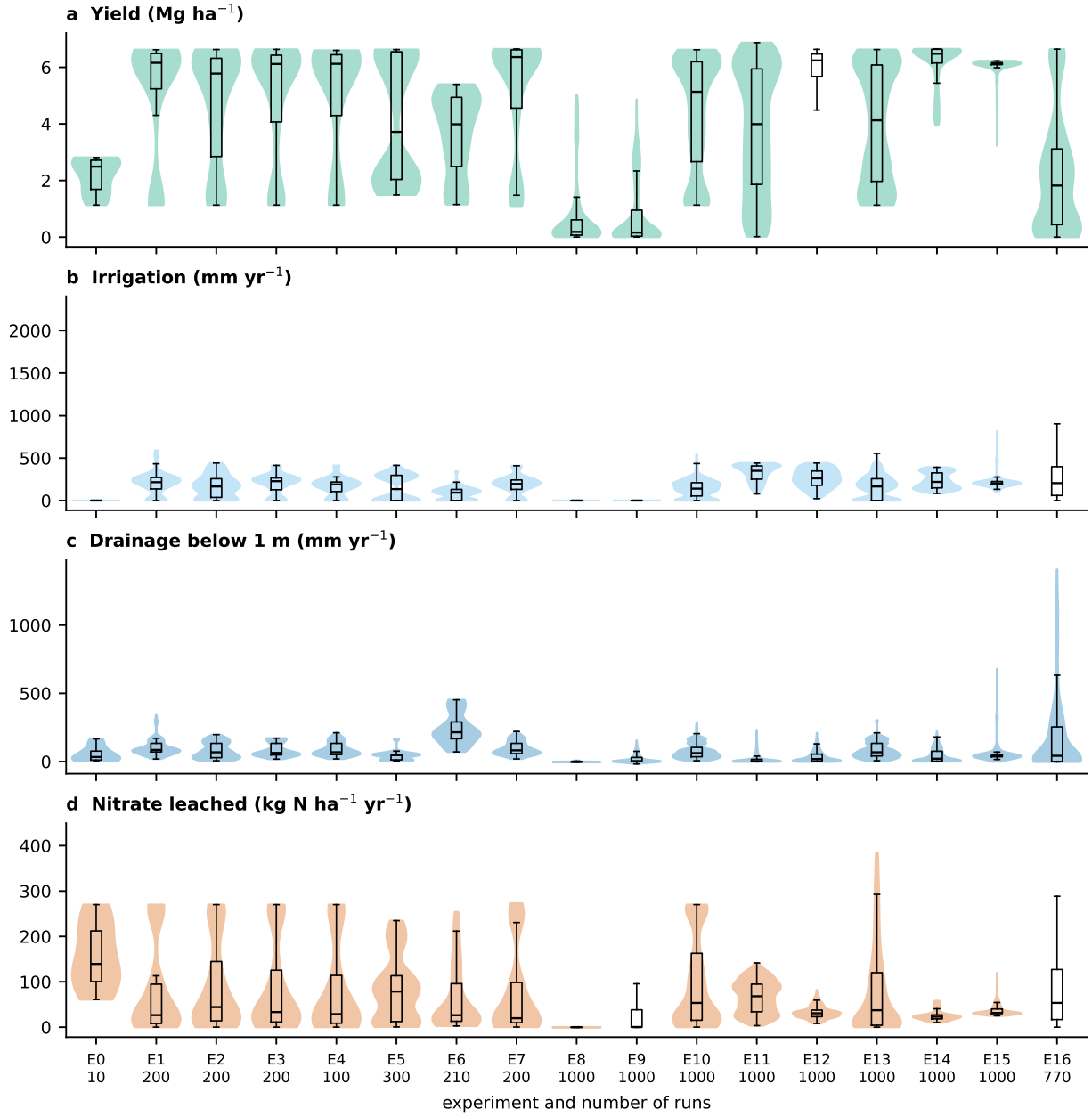


Figure 4: The data of the ensemble: yield, irrigation, drainage and nitrate leaching of the 10 041 valid runs, by experiment (violins: distribution; boxes: quartiles). Source: `00_summary_all_simulations.csv.gz` in `Data/Comprehensive_Summary`; script `reproducibility/analysis/data_overview.py`.

Table 5: Descriptive statistics of the 10 041 valid runs (mm yr^{-1}; yield in Mg ha^{-1} not shown).

Variable	Mean	SD	25 %	Median	75 %
Rain P_r	417	143	297	464	464
Irrigation I	180	191	0	178	268
Water input $W = P_r + I$	597	251	464	654	722
Evapotranspiration ET	546	196	386	611	671
Drainage D (1 m)	60	106	3	34	78
Unattributed R (Eq. (11))	164	96	103	129	223

Table 6: Soil parameters used as predictors (effective van Genuchten values from the project documentation; *TAW* in mm to 1 m).

ID	Texture	α (cm^{-1})	n	*TAW*
S01	Coarse loam	0.0961	1.353	130
S02	Coarse sand	0.0536	1.600	105
S03	Sand	0.1450	2.680	60
S04	Loamy sand	0.1240	2.280	75
S05	Sandy loam	0.0750	1.890	95
S06	Loam	0.0360	1.560	150
S07	Silt loam	0.0200	1.410	220
S08	Sandy clay loam	0.0590	1.480	160
S09	Clay loam	0.0190	1.310	175
S10	Silty clay loam	0.0100	1.230	195

Failed runs and why they are excluded. Of the 10 190 runs, 149 (1.5 %) are flagged as failed in the ensemble and are excluded from every analysis. Because exclusion that is not random can bias a discovery study, the pattern of failure was examined before the runs were dropped, and the argument is set out in four parts.

Failures are concentrated in three soils and in specific experiments. Of the 149 failures, 129 (87 %) occur in the coarse sand S02, 13 in the sand S03 and 7 in the silty clay loam S10; the other seven soils have none (Fig. 5a). In S02, 24.9 % of the 519 runs fail. In S03 all 13 failures belong to the initial-moisture experiment E9, and in S10 all 7 belong to the nitrogen–water and dose experiments (E13, E1). The three soils lie at the extremes or in the anomalous part of the van Genuchten range ($n = 1.60$ for S02, $n = 2.68$ for S03, the steepest retention curve, and $n = 1.23$ for S10, the flattest), which is where the unsaturated conductivity function changes most rapidly with suction.

Within a failing soil, failure follows the size of the water pulse. In S02, the fraction of failed runs rises with the irrigation depth applied per event in experiment E10: none of the 20 runs at 1 and 2 mm fails, 10 % at 3 mm, 50 % at 8–10 mm and all of the 30 runs at 15, 20 and 30 mm (Fig. 5b). It also depends on the rain scale: 69 % of the S02 runs at a scale of 1.5 fail (22 of 32) and 12 % at 0.5 (5 of 42), against none of the 90 runs at intermediate scales 0.6–1.4 (Fig. 5c). Runs that share the same weather, the same soil and the same code, and differ only in how much water arrives at once, therefore fail or succeed in a systematic order. This is the behaviour of a state-dependent numerical difficulty and not of a corrupted input, which would hit runs irrespective of their forcing.

The outputs of failed runs are physically impossible. In valid runs the annual drainage at 1 m is never below -17.5 mm (median in valid S02 runs, 134 mm). In failed runs it is negative in 65 % of cases and falls to -882 mm (Fig. 5d), and the ratio of evapotranspiration to water input ($P + I$) ranges from 0.54 to 4.34, with a median of 1.09, so that in some runs more water leaves the profile than entered it. A net upward flux of hundreds of millimetres through the lower boundary of a free-draining column, or evapotranspiration several times the water input, cannot be produced by the physics of the model; it is the signature of a solver that has not converged, in which a mass error is absorbed in the boundary or storage terms.

The alternatives are less consistent with the pattern. A fault in the weather or in the run definition would not depend on soil and on dose in this ordered way; a failure from over-wetting in the field, such as ponding or runoff, is a physical process that the model is built to represent and would not give negative drainage; and a random job failure would spread evenly over soils. Numerical non-convergence of the Richards-equation solver in a stiff regime (a very permeable, quickly draining column that receives a large input, or a column with an extreme retention curve) accounts for the concentration in soils, the order with water input and the impossible outputs together.

Treatment of the excluded runs. The failed runs are interpreted as numerical failures of the soil-water solver under stiff conditions, not as information about soil hydrology, and they are excluded for that reason; the interpretation follows from the pattern of failure described above. Two consequences follow. The exclusion removes a quarter of the runs of S02, preferentially in the wettest and driest forcing and at the largest doses, so results for this soil, and any law fitted to it, apply to the part of the factor space in which the solver succeeded. The other nine soils are essentially unaffected (at most 2.5 % of runs in S03), and the conclusions that rest on across-soil comparisons were tested for sensitivity to the exclusion by re-analysis. With S02 removed (9 651 runs) the three-term ET law is re-estimated with coefficients 0.652 for $\ln W$, -0.297 for n and 0.029 for $\ln(1 + I)$, against 0.646, -0.299 and 0.029 for the full ensemble, with LOEO $R^2 = 0.897$ and LOSO $R^2 = 0.904$ (full ensemble 0.898 and 0.904), and the unattributed term R is 30 % of ET (163 mm yr^{-1}; full ensemble 30 % and 164 mm yr^{-1}). With S02 and S03 both removed (9 145 runs) the scores are 0.893 and 0.898 and R is 30 % (165 mm yr^{-1}). The exclusion therefore does not drive the across-soil results.

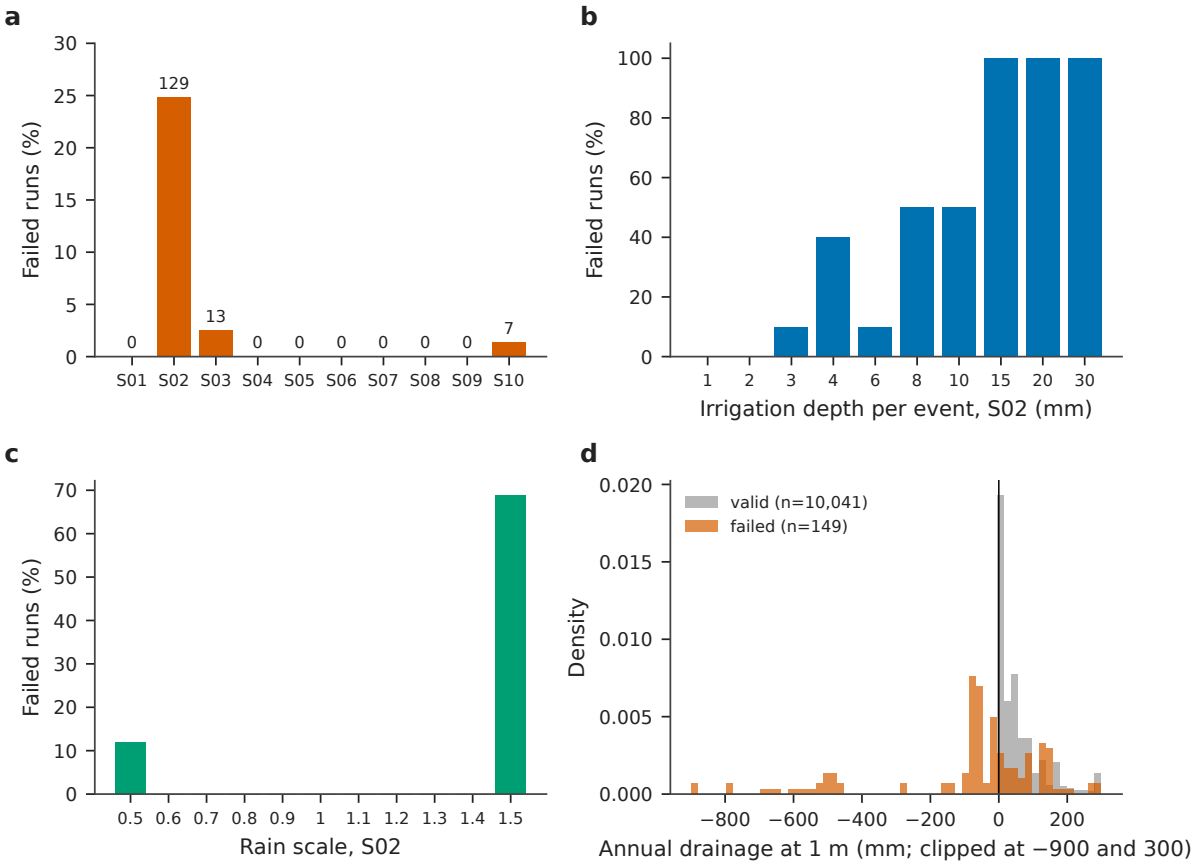


Figure 5: Pattern of the 149 failed runs. (a) Failed runs per soil, as a percentage of the 519 runs per soil (counts above bars). (b) Failed fraction in S02 against irrigation depth per event in E10. (c) Failed fraction in S02 against rain scale. (d) Annual drainage at 1 m for valid and failed runs, normalised as densities; negative drainage of several hundred millimetres occurs only in failed runs.

Governing balance and the flux matrix. For each run the mean annual water balance is

$$P_r + I \;=\; ET + D + \Delta S + Q_r, \qquad ET = T + E_s + E_c + R, \tag{10}$$

where ΔS is storage change, Q_r runoff, and R is defined as the remainder of ET after the three reported components,

$$R \;=\; ET - T - E_s - E_c. \tag{11}$$

Equation (11) defines R without ascribing a process. Let $\mathbf{F} \in \mathbb{R}^{N\times q}$ be the matrix whose rows are runs and whose columns are the fluxes (rain, irrigation, ET, T, E_s, E_c, D). If a linear invariant $\sum_j c_j F_{ij} = 0$ holds for all runs, then $\mathbf{c}$ lies in the null space of $\mathbf{F}$. Define $W = P_r + I$ and the closure residual

$$\varepsilon \;=\; \frac{W - ET - D}{W} \;\approx\; \frac{\Delta S + Q_r}{W}. \tag{12}$$

Linear-invariant search by SVD with a permutation null. The singular-value decomposition $\mathbf{F} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with singular values $\sigma_1 \geq \cdots \geq \sigma_q$ identifies near-null directions as the right singular vectors $\mathbf{v}_k$ with small σ_k. Each column is first divided by its root-mean-square value so that no flux dominates through its units; $\mathbf{v}_k$ is mapped back to mm units and normalised so that the ET coefficient equals -1. Because singular values depend on the scale of the columns, significance is judged against a *column-permutation null*: each column of $\mathbf{F}$ is permuted independently across runs (destroying cross-column dependence but preserving each marginal), and the smallest singular value under the null is recorded; an invariant is accepted only if $\sigma_{\min}$ lies far below the null value. Uncertainty of the coefficients is estimated by a nonparametric bootstrap over runs (resampling 10 041 rows; 95 % percentile intervals). The test is repeated (a) on the four-flux subsystem (P_r, I, ET, D), which is the set that appears in the balance; (b) on the seven-column matrix; and (c) excluding E8 and E9, whose initial states are deliberately away from equilibrium. It is emphasised that this is a *positive control*: the expected answer is $\mathbf{c} = (1, 1, -1, -1)$ up to scale and the pipeline is judged by how closely it recovers it, and by whether it fails in a diagnosable way.

Any decomposition that should sum to a total is audited by computing the residual. The audit covers (a) Eq. (11), reporting R, its share R/ET by experiment, and its dependence on drivers by regression of $\ln R$ on candidate drivers (irrigation, soil, mulch capacity, temperature, evaporation factor); and (b) the storage closure ε of Eq. (12) by experiment. The audit is descriptive: it identifies where the budget does not close, and tests candidate attributions by whether R responds to the drivers that would be expected under each.

The partition of water input into fates (transpiration, soil evaporation, canopy evaporation, unattributed R, drainage) is compositional: components are positive and sum to a constant, so Euclidean statistics are inappropriate (Aitchison, 1986; Egozcue et al., 2003). Writing the composition as $\mathbf{x} = (x_1, \ldots, x_5)$ with $x_i = \text{flux}_i / \sum_j \text{flux}_j$, the centred log-ratio (clr) transform is applied

$$\text{clr}(\mathbf{x})_i = \ln x_i - \frac{1}{5}\sum_{j=1}^{5} \ln x_j, \tag{13}$$

and PCA (Wold et al., 1987) to the transformed data. Drainage is zero or negative in 14 % of runs, so zeros are replaced by a floor δ (in mm yr^{-1}) before the closure to unit sum; because the results depend strongly on δ, results are reported for $\delta \in \{10^{-3}, 0.1, 1, 5, 10\}$. Because log-transformations are inherently sensitive to near-zero values, the sensitivity to the floor is mapped across four orders of magnitude and not fixed at a single value, so that the stability of the drainage

axis can be judged independently of the floor selected. The number of components needed to explain the variance is a data-driven estimate of the dimension of the partition manifold.

For a target y (log of a flux), a library $\boldsymbol{\Theta}$ is built of dimensionless or log-transformed candidates (Table 7). The model is

$$\ln y \;=\; \beta_0 + \sum_{k\in\mathcal{S}} \beta_k\, \theta_k + \epsilon, \qquad \mathcal{S} \subseteq \{1,\dots,p\}, \tag{14}$$

with the active set $\mathcal{S}$ chosen by forward selection: at each iteration the candidate that maximises the leave-one-experiment-out cross-validated R^2 is added, and the search stops when the improvement is below 0.003. The cross-validated R^2 is

$$R^2_{\mathrm{cv}} = 1 - \frac{\sum_i (\ln y_i - \widehat{\ln y}_i^{(-g(i))})^2}{\sum_i (\ln y_i - \overline{\ln y})^2}, \tag{15}$$

where $\widehat{\ln y}_i^{(-g(i))}$ is the prediction for run i from a model fitted with all runs of its group $g(i)$ held out. Groups are experiments (LOEO, 17 folds) or soils (LOSO, 10 folds). LOEO is the demanding test: it asks whether a law fitted on 16 experimental designs predicts the 17th, which differs in which factors vary and in what range. The root-mean-square error in mm and the mean absolute percentage error after back-transformation are also reported. The benchmark is a histogram-based gradient-boosting regressor (Friedman, 2001; Pedregosa et al., 2011) with 250 iterations on the whole library, validated under the same groupings. Caveat: because forward selection uses the LOEO score, the reported LOEO R^2 of the selected model is mildly optimistic (selection bias); LOSO values, which were not used for selection, provide a partly independent check, and coefficient stability across folds is reported.

Table 7: Candidate library $\boldsymbol{\Theta}$ (log-transformed or dimensionless).

Symbol	Definition	Role
$\ln P_r$	log rainfall	climate input
$\ln(1+I)$	log irrigation	management input
$\ln W$	log water input $P_r + I$	total supply
$\ln E_p$	log soil-evaporation factor	evaporative demand
ΔT	temperature shift (°C)	warming
n	van Genuchten shape parameter	pore-size distribution
$\ln TAW$	log total available water	storage capacity
$\ln \alpha$	log van Genuchten scaling	air-entry scale
mulch	VFF×CAP (mulch lattice)	residue layer
I/W	irrigation fraction	managed share
$\ln(W/TAW)$	dimensionless supply-to-storage	wetness index

Suppose the true model is $\mathbf{Fc} = \boldsymbol{\eta}$ with $\boldsymbol{\eta}$ a small run-wise residual (here $-\Delta S - Q_r$ for $\mathbf{c} = (1, 1, -1, -1)$). The SVD gives the best rank-$(q-1)$ approximation of $\mathbf{F}$ (Eckart–Young), and the right singular vector $\mathbf{v}_q$ minimises $\|\mathbf{Fv}\|_2$ over unit vectors; the corresponding singular value is $\sigma_q = \|\mathbf{Fv}_q\|_2$. Because the columns are all measured with error (here, the model's own discretisation and storage noise), $\mathbf{v}_q$ is a total-least-squares solution, in which errors in all variables are treated symmetrically; this is preferable to regressing one flux on the others, whose coefficients would be attenuated and depend on the choice of dependent variable. When η is uncorrelated with the columns, $\sigma_q^2 \approx \|\boldsymbol{\eta}\|^2$, so $\sigma_q/\sqrt{N}$ is the root-mean-square residual per run in the units of the scaled columns. Before the decomposition each column is divided by its root-mean-square value (so that no flux dominates by its units), and the singular vector is mapped back to mm units afterwards; the singular values are therefore dimensionless, and for the four-flux subsystem $\sigma_q/\sqrt{N} = 5.9/\sqrt{10041} = 0.059$, i.e. a residual of about 6 % of a typical column scale. A caveat is that the matrix is not mean-centred: centring would remove the constant direction, but a conservation law $\sum c_j F_j = 0$ has no intercept, so the uncentred matrix is the correct object.

Permuting each column independently preserves marginal distributions and destroys all linear relations between columns, so the smallest singular value under permutation, σ_q^{perm}, measures how small σ_q would be if the columns were unrelated but had the same scales. A ratio $\sigma_q/\sigma_q^{\text{perm}} = 5.9/33.3 = 0.18$ for the four-flux matrix shows that the null direction reflects genuine dependence. Because fluxes are positively skewed and share a common experiment structure (for example rainfall is identical across many runs), the permutation null is conservative in one respect (it breaks the experimental clustering) and liberal in another (it does not respect the clustering when permuting). Runs are therefore also bootstrapped and intervals reported.

1. Initialise $\mathcal{S} = \emptyset$, best score $s^* = -\infty$.
2. For each candidate $k \notin \mathcal{S}$, fit Eq. (14) on the training groups and compute R^2_{cv} (Eq. (15)) for $\mathcal{S} \cup \{k\}$ with 17 LOEO folds.
3. Let k^* be the maximiser; if $R^2_{\text{cv}}(\mathcal{S} \cup \{k^*\}) > s^* + 0.003$, add k^* and update s^*; otherwise stop.
4. Refit on all data to report coefficients; evaluate LOSO using the same terms.

Table 19 shows the path for the ET law: $\ln W$ alone gives $R^2_{\text{cv}} = 0.801$, adding n raises it to 0.895 and adding $\ln(1 + I)$ to 0.898. The third term improves the score by only 0.003, at the stopping threshold, so a two-term law $\ln ET = f(\ln W, n)$ is almost as good and is the safer summary.

The compensation coefficient κ measures the share of a rainfall change that is offset by irri-

gation. For rainfall change ΔP_r with other factors fixed,

$$\kappa = -\frac{\Delta I}{\Delta P_r}. \tag{16}$$

In single-soil lattices $\mathrm{d}W/\mathrm{d}P_r = 1 - \kappa$ is estimated by ordinary least squares; in E16, paired differences are formed of rain scale 0.5 and 1.5 across the 32 combinations of other factors within each soil, and bootstrap the pooled estimate (500 resamples).

All analyses are scripted in Python (Harris et al., 2020; Hunter, 2007; Pedregosa et al., 2011; Virtanen et al., 2020); scripts and intermediate results are in the project repository (Data, Code) and mapped in Appendix D. Model evaluation follows the recommendations of Moriasi et al. (2007) and Bennett et al. (2013); R^2 is reported on the log scale unless stated. No test data were used for model selection other than through the declared cross-validation.

3. Results

3.1 Memory analysis

Antecedent storage release drains differently in different soils. Across the 10×10 E9 lattice the storage release $\bar{R}$ varies strongly with initial suction (Fig. 6). In sand (S03, $n = 2.68$) the release saturates near $-25\,\mathrm{mm\,yr^{-1}}$ once h_0 exceeds about 100 cm, whereas in the fine-textured soils (S07, S01, S09, S10) $\bar{R}$ continues to change across the full range of h_0 and the curves show a pronounced knee near 300–800 cm. The rank order of the panels in Fig. 6 (by decreasing n) therefore corresponds to a visible change in the shape of the memory curve from a sharp, early-saturating response to a gradual, extended one.

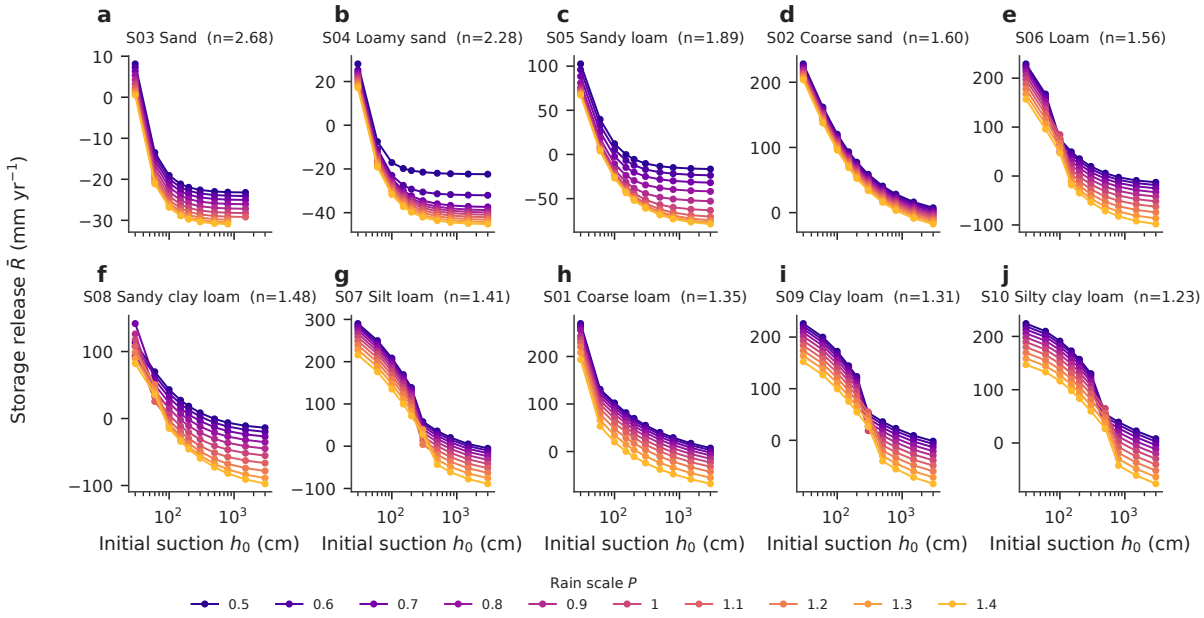


Figure 6: Storage release $\bar{R}$ (mm yr^{-1}) versus initial suction h_0 in the E9 lattice, one panel per soil (a–j, ordered by decreasing van Genuchten n). Lines are rain scales $P = 0.5$–1.4 (colour bar). Wetter starts (small h_0) release more storage; the response saturates early in coarse soils and extends over the full h_0 range in fine soils.

The memory fates (Eq. (5); Fig. 7a, Table 8) differ by more than an order of magnitude among soils. Drainage takes 85 % (95 % CI 0.80–0.89) of released antecedent storage in sand, 74 % in loamy sand and 64 % in sandy loam, but only 2–3 % in silty clay loam and coarse loam and 9 % in clay loam. In the finer soils transpiration takes most of the released storage ($\phi_T = 0.75$–0.84 in S01 and S10); in coarse soils evaporation and transpiration together are small ($\phi_{E_s} + \phi_T = 0.15$–$0.33$ in S03–S05).

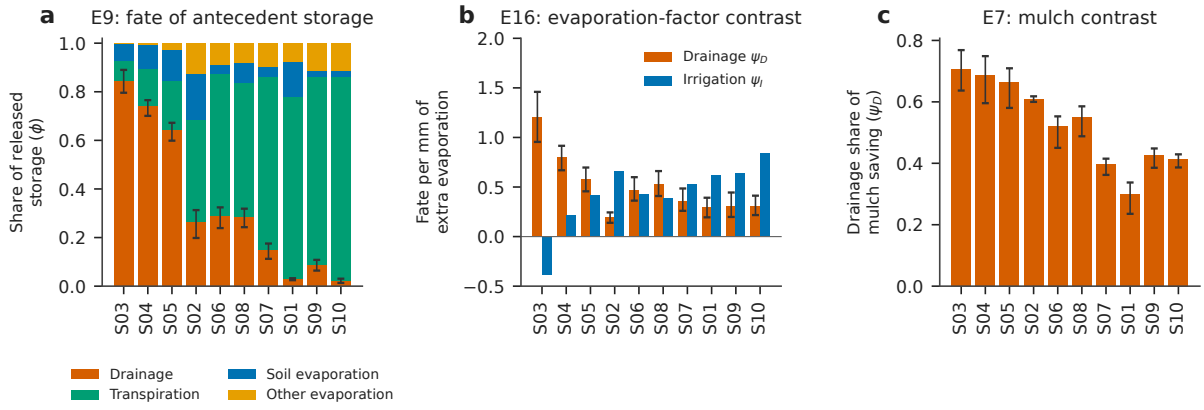


Figure 7: (a) Memory fates ϕ of released antecedent storage (E9; error bars: bootstrap 95 % CI for drainage). (b) Loss fates ψ_D (drainage) and ψ_I (irrigation) of an evaporation-factor increase (E16). (c) Drainage share of the evaporation saving from mulch (E7). Soils are ordered by decreasing ϕ_D.

Table 8: Memory fates (E9) and loss fates (E16, E7) by soil. $\phi_D, \phi_T, \phi_{E_s}$: drainage, transpiration and soil-evaporation shares of released antecedent storage; ψ_D, ψ_I: drainage and irrigation shares of an evaporation change; $\rho_R^{16} = 1 - \psi_D^{E16}$.

Soil	ϕ_D	ϕ_T	ϕ_{E_s}	cap$_{FC}$ (mm)	ψ_D^{E16}	ψ_I^{E16}	ρ_R^{16}	ψ_D^{E7}	ψ_I^{E7}
S01	0.029	0.750	0.145	82.8	0.295	0.619	0.705	0.298	0.583
S02	0.265	0.419	0.193	104.0	0.193	0.658	0.807	0.607	0.339
S03	0.846	0.080	0.074	4.2	1.210	−0.384	−0.210	0.706	0.000
S04	0.741	0.153	0.100	13.2	0.799	0.219	0.201	0.687	−0.023
S05	0.643	0.204	0.128	41.3	0.582	0.415	0.418	0.663	−0.027
S06	0.289	0.582	0.040	131.8	0.469	0.433	0.531	0.521	0.379
S07	0.147	0.716	0.042	208.1	0.362	0.528	0.638	0.396	0.605
S08	0.285	0.555	0.078	57.0	0.533	0.386	0.467	0.549	0.332
S09	0.087	0.774	0.027	167.5	0.312	0.645	0.688	0.426	0.584
S10	0.022	0.842	0.025	173.0	0.310	0.838	0.690	0.414	0.629

Loss fates and the share of an evaporation change that is drainage. In E16 the drainage share of the extra evaporation ψ_D ranges from 0.19 (coarse sand) to 1.21 (sand; Table 8). In sand a millimetre of additional non-transpirational evaporation removes more than a millimetre of drainage ($\psi_D = 1.21$) while irrigation changes in the opposite direction ($\psi_I = -0.38$) and transpiration falls by 0.18; the three terms still sum to 1.00 (Table 28). The cause was not isolated of the negative irrigation response in sand; it may reflect the interaction of the suction trigger with the very small *TAW* (60 mm), but this is a conjecture, and it is the reason $\psi_D > 1$ is not interpreted as a literal fraction. In contrast, in silty clay loam (S10) only 31 % of the evaporation change is drainage and 84 % is offset by irrigation. In the mulch lattice E7, ψ_D ranges from 0.30 (S01) to 0.71 (S03) and falls monotonically from sandy to fine-textured soils (Fig. 7c), with the exception of the coarse sand S02, where mulch effects on drainage are larger than the E16 contrast suggests (0.61 versus 0.19). The recoverable fraction $\rho_R = 1 - \psi_D$ is therefore small in sandy soils (0.20–0.42 for S04–S05; ≤ 0 for S03) 0.64–0.71 in the coarse loam and the fine soils (S01, S07, S09, S10), 0.47–0.53 in the loam and sandy clay loam, and 0.81 in the coarse sand (Table 8).

Hydraulic proxies of pore connectivity predict the memory fate. ϕ_D is strongly ordered by n ($\rho = 0.95$, permutation $p = 1.5 \times 10^{-4}$, bootstrap CI 0.69–1.00), by $S_e(100)$ ($\rho = -0.85$), by *TAW* ($\rho = -0.77$) and by $\log_{10} K_s$ over the eight soils with literature K_s ($\rho = 0.97$; Fig. 8). The ordering by n is consistent with the idea that a narrow pore-size distribution implies a strongly connected network of large pores that drains quickly once the profile is wetted, while the broad distribution of fine soils provides a continuum of small pores that retain water at the tension states visited by the crop. Because n is also correlated with K_s in texture-based datasets (Carsel and

Parrish, 1988; Schaap et al., 2001; Weber et al., 2024), the two rankings cannot be separated here (Section 4.1).

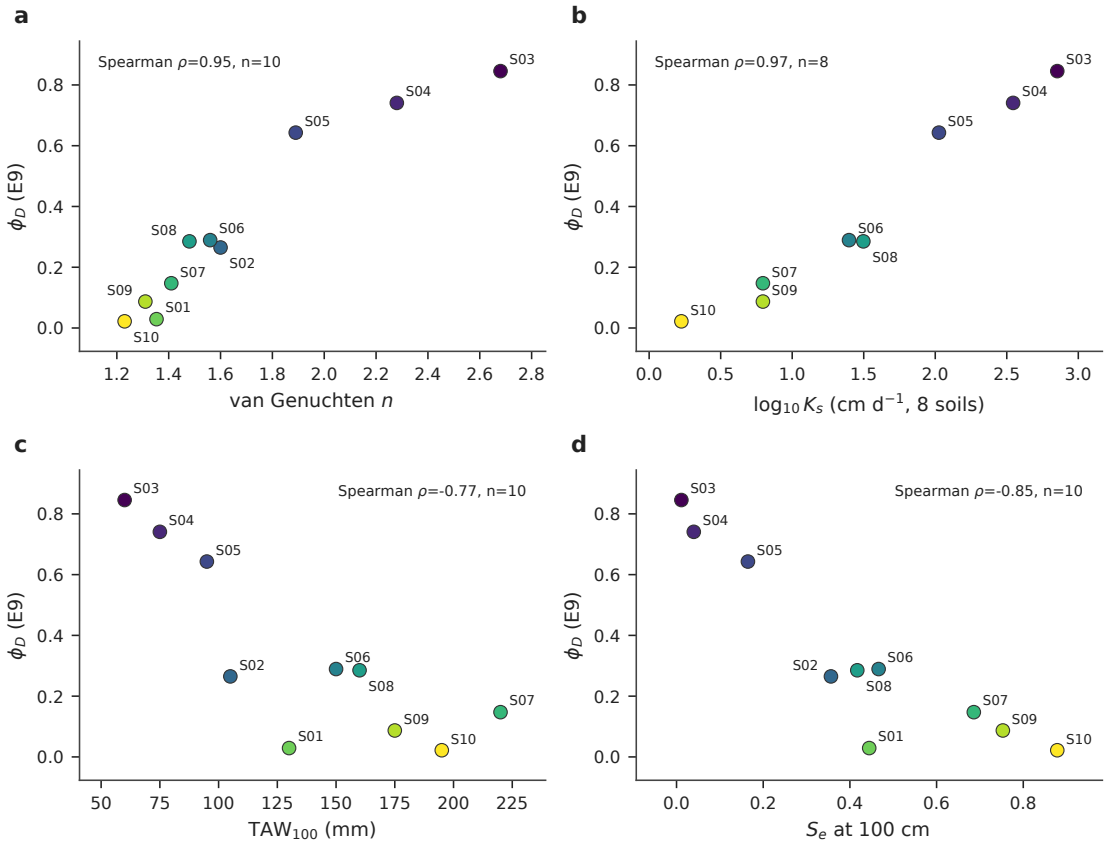


Figure 8: Relationship between the memory fate ϕ_D (E9) and four soil-level predictors: (a) van Genuchten n, (b) $\log_{10} K_s$ (eight soils), (c) *TAW* to 1 m, (d) effective saturation at 100 cm. Spearman ρ given in each panel.

Rate-competition law and out-of-sample skill. Fitting Eq. (8) to the eight soils with literature K_s gives $q^* = 71.5\,\mathrm{cm\,d^{-1}}$ with $R^2 = 0.971$ (Fig. 9a). With eight points the parameter is only moderately well determined: a bootstrap over soils (5 000 resamples, of which 4 999 contained at least three distinct soils and were used) gives a 95 % interval of 58–105 $\mathrm{cm\,d^{-1}}$, and omitting each soil in turn gives values between 66 and 80 $\mathrm{cm\,d^{-1}}$, so the shape of the law is more secure than the exact value of q^*. Leave-one-soil-out RMSE is 0.058 on the eight soils with literature K_s. On the same eight soils the linear models give 0.100 for n, 0.064 for $S_\mathrm{e}(100)$ and 0.088 for *TAW*, and the mean-only benchmark gives 0.338; on all ten soils the corresponding values are 0.097, 0.136, 0.183 and 0.318 (Fig. 9c; Table 9). On like-for-like soils the K_s law therefore reduces the out-of-sample error by 83 % relative to the mean and by 42 % relative to n, but by only 9 % relative to $S_\mathrm{e}(100)$, so the advantage of the law over the best simple alternative is small and is not distinguishable with eight points; its value lies in the mechanism and in having a single

parameter. Two cautions apply. First, the comparison on ten soils involves a different soil set from the K_s model; an earlier logistic fit in n on all ten soils gave an LOO RMSE of 0.111. Second, $q^* = 71.5\,\mathrm{cm\,d^{-1}}$ is orders of magnitude larger than the mean evapotranspiration rate ($\approx 0.2\,\mathrm{cm\,d^{-1}}$); it is an effective scale for the competition between drainage and withdrawal at near-saturated states and should not be read as a physical flux. The law implies $K_s \approx 2.1\,\mathrm{cm\,d^{-1}}$ for S01 and $\approx 25.8\,\mathrm{cm\,d^{-1}}$ for S02, which are plausible for a coarse loam and a coarse sand respectively but remain untested predictions (the markers in Fig. 9a).

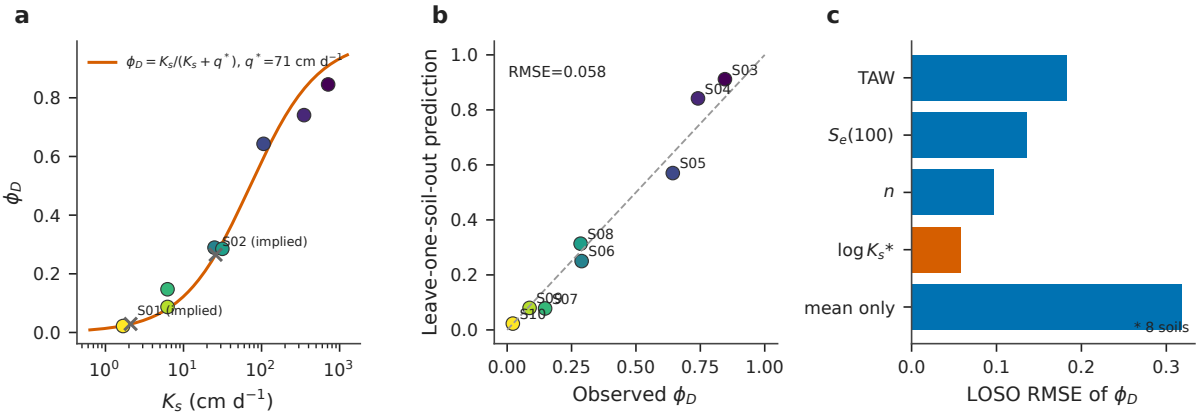


Figure 9: Rate-competition law. (a) ϕ_D against K_s with the fitted curve $K_s/(K_s + q^*)$; crosses mark implied K_s for S01 and S02. (b) Leave-one-soil-out predictions against observations (RMSE 0.058). (c) LOSO RMSE of each predictor of ϕ_D (asterisk: eight soils; the other predictors are shown for all ten soils, and like-for-like eight-soil values are given in Table 9).

Table 9: Leave-one-soil-out (LOSO) predictive error of ϕ_D by predictor.

Predictor	Soils	LOSO RMSE	Model
Mean only	10	0.318	constant
Mean only	8	0.338	constant
TAW	8	0.088	linear, clipped
$S_e(100)$	8	0.064	linear, clipped
n	8	0.100	linear, clipped
TAW	10	0.183	linear, clipped
$S_e(100)$	10	0.136	linear, clipped
n	10	0.097	linear, clipped
$\log_{10} K_s$ (linear)	8	0.062	linear, clipped
$K_s/(K_s + q^*)$	8	0.058	Eq. (8), $q^* = 71.5$

Memory fate predicts the loss fate in independent experiments. The key test of transferability is whether ϕ_D, measured with antecedent moisture, predicts ψ_D, measured in experiments that never varied antecedent moisture. It does: ϕ_D against ψ_D^{E16} gives $\rho = 0.85$ ($p = 0.002$) and against ψ_D^{E7} gives $\rho = 0.89$ ($p = 0.0014$; Fig. 10). The two loss-fate experiments agree with each other less

well ($\rho = 0.71$). The van Genuchten n predicts ψ_D^{E7} well ($\rho = 0.89$, $p = 0.001$) but ψ_D^{E16} only moderately ($\rho = 0.70$, $p = 0.03$, bootstrap CI -0.06 to 1.00), and the bootstrap interval includes zero. Fig. 11 shows the permutation distribution for n versus ϕ_D and the bootstrap intervals of all five associations. *An important qualification*: n predicts ψ_D as well as ϕ_D does, so the data do not show that the memory fate is a better predictor of recoverability than the soil's own hydraulic shape parameter. What ϕ_D adds is an operational definition and a demonstration that the single mechanism (drainage competition) shows up under three different perturbations.

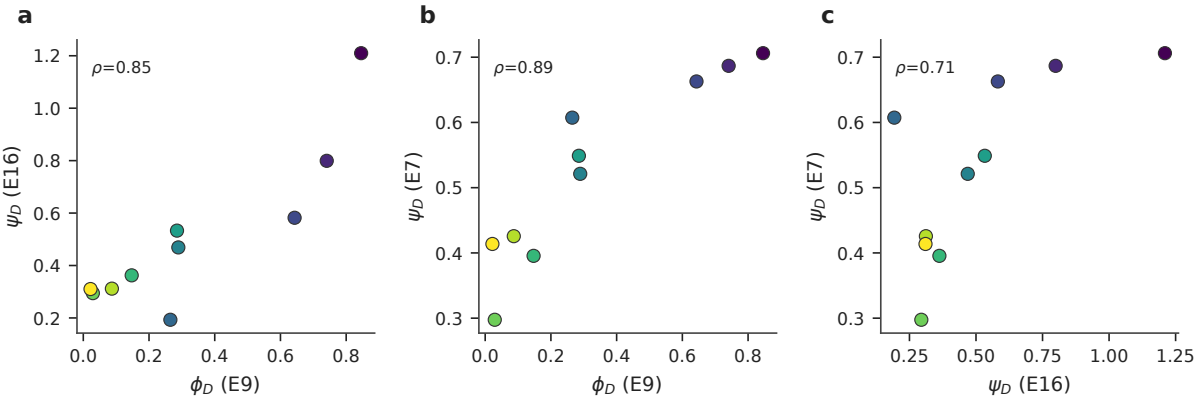


Figure 10: Cross-experiment transfer. (a) ψ_D from E16 against ϕ_D from E9; (b) ψ_D from the mulch lattice E7 against ϕ_D; (c) E7 against E16. Each point is a soil, coloured by n (as in Fig. 8).

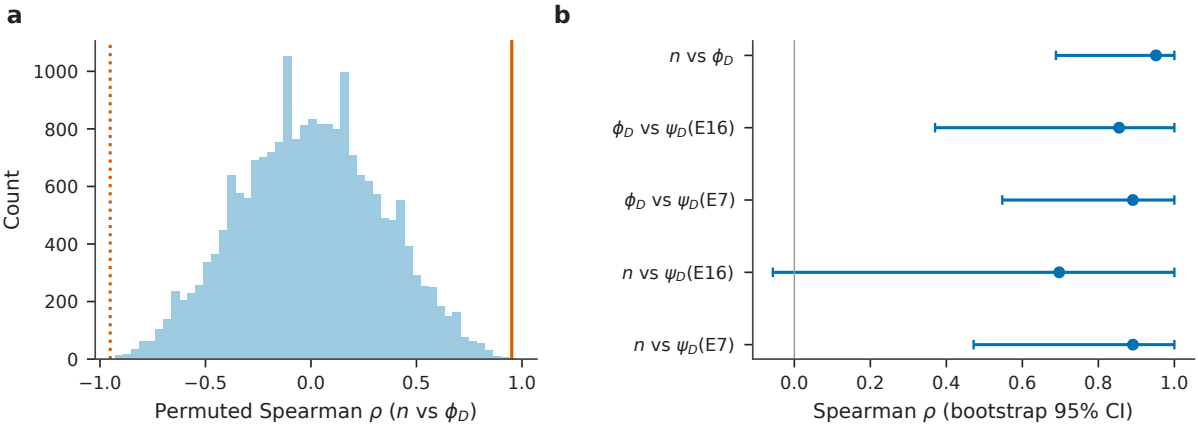


Figure 11: Robustness of the key associations. (a) Permutation distribution of Spearman ρ between n and ϕ_D (solid line: observed; dotted: its mirror). (b) Bootstrap 95 % intervals of five rank associations (10 000 resamples).

Table 10: Rank associations among soil-level diagnostics ($n = 10$ soils). CI: percentile bootstrap.

Pair	Spearman ρ	Permutation p	Bootstrap 95 % CI
n vs. ϕ_D	0.95	0.00015	0.69–1.00
ϕ_D vs. ψ_D (E16)	0.85	0.002	0.37–1.00
ϕ_D vs. ψ_D (E7)	0.89	0.0014	0.55–1.00
n vs. ψ_D (E16)	0.70	0.032	−0.06–1.00
n vs. ψ_D (E7)	0.89	0.001	0.47–1.00

Climate and management modify the recoverable share. Because ψ_D was computed at each of the 32 combinations of the other factors in E16, it can be recomputed at each level of a second factor, and the paired change across soils tested (Fig. 12, Table 11). Warming from $+2$ to $+6$ °C lowers ψ_D in all ten soils (mean change -0.16; sign test $p = 0.002$), so a larger share of the extra evaporation is offset by irrigation or transpiration rather than drainage, and the recoverable share rises. In S01 ψ_D falls from 0.41 to 0.17. Raising rainfall from 0.5 to 1.5 times the baseline raises ψ_D in nine of nine soils with data in both levels (mean change $+0.42$; $p = 0.004$). Wetter climates leave more water to drain, so the same evaporation saving is more likely to be a drainage saving. Trigger suction (+0.14, 7/10 soils up, $p = 0.16$), dose (-0.04, 2/10 up, $p = 0.32$) and nitrogen rate (-0.008, $p = 0.064$, a negative control that should not influence the water partition) are not significant at $\alpha = 0.05$.

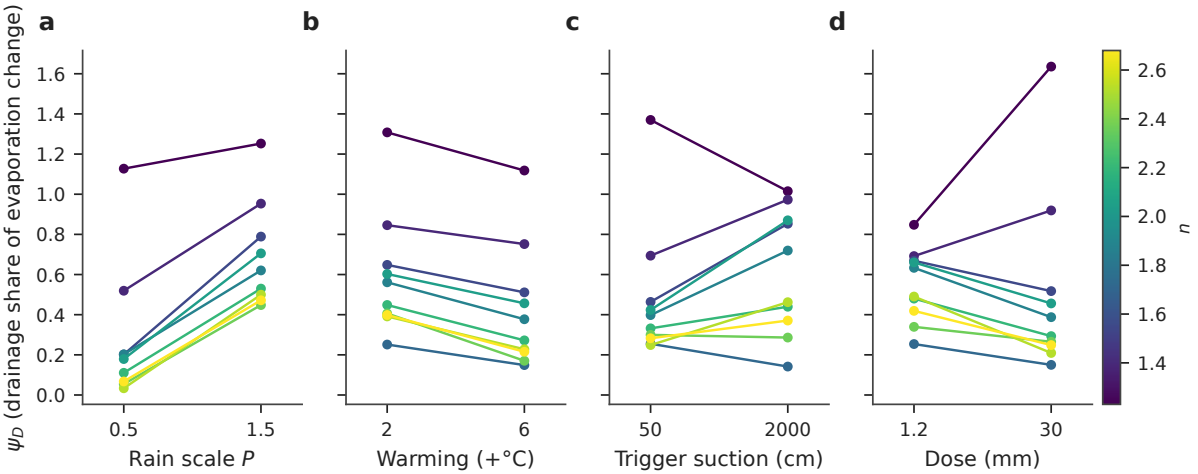


Figure 12: Modifiers of the drainage share ψ_D in E16. Each line is a soil (colour: n); (a) rainfall scale, (b) warming, (c) trigger suction, (d) irrigation dose.

Table 11: Paired effects of E16 factors on the drainage share ψ_D of an evaporation change. "Up" counts soils in which ψ_D is higher at the second level. p: exact two-sided sign test across soils.

Factor (low → high)	Mean change in ψ_D	Soils up / n	p	Verdict
Rain scale 0.5 → 1.5	+0.419	9/9	0.004	supported
Warming +2 → +6 °C	−0.161	0/10	0.002	supported
Trigger 50 → 2000 cm	+0.136	7/10	0.160	not significant
Dose 1.2 → 30 mm	−0.040	2/10	0.322	not significant
N rate 25 → 150	−0.008	2/10	0.064	negative control

The E16 contrasts use only two levels per factor. Experiment E11 provides a finer test for coarse loam (S01): a $10 \times 10 \times 10$ lattice of warming $\Delta T \in [-1, 3.5]$ °C, evaporation factor $E_p \in [0.3, 1.2]$ and rain scale $P \in [0.5, 1.4]$. For each $(\Delta T, P)$ pair, drainage is regressed on ET_{nt} across the ten E_p levels, giving $\psi_D = -\mathrm{d}D/\mathrm{d}ET_{\mathrm{nt}}$ (Fig. 13). The surface is smooth and monotone: ψ_D rises with rainfall (Spearman $\rho = 0.88$ with P over the 100 cells) and falls with warming ($\rho = -0.40$), ranging from 0.01 in the driest, warmest corner to 0.61 at $P = 1.4$ and $\Delta T = -1$ °C. Averaged over ΔT, the drainage share rises from 0.013 at $P = 0.5$ to 0.367 at $P = 1.4$; averaged over P, it falls from 0.283 at $\Delta T = -1$ °C to 0.036 at +3.5 °C. This replicates, within a single soil and over a continuous range, the signs found across soils in E16 (Table 11). It also shows that the single number $\psi_D = 0.295$ reported for S01 in Table 8 is a point on a surface that spans a factor of about 50, so any soil-level value of the recoverable fraction should be quoted together with the climate for which it was computed. The magnitude of the warming effect depends on P: at $P = 0.5$ the drainage share is near zero everywhere (0.01–0.02), so the recoverable share is close to one regardless of temperature, whereas at $P = 1.4$ it falls from 0.61 to 0.09 across the warming range.

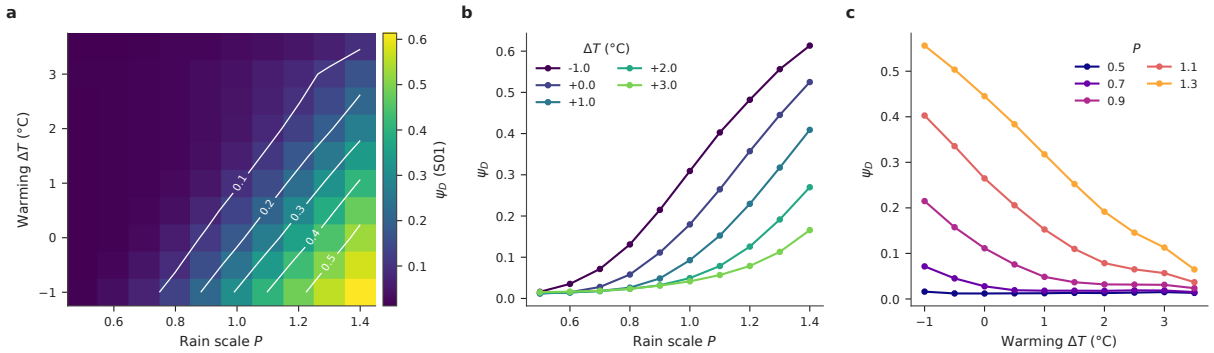


Figure 13: Climate surface of the drainage share ψ_D for coarse loam (S01) from the E11 lattice. (a) ψ_D over rain scale and warming (white contours), (b) ψ_D against P for selected warming levels, (c) ψ_D against warming for selected rain scales.

Alternative formulations and what they establish. Four alternative formulations were tested against the ensemble, and each outcome sharpens the definition of the memory fate. (A1) A universal curve of D/W against W/TAW that collapses all soils does not exist: the curves remain ordered by soil, which establishes that recoverability must be defined per soil and is the reason ϕ_D is a soil-level diagnostic. (A2) Nitrogen leaching does not track drainage with a soil-independent coefficient; it depends on soil and fertiliser timing, so the water diagnostics are the transferable quantities and nitrogen is a soil-conditional downstream response. (A3) Trigger suction and irrigation dose are not first-order controls of recoverability (neither is significant), so the recoverable share is set by soil and climate and is robust to scheduling detail. (A4) Literature conductivities are available for eight of the ten soils, so the conductivity law is calibrated and validated on those eight, with the pore-size parameter n providing the ten-soil comparison. The pore-size parameter n and the memory fate carry comparable information on the loss fate; this is expected, because both are expressions of one pore structure, and Section 4.1 sets out what the flux-based definition adds.

Soil-group profiles of the three fates.

Sand (S03). $\phi_D = 0.85$, the highest of the set. With $TAW = 60$ mm and $\text{cap}_{FC} = 4$ mm the profile cannot hold antecedent water: release saturates at $h_0 \approx 100$ cm (Fig. 6a). Both loss fates are extreme ($\psi_D^{E16} = 1.21$, $\psi_I^{E16} < 0$), and the mulch lattice returns $\psi_D^{E7} = 0.71$ with $\psi_I^{E7} = 0$, that is, none of the mulch saving appears as reduced irrigation. In this soil an evaporation saving is effectively an increase in deep drainage.

Loamy sand and sandy loam (S04, S05). $\phi_D = 0.74$ and 0.64. $\psi_D^{E7} = 0.69$ and 0.66 with slightly negative ψ_I^{E7}. These two soils lie between sand and the loams in n (2.28, 1.89) and also in behaviour; they are the soils for which the K_s law has its largest absolute residuals after sand (Fig. 9b).

Intermediate soils (S06, S08, S02). $\phi_D \approx 0.26$–0.29, with $\psi_D^{E16} = 0.47$, 0.53, 0.19. The coarse sand S02 is anomalous in two ways: it has $n = 1.60$, close to the loams, and its drainage share is therefore similar ($\phi_D = 0.265$), yet its ψ_D^{E16} is the smallest of the set while its ψ_D^{E7} (0.61) is among the largest. This is attributed to the reduced E16 design for S02 (37 valid combinations), and the S02 loss fates are read as indicative.

Fine and structured soils (S07, S01, S09, S10). $\phi_D = 0.02$–0.15; $\psi_D^{E16} \approx 0.30$–$0.36$; $\rho_R \approx 0.64$–0.71. These are the soils in which evaporation savings are most recoverable and in which irrigation compensates most strongly ($\psi_I^{E16} = 0.53$–0.84).

Sensitivity of the conclusions to analytical choices. Three analytical choices could in principle drive the results. (i) *Pooling within P in Eq. (5)* is essential, not cosmetic. A single regression of drainage on $\bar{R}$ over all 100 cells, ignoring P, gives values that differ from the pooled ϕ_D by 0.003–0.04 in most soils but by -0.23, -0.48 and -0.93 in S05, S04 and S03 (the rank correlation between the two estimates is only 0.43), because in free-draining soils rainfall changes both $\bar{R}$ and drainage strongly. The memory fate is therefore defined as the response to antecedent state *at fixed rainfall*, and the results above should not be compared with unpooled slopes. (ii) *Ratio-of-sums in Eq. (6)*: a ratio of means is more stable than a mean of ratios because individual paired ΔET_{nt} can be small when P is low; the bootstrap intervals in Fig. 7b reflect this. (iii) *Soil-level correlations treat each soil as independent*: this inflates significance if soils are correlated through texture, which is why both permutation p values and bootstrap intervals are reported and the interval is emphasised for n versus ψ_D^{E16}, whose lower bound is below zero.

Table 12: Hypothesis register. the test and outcome of every formulation examined. Verdicts refer to the ten-soil simulated ensemble.

ID	Hypothesis	Verdict	Evidence
H0	ϕ_D differs among soils and is ordered by n	supported	$\rho = 0.95$; Table 10
H1	ϕ_D transfers to ψ_D (E16, E7)	supported	$\rho = 0.85$, 0.89
H2	One-parameter K_s law beats structural predictors out of sample	supported (8 soils)	LOSO RMSE 0.058 vs. 0.097–0.183
H3	ϕ_D predicts recoverability better than n itself	comparable to n	n vs. ψ_D^{E7}: $\rho = 0.89$
H4	Warming raises recoverable share	supported	10/10 soils; E11 surface
H5	Wetter climate lowers recoverable share	supported	9/9 soils; E11 surface
H6	Trigger or dose is a first-order modifier	not first-order	$p = 0.16$, 0.32
H7	Nitrogen has no effect (negative control)	as expected	-0.008
H8	Universal D/W–W/TAW collapse	soil-specific	curves remain soil-ordered
H9	N leaching tracks drainage with a common coefficient	soil-specific	soil-dependent

3.2 Accounting audit

The pipeline recovers the water balance (positive control). The singular spectrum of the seven-flux matrix falls steeply, with a smallest singular value of 5.5 compared with about 33 under column permutation (Fig. 14a), and the corresponding right singular vector places weight on rain, irrigation, ET and drainage with opposite signs for inputs and outputs (Fig. 14b). For the four-flux subsystem the null direction, scaled to $c_{ET} = -1$, is

$$\mathbf{c} = (c_{P_r}, c_I, c_{ET}, c_D) = (1.037,\ 0.957,\ -1,\ -1.036), \tag{17}$$

with bootstrap 95 % intervals 1.031–1.045 for rain, 0.940–0.974 for irrigation and -1.052 to -1.023 for drainage, and a smallest singular value of 5.9 against 33.3 under permutation. The pipeline therefore recovers the conservation law $P_r + I - ET - D \approx 0$ with every coefficient within 5 % of the exact value, and the permutation null shows that the null direction is not an artefact of scale. *However, the bootstrap intervals do not contain the exact values*: rain $\pm$ 1 lies below the rain interval, and irrigation lies above its interval. The deviation reflects the nonzero storage term and other unreported fluxes: excluding E8 and E9, whose initial moisture is deliberately unbalanced, gives $(1.021,\ 0.969,\ -1,\ -1.019)$ with a smallest singular value of 3.0, closer to the exact balance but still not exact.

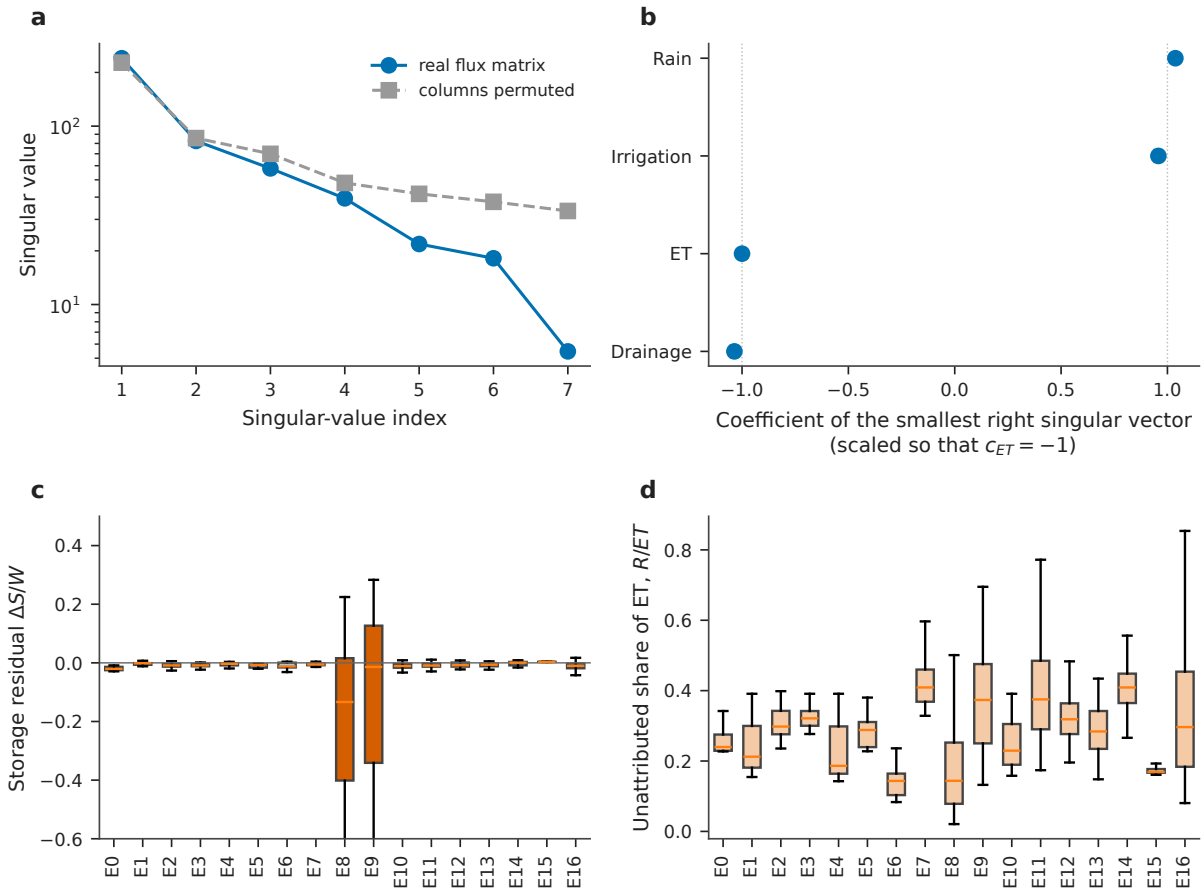


Figure 14: Linear invariants. (a) Singular values of the seven-flux matrix for the real data and for column-permuted data. (b) Coefficients of the smallest right singular vector (scaled so $c_{ET} = -1$) with bootstrap intervals. (c) Storage residual $\Delta S/W$ by experiment: E8 and E9 do not close. (d) Unattributed share of ET, R/ET, by experiment.

The result illustrates both the promise and the discipline needed for law discovery. The method finds the law that is there, without being told which variables are inputs and which are outputs; but its coefficients should not be over-interpreted, because the third decimal of a coefficient reflects what the data do *not* close, not what the law is.

Fitting the four-flux null vector separately within each experiment (Table 13) tests whether the pooled vector Eq. (17) hides heterogeneity. In 11 of the 14 experiments with irrigation all coefficients are within about 5 % of the exact values $(1, 1, -1, -1)$ (E11 is at 5.1 % for drainage; E16, with the extreme corners of the factorial, gives 0.901 for irrigation and -0.919 for drainage, E6 gives 0.961, 0.927 and -0.869; and E12 gives drainage -1.116 are the exceptions). The residual scale $\sigma_q/\sqrt{N}$ is 0.001–0.017 in the well-closing experiments and 0.081 in E16, 0.030 in E6, 0.166 in E8 and 0.219 in E9. E0 (no irrigation variation within the 10 baseline runs), E8 and E9 (rainfed) have a constant zero irrigation column and the invariant is formed from three fluxes; for E8 the drainage coefficient is $+15.5$, a sign of an ill-conditioned decomposition in an experiment

in which storage dominates the balance, which is a cautionary example of how an unconstrained search can return a numerically valid but physically meaningless vector. The method therefore identifies its own failure by the magnitude of $\sigma_q/\sqrt{N}$ and by physical-plausibility checks on the coefficients, which is recommended as a rule.

Table 13: Four-flux null vector fitted within each experiment (scaled to $c_{ET} = -1$); "–" marks a flux with no variation in the experiment (irrigation in E0, E8, E9). $\hat{\sigma} = \sigma_q/\sqrt{N}$ is the residual scale in units of the column root-mean-square.

Experiment	Runs	c_{P_r}	c_I	c_D	$\hat{\sigma}$
E0	10	1.025	–	−1.041	0.004
E1	189	1.009	0.969	−0.979	0.017
E2	197	1.022	0.983	−1.027	0.004
E3	200	1.016	0.995	−1.012	0.004
E4	96	1.006	0.989	−0.983	0.003
E5	300	1.016	0.993	−1.027	0.002
E6	193	0.961	0.927	−0.869	0.030
E7	200	1.008	0.992	−0.990	0.003
E8	1000	1.241	–	15.496	0.166
E9	987	1.182	–	−1.249	0.219
E10	954	1.026	0.972	−1.038	0.010
E11	1000	0.998	1.024	−1.051	0.004
E12	1000	1.016	1.005	−1.116	0.005
E13	975	1.016	0.984	−1.008	0.004
E14	1000	0.982	1.035	−0.974	0.001
E15	1000	1.009	0.966	−0.971	0.003
E16	740	1.018	0.901	−0.919	0.081

The closure residual ε (Eq. (12)) has a median absolute value of 0.9 % of water input over all valid runs and 0.65 % excluding E8 and E9 (95th percentile 2.3 %), so for 15 of the 17 experiments the mass balance holds to within a few percent. In E8 and E9, which vary initial moisture, the median absolute residual is 16 % of W, with the residual reaching ± 0.3–0.6 in the tails (Fig. 14c). This is expected, because these experiments deliberately impose a non-equilibrium initial storage that is released over the 20-year window, but it matters for discovery: a conservation law fitted without storage will appear to be violated in exactly the experiments where hydraulic memory is strongest. A global law should therefore include storage explicitly or be fitted only on near-equilibrium experiments, and the two cases give different coefficients (Eq. (17) and the text following it).

A unattributed term in the ET partition. The ET components that Daisy reports do not add up to ET. The remainder R (Eq. (11)) averages 164 mm yr^{-1}, with an interquartile range of 103–223 mm yr^{-1}, and equals 30 % of ET (median 29 %; range 2–88 %). Its share varies between

experiments from 14 % (E6, E8) to 41 % (E7, E14; Fig. 14d, Table 14). The dependence of R on management is informative (Fig. 15). In E4 (methods), R is 105 mm yr^{-1} for every subsurface-drip configuration (5–60 cm), 183 for surface and 166 for overhead irrigation, so that *where* water is applied matters while the drip depth does not (Fig. 15a). In the rainfed E9 lattice, $R = 50.9 + 0.121\,P_r$ (Fig. 15b), i.e. about 12 % of rainfall, and in E11 (coarse loam)

$$\ln R = 5.75 - 0.355 \ln P + 0.863 \ln E_p + 0.136\,\Delta T, \qquad R^2 = 0.979, \tag{18}$$

where P is the rain scale, E_p the soil-evaporation factor and ΔT the temperature shift, so that R scales almost linearly with the soil-evaporation factor, decreases with rain scale (elasticity -0.36) and increases with warming. The mulch experiments E7 and E14 have the highest median shares (41 %) and the cross-experiment sparse model for $\ln R$ selects $\ln(1 + I)$, n, the mulch capacity and ΔT.

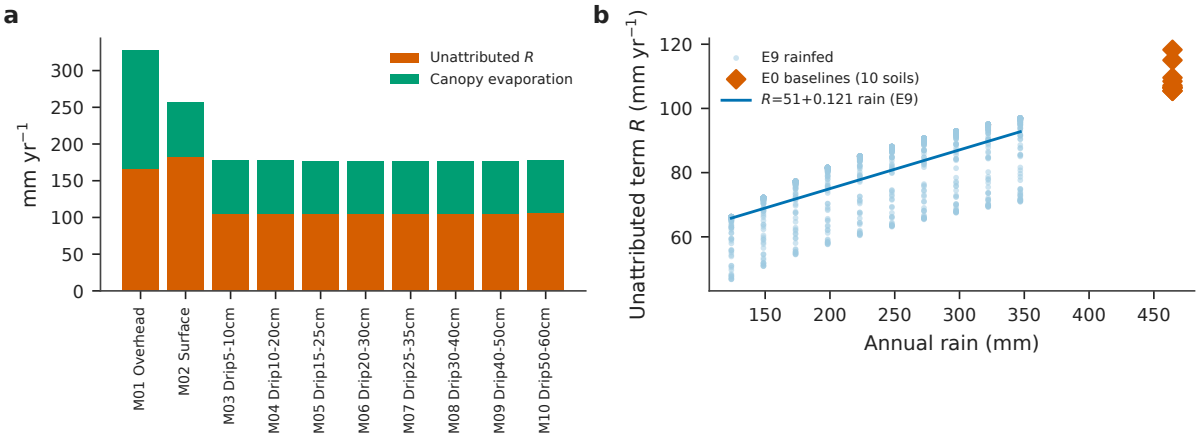


Figure 15: The unattributed term R. (a) R and canopy evaporation in E4 by irrigation method. (b) R against annual rain in E9 (rainfed, light points), with the linear fit and the ten E0 baselines.

Table 14: Unattributed term R by experiment. median R (mm yr^{-1}), median share of ET, and median storage residual $\Delta S/W$.

Experiment	Runs	R (mm)	R/ET	$\Delta S/W$
E0 Baselines	10	107	0.24	−0.021
E1 Doses	189	118	0.21	−0.001
E2 Trigger pressure	197	168	0.30	−0.009
E3 Sensor depth	200	202	0.32	−0.009
E4 Methods	96	103	0.19	−0.003
E5 SHP dynamics	300	162	0.29	−0.007
E6 Mediterranean	193	70	0.14	−0.012
E7 Mulching	200	226	0.41	−0.006
E8 Rain intensity×moisture	1000	39	0.14	−0.133
E9 Soil moisture×rain	987	81	0.37	−0.014
E10 Amount, timing, soil	954	118	0.23	−0.011
E11 Temperature×E_p×rain	1000	296	0.38	−0.009
E12 Sensor, trigger, rain	1000	218	0.32	−0.008
E13 N×water	975	125	0.28	−0.006
E14 Mulch×rain	1000	262	0.41	0.001
E15 Method, depth, amount	1000	103	0.17	0.003
E16 Edge cases	740	157	0.30	−0.010

What is R? The data do not identify it, but they constrain candidates. A process that is independent of drip depth, larger under surface and overhead application, proportional to the soil-evaporation factor, increased under mulch and weakly increasing with warming is consistent with evaporation from the *surface layer* (mulch or litter and ponded water) that is not reported in the soil-evaporation or canopy-evaporation columns. That reading is a hypothesis; the model code has not been inspected to confirm which Daisy state variables sum into the model's ET output, and a software-level audit of the output definitions would be required to turn the hypothesis into a statement. The practical consequence is nonetheless clear: any analysis that treats $T + E_s + E_c$ as ET (as many reduced-form ET-partitioning analyses do) omits about 30 % of the flux, and the partition fractions reported for the ensemble depend on how R is allocated.

A sparse model for R across all experiments has limited out-of-sample skill. The forward-selected model $\ln R = 3.35 + 0.194 \ln(1 + I) + 0.492\, n + 0.145\, \text{cap} + 0.238\, \Delta T$ has an in-sample $R^2 = 0.61$ on the log scale but a leave-one-experiment-out error of 0.51 in $\ln R$, equal to a level-scale R^2 of only 0.30 (RMSE 81 mm) versus 0.50 (RMSE 69 mm) for gradient boosting. Within a single experiment, Eq. (18) is excellent; across experiments it is not, which implies that R depends on interactions among irrigation method, mulch state and demand that the library does not contain.

A dominant drainage axis, with its variance share quantified by bootstrap. The log-ratio transform requires strictly positive components, and drainage is zero or negative (net capillary rise) in 1 429 of the 10 041 runs (14 %) and below 1 mm yr^{-1} in 1 947. The treatment of these zeros is therefore not a detail. The first analysis replaced all components below 10^{-3} mm yr^{-1} by that floor, which gave explained variances of 88.6 %, 7.3 % and 3.3 % for the first three components. Because $\ln(10^{-3})$ lies far from the logs of the other components, the runs with zero drainage then dominate the first axis. Repeating the analysis with floors of 0.1, 1, 5 and 10 mm yr^{-1} gives a first-component share of 74.3, 62.7, 56.5 and 52.3 % (Fig. 16a, Table 15). The result that a single drainage axis dominates is robust, since the loading of drainage on PC1 stays between $+0.82$ and $+0.88$ for every floor and also when only runs with drainage above 5 or 20 mm are used (PC1 share 57 % and 52 %), but its *magnitude* is a function of the data-handling choice and the figure of 88.6 % should not be quoted as a property of the system. The number of components needed to explain 90 % of the variance is two to three in all cases (three components carry 92–98 % for floors from 10^{-3} to 10 mm, with 97.3 % at the 1 mm floor adopted for the loadings, Fig. 16b,c).

With the 1 mm floor, the first component (62.7 %) is a drainage-versus-evaporation axis: drainage loads $+0.84$, soil evaporation -0.37, unattributed R -0.34, transpiration -0.20 and canopy evaporation $+0.06$. Its scores correlate with the drainage fraction D/W at Spearman $\rho = 0.98$, with $\ln(W/TAW)$ at 0.48 and with n at 0.45. The second component (24.8 %) separates canopy evaporation ($+0.77$) from soil evaporation (-0.56), and the third (9.8 %) opposes R ($+0.83$) to soil evaporation (-0.43) and transpiration (-0.26). The choice of the 1 mm floor was made *after* seeing the sensitivity and is justified only as a value close to the resolution of annual flux accounting; the full sensitivity is reported so that readers can judge. The practical conclusion is that the drainage fraction is, to a first approximation, the leading summary of water partitioning (its score correlation is 0.98), and that a second axis, the canopy-versus-soil evaporation split, is needed for a complete description.

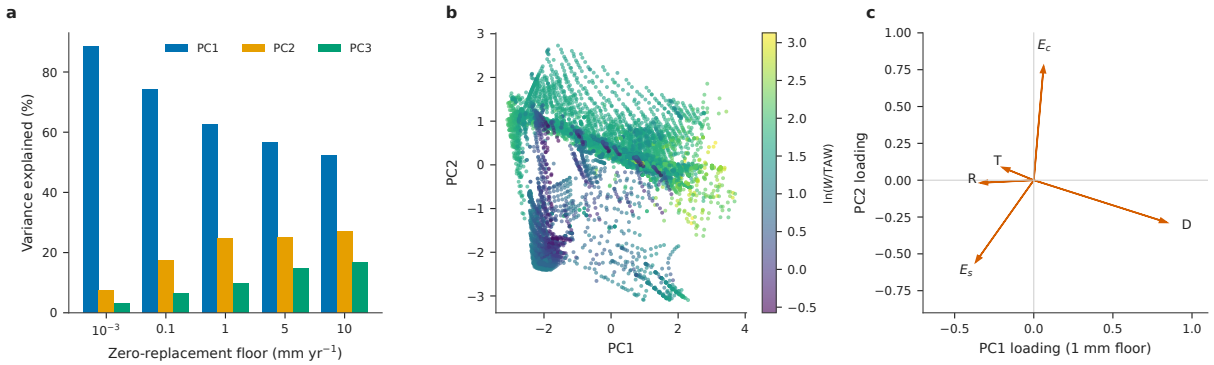


Figure 16: Partition manifold and its sensitivity to zero handling. (a) Variance explained by the first three clr-PCA components for five zero-replacement floors. (b) Scores on the first two components (1 mm floor) coloured by $\ln(W/TAW)$. (c) Loadings of the five fates on the first two components.

Table 15: Sensitivity of the clr-PCA to the zero-replacement floor (variance explained, %).

Floor (mm yr^{-1})	PC1	PC2	PC3	PC4
10^{-3}	88.6	7.3	3.3	0.8
0.1	74.3	17.3	6.6	1.7
1	62.7	24.8	9.8	2.6
5	56.5	25.1	14.8	3.6
10	52.3	27.1	16.8	3.8

Averaged over the 15 near-equilibrium experiments, the five fates (Fig. 17a) show the expected gradient: drainage takes 34.5 % of water input in sand (S03), 28.5 % in loamy sand and 5.4 % in coarse loam (S01), 8.5 % in silty clay loam, while transpiration rises from 19 % in sand to 31–33 % in the loams. The unattributed term, in contrast, is remarkably constant across soils: R takes 22–26 % of W in nine of ten soils (31 % in coarse loam, which has the mulch and temperature lattices) and its share of ET is 23–39 % (Fig. 17c), the largest in sand, where there is least else to evaporate. Since R is almost soil-independent in share of W, it behaves as a surface-driven term rather than a soil-hydraulic one, consistent with the surface-layer hypothesis. The stacked fractions sum to 0.96–1.01 of W (the remainder being storage change and runoff). Panel b is a reminder that coarse loam contributes 4 419 of the near-equilibrium runs while each other soil contributes 290–419.

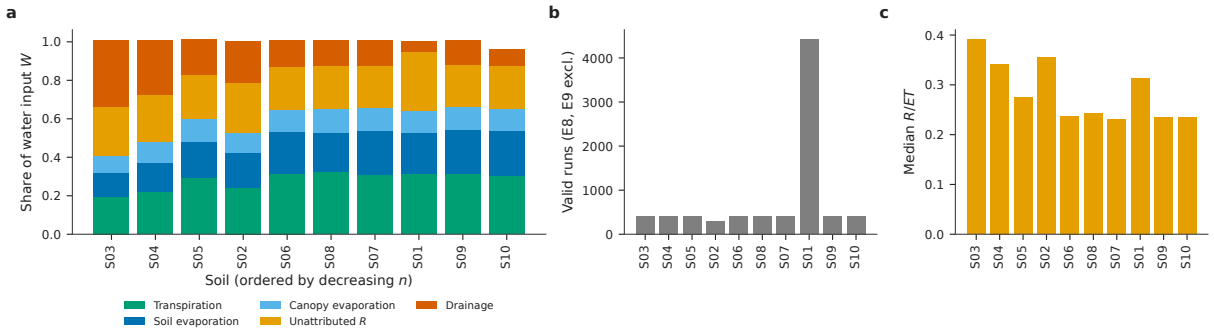


Figure 17: Water fates by soil in the 15 near-equilibrium experiments (E8, E9 excluded). (a) Mean fate shares of water input W, soils ordered by decreasing n. (b) Number of valid runs per soil. (c) Median share of the unattributed term in ET.

A compact evapotranspiration law with out-of-sample skill. Forward selection over the library selects three terms for ET (Table 21): $\ln W$, n and $\ln(1+I)$, giving

$$\ln ET = 2.503 + 0.646\,\ln W - 0.299\,n + 0.029\,\ln(1+I), \tag{19}$$

with ordinary-least-squares standard errors 0.027, 0.0051, 0.0045 and 0.0011 (which understate uncertainty because runs are not independent) and in-sample $R^2 = 0.908$. Leave-one-experiment-out $R^2 = 0.898$ and leave-one-soil-out $R^2 = 0.904$, against 0.876–0.879 (LOEO) and 0.881–0.894 (LOSO) for gradient boosting in two independent runs, so the three-term law matches or exceeds the benchmark under both groupings (Fig. 18a,b). The RMSE in leave-one-experiment-out prediction is 68.8 mm yr^{-1} (MAPE 9.2 %). The fitted coefficients are stable across the 17 folds (Fig. 18c; coefficient means 2.51, 0.645, -0.298, 0.029 with standard deviations of 0.077, 0.017, 0.017, 0.0023).

Equation (19) has a straightforward physical reading. The exponent on W (0.646) means that ET rises sublinearly with water supply, the signature of an energy- or storage-limited regime also encoded in Budyko-type curves (Budyko, 1974; Milly, 1994; Zhang et al., 2001): a 10 % increase in supply raises ET by 6.5 %, and the remainder goes to drainage, runoff and storage. The coefficient on n (-0.299) implies a factor $\exp(-0.299 \times 1.45) = 0.65$ in ET between the most uniform (S03, $n = 2.68$) and the most graded (S10, $n = 1.23$) soil at the same W: soils that drain freely retain less water for evaporation, in agreement with the hydraulic-memory results of the memory analysis (Section 3.1). The small positive coefficient on $\ln(1+I)$ (0.029) indicates that, at equal total input, a larger irrigated share raises ET slightly, plausibly through the wetting of the surface that increases R and E_s. The law is *not* a fundamental conservation law, nor a model of

mechanism; it is a compact summary of how the model's ET responds across 17 designs.

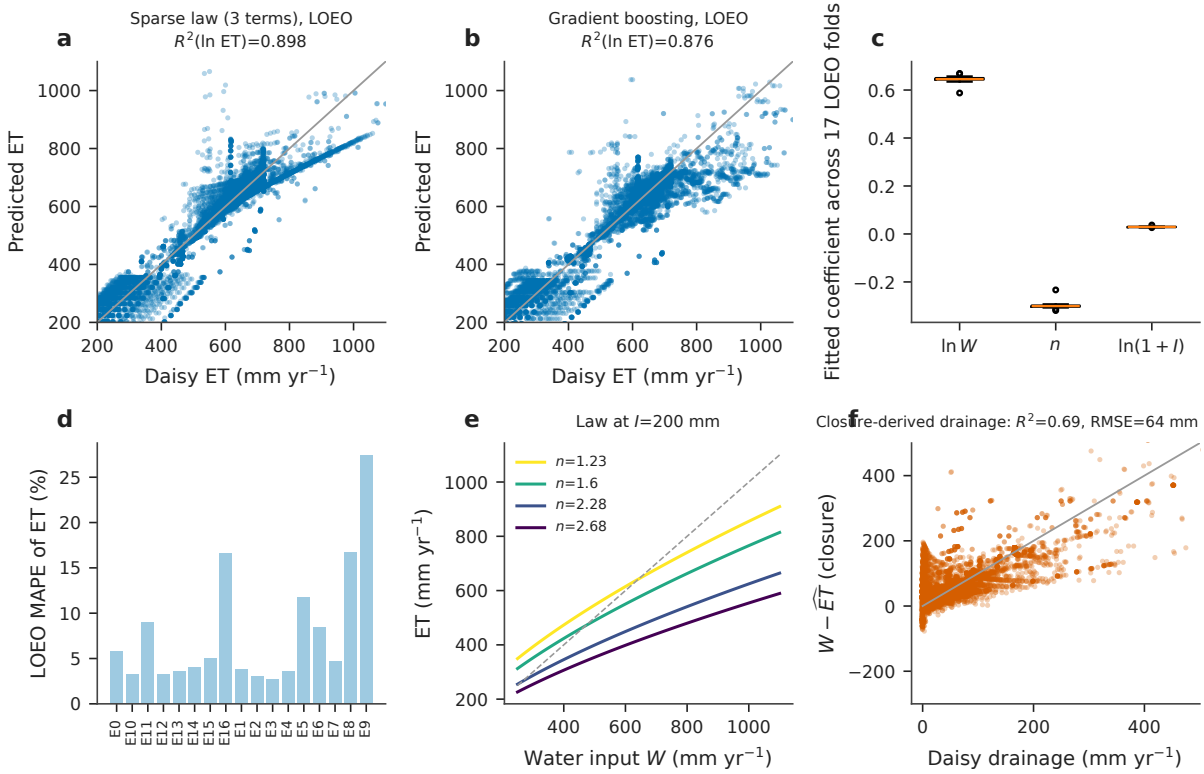


Figure 18: Compact ET law. (a) Leave-one-experiment-out predictions of the three-term law versus Daisy ET. (b) Gradient-boosting benchmark. (c) Fitted coefficients across the 17 LOEO folds. (d) LOEO MAPE by experiment. (e) The law at $I = 200$ mm for four soils (grey dashed: ET $= W$). (f) Drainage derived from closure, $W - \widehat{ET}$, versus Daisy drainage ($R^2 = 0.69$, RMSE 64 mm; E8 and E9 excluded).

The 17 experiments were resampled with replacement (2 000 cluster-bootstrap resamples) and the cross-validated predictions of the law were compared with those of a gradient-boosting model that uses the same three inputs ($\ln W$, n, $\ln(1 + I)$; LOEO $R^2 = 0.866$). The paired difference in R^2 is $+0.032$ with a 95 % interval of 0.004–0.087, so the law is better than a flexible learner given the same information in this resampling, with the caveat that the law's terms were selected with the same folds. With the *full* library the benchmark reaches 0.876–0.879, and the two are regarded as comparable rather than claiming superiority.

The error is not uniform (Table 16). Mean absolute percentage errors are 2.6–5.8 % in most one- and two-factor experiments, but 16.6 % in E16 (edge cases: extreme combinations), 16.7 % in E8 and 27.4 % in E9, the two experiments that violate storage closure, and 11.7 % in E5 (time-varying soil hydraulic properties, where a single static n is wrong by construction). The law is therefore valid where the water balance is near equilibrium and soil hydraulics are static; it should not be extrapolated to initial-state perturbations or to dynamic soil structure.

Table 16: Leave-one-experiment-out performance of Eq. (19) by experiment: MAPE (%) and mean error (predicted minus Daisy, mm yr^{-1}).

Experiment	Runs	MAPE (%)	Mean error (mm)
E0 Baselines	10	5.8	−25.2
E1 Doses	189	3.8	9.8
E2 Trigger pressure	197	3.0	−6.6
E3 Sensor depth	200	2.6	−8.0
E4 Methods	96	3.6	9.4
E5 SHP dynamics	300	11.7	−63.8
E6 Mediterranean	193	8.5	41.5
E7 Mulching	200	4.7	10.6
E8 Rain intensity	1000	16.7	−11.3
E9 Soil moisture	987	27.4	12.9
E10 Amount, timing	954	3.3	1.3
E11 Temperature, E_p, rain	1000	9.0	−62.6
E12 Sensor, trigger, rain	1000	3.3	−15.2
E13 N, water	975	3.5	−3.7
E14 Mulch, rain	1000	4.0	7.5
E15 Method, depth, amount	1000	5.0	29.0
E16 Edge cases	740	16.6	34.5

Table 17 gives the closure residual ε for every experiment. In 15 experiments the median absolute residual is below 2.1 % of W and the 95th percentile below 4 % (E6, the Mediterranean-pollution scenarios, is the exception at 10.7 %); in E8 and E9 the median absolute residual is about 16 % and the 95th percentile exceeds 100 % of W. For E8 the mean (signed) residual is negative, −13 %: the profile is gaining water, that is, the run begins drier than the equilibrium state and refills. The law Eq. (19) was fitted on water inputs from 124 to 2 762 mm yr^{-1} (ET of 95–1 600 mm yr^{-1}), n from 1.23 to 2.68, in a temperate climate, and it should be applied only within this domain and with near-equilibrium storage.

Table 17: Closure residual $\varepsilon = (W - ET - D)/W$ by experiment (per cent of water input): median signed, median absolute and 95th percentile of absolute residual.

Experiment	Runs	Median ε	Median $\lvert\varepsilon\rvert$	95th pct. $\lvert\varepsilon\rvert$
E0	10	-2.06	2.06	2.9
E1	189	-0.12	0.33	1.2
E2	197	-0.87	0.87	2.3
E3	200	-0.87	0.87	2.0
E4	96	-0.33	0.34	1.2
E5	300	-0.74	0.74	1.9
E6	193	-1.19	1.57	10.7
E7	200	-0.57	0.57	1.2
E8	1000	-13.35	15.76	105.8
E9	987	-1.40	16.87	108.5
E10	954	-1.08	1.17	2.5
E11	1000	-0.93	0.94	1.8
E12	1000	-0.82	0.83	1.8
E13	975	-0.64	0.64	2.3
E14	1000	0.09	0.40	1.5
E15	1000	0.34	0.34	0.4
E16	740	-1.01	1.22	3.8

The ensemble is heavily imbalanced: 5 519 of the 10 041 runs (55 %) are coarse loam (S01) because the large lattices E8, E11, E12, E14 and E15 use only that soil, so soil-level behaviour in the other nine soils rests on about 400–520 runs each. The LOEO residuals of Eq. (19) by soil (Table 18) show that the law over-predicts ET in the fine soils (mean error $+22$, $+20$ and $+31$ mm in S08, S09 and S10, where positive means over-prediction relative to Daisy, since errors are predicted minus Daisy) and has the largest percentage errors in sand (15.1 %) and loamy sand (13.7 %), the soils for which n is most extreme. A single shape parameter therefore does not remove all soil dependence, and the MAPE of about 7.5–15 % by soil is consistent with the overall 9.2 %.

Table 18: LOEO residuals of the ET law by soil (predicted minus Daisy, mm yr^{-1}). MAPE: mean absolute percentage error; RMSE in mm yr^{-1}.

Soil	Texture	Runs	Mean error	MAPE (%)	RMSE
S01	Coarse loam	5519	-10.7	7.5	58.9
S02	Coarse sand	390	15.8	8.6	69.8
S03	Sand	506	-13.6	15.1	63.8
S04	Loamy sand	519	-4.8	13.7	57.9
S05	Sandy loam	519	-3.8	10.9	56.1
S06	Loam	519	-0.2	9.9	77.5
S07	Silt loam	519	0.4	10.6	106.2
S08	Sandy clay loam	519	22.2	10.2	57.8
S09	Clay loam	519	19.7	9.9	92.8
S10	Silty clay loam	512	31.2	11.6	107.4

Table 19: Forward-selection path for the ET law. LOEO R^2_{cv} after each added term; and for other targets (first terms).

Target	Term added	R^2_{cv} (LOEO)
ln *ET*	1 ($\ln W$)	0.801
	2 (n)	0.895
	3 ($\ln(1+I)$)	0.898
$\ln T$	1 ($\ln W$)	0.581
	2 (n)	0.641
$\ln E_s$	1 (n)	0.070
	2 (mulch)	0.237
	3 ($\ln P_r$)	0.308
$\ln E_c$	1 ($\ln W$)	0.641
	2 ($\ln P_r$)	0.651
	3 (I/W)	0.668
ln yield	1 ($\ln W$)	0.531
	2 ($\ln(1+I)$)	0.549
	3 ($\ln P_r$)	0.599
	4 (mulch)	0.604
ln DPF	1 ($\ln(W/TAW)$)	0.193
	2 ($\ln \alpha$)	0.260
	3 (I/W)	0.422
	4 (ΔT)	0.433

If E8 and E9, which break closure, are removed from both fitting and validation (15 experiments, 8 054 runs), the same three terms give LOEO $R^2 = 0.882$, LOSO $R^2 = 0.881$, RMSE 65.9 mm and MAPE 5.8 %, so the law is more accurate (relative error 5.8 % versus 9.2 %) in the domain for which it is intended. The marginal contribution of the terms is clearer in this subset: $\ln W$ alone gives LOEO $R^2 = 0.600$ (RMSE 138 mm), adding n raises it to 0.857 (RMSE 74 mm) and adding $\ln(1+I)$ to 0.882. A gradient-boosting model restricted to the same three

inputs reaches LOEO $R^2 = 0.778$, below the law, so the advantage of the compact form is not that it uses more information but that its functional form (power law in W, exponential in n) extrapolates across experiments better than piecewise-constant trees, which cannot extrapolate beyond the range of the training experiments. (This comparison uses three inputs and 15 folds and is not directly comparable with the full-library benchmark of Table 21.)

Table 20 applies Eq. (19) to three combinations of water input and irrigation in three soils. At equal input, ET is about 35 % lower in sand than in silty clay loam (263 versus 406 mm at $W = 400$ mm), because water not held in the root zone drains; the implied drainage plus storage change $W - ET$ is 137 mm in sand and negative-to-small in the fine soils, in line with the memory results. The example is illustrative; its uncertainty is the RMSE of 66–69 mm.

Table 20: ET (mm yr^{-1}) predicted by Eq. (19) for three water inputs W with irrigation I.

	$W = 400$, $I = 0$	$W = 650$, $I = 200$	$W = 1000$, $I = 300$
S03 sand ($n = 2.68$)	263	420	561
S01 coarse loam ($n = 1.353$)	391	624	834
S10 silty clay loam ($n = 1.23$)	406	648	865

Table 21 collects all sparse laws that were fitted. Transpiration is moderately predictable by a two-term law ($\ln T = -1.329 + 1.127 \ln W - 0.501\, n$; LOEO $R^2 = 0.641$, equal to gradient boosting 0.642). Canopy evaporation is predicted at LOEO 0.668 (gradient boosting 0.691) by $\ln E_c = -13.3 - 0.470 \ln W + 3.157 \ln P_r + 4.240\, I/W$, whose large coefficients reflect the narrow range of E_c and should not be interpreted mechanistically. Crop yield is predicted at LOEO 0.604 (gradient boosting 0.650). In contrast, soil evaporation is poorly predicted by sparse terms ($\ln E_s = 3.271 - 0.770\, n - 0.380$ mulch $+ 0.462 \ln P_r$; LOEO 0.308 versus 0.547 for gradient boosting; LOSO 0.348 versus 0.476), indicating that interactions that the library does not contain control E_s. The drainage fraction (DPF $= D/W$), the natural target given the PCA result, is the worst: the best sparse model with exogenous predictors has a LOEO R^2 of 0.43 and a LOSO R^2 of 0.22, against 0.50 (LOEO) for gradient boosting, and allowing partition predictors in the library raises gradient boosting to 0.59 but does not change the sparse law. The shortfall is a negative result for compact drainage laws.

Table 21: Sparse laws and benchmark. cross-validated R^2 on the log scale. Selected terms in order of selection.

Target	Selected terms	Sparse LOEO	Sparse LOSO	GBM LOEO	GBM LOSO
$\ln ET$	$\ln W$, n, $\ln(1+I)$	0.898	0.904	0.876–0.879	0.881–0.894
$\ln T$	$\ln W$, n	0.641	0.652	0.642	0.659
$\ln E_s$	n, mulch, $\ln P_r$	0.308	0.348	0.547	0.476
$\ln E_c$	$\ln W$, $\ln P_r$, I/W	0.668	0.669	0.691	0.806
ln yield	$\ln W$, $\ln(1+I)$, $\ln P_r$, mulch	0.604	0.617	0.650	0.743
ln DPF	$\ln(W/TAW)$, $\ln\alpha$, I/W, ΔT	0.433	0.217	0.498	–
$\ln R$	$\ln(1+I)$, n, cap, ΔT	0.41[†]	0.43[†]	0.43–0.47[†]	–

[†] R^2 on the log scale for R; level-scale R^2 is 0.30 (sparse) and 0.50–0.51 (gradient boosting).

Because the balance holds to within a few percent in 15 of 17 experiments, drainage can be derived by closure as $\hat{D} = W - \widehat{ET}$, which converts a hard prediction problem (drainage) into one with a good compact law (ET). On the 8 054 runs outside E8 and E9 this yields $R^2 = 0.69$ and RMSE 64 mm against Daisy drainage (Fig. 18f), and negative predicted drainage in 4.8 % of runs. The accuracy is limited by the relative size of the quantities and is not uniform (Table 22): the mean absolute error is smallest, 21–28 mm, for Daisy drainage between 10 and 200 mm (relative error 19–66 %), but the closure-derived value overpredicts by 32 mm on average for drainage below 10 mm (1 944 runs) and *underpredicts* by 69 mm and 308 mm for drainage of 200–500 and above 500 mm. The ET law is smooth in W and cannot represent the large drainage events, so closure-derived drainage is a useful consistency check for moderate drainage and not a predictor of extremes.

Table 22: Closure-derived drainage $W - \widehat{ET}$ (LOEO ET law) against Daisy drainage by drainage class (E8, E9 excluded). Bias = predicted minus Daisy.

Daisy drainage (mm yr^{-1})	Runs	Bias (mm)	Mean absolute error (mm)
< 10	1944	+31.7	39.7
10–50	2631	+8.2	21.0
50–100	1736	−2.2	25.6
100–200	1284	−13.7	28.3
200–500	377	−68.8	104.0
> 500	82	−308.1	318.6

Irrigation compensation is regime-specific. A natural candidate invariant is the compensation of rainfall changes by irrigation, κ (Eq. (16)). It is not universal (Fig. 19, Table 23). In single-soil lattices the slope of W on rain is 0.58 ± 0.02 in E11 ($\kappa = 0.42$) and 0.33 in E12 and E14 ($\kappa = 0.67$), whereas in E16 the pooled κ across the ten soils is 0.159 (bootstrap 95 % interval 0.134–0.181) and the per-soil values range from -0.014 (sand) to 0.303 (coarse loam). Within

E16, κ falls from 0.193 to 0.123 as E_p is raised from 0.3 to 1.2, from 0.188 to 0.129 with warming from $+2$ to $+6$ °C, and from 0.211 to 0.107 with trigger suction from 50 to 2000 cm, but rises from 0.079 to 0.241 with irrigation dose from 1.2 to 30 mm. Compensation therefore depends on the scheduling rule and soil, and a single value of κ should not be used in water-allocation arguments.

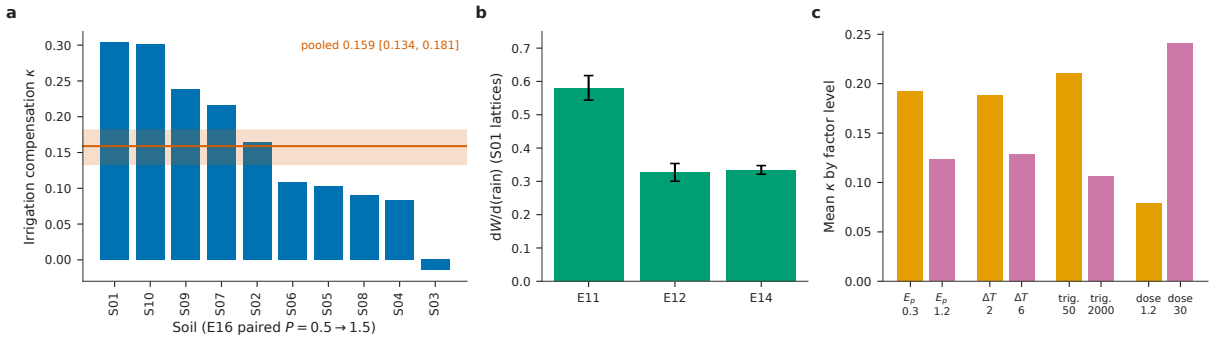


Figure 19: Irrigation compensation κ is not universal. (a) Per-soil κ in E16 (paired $P = 0.5 \rightarrow 1.5$) with the pooled estimate and bootstrap interval. (b) Slope of water input on rain in single-soil lattices (95 % intervals). (c) Mean κ by factor level in E16.

Table 23: Compensation $\kappa = -\Delta I/\Delta P_r$ and the response of ET and drainage to rain, per soil in E16.

Soil	κ	$\Delta ET/\Delta P_r$	$\Delta D/\Delta P_r$	Pairs
S01	0.303	0.512	0.186	32
S02	0.164	0.346	0.495	8
S03	−0.014	0.473	0.550	32
S04	0.084	0.473	0.453	32
S05	0.102	0.551	0.358	32
S06	0.108	0.619	0.281	32
S07	0.216	0.542	0.246	32
S08	0.090	0.613	0.308	32
S09	0.238	0.515	0.253	32
S10	0.302	0.477	0.237	32

The unattributed term and the recoverable share. The memory analysis measures an evaporation change by the non-transpiration term $E_{nt} = ET - T$, which carries the unattributed term R with it. A user who read the reported soil evaporation E_s alone would measure a different quantity. For the mulch experiment E7 the slope of drainage on E_s was computed in place of the slope on E_{nt}, with the same regression that gives ψ_D in the memory analysis (which reproduces the archived values for all ten soils; Table 24). The change of E_s carried a median of 47 % of the change of E_{nt} (range -0.40 to 0.56), the drainage share differed by more than 0.1 in 6 of 10 soils, and in 2 coarse soils (S03, S04) it changed sign (-0.93 and -0.35 against 0.71 and 0.69). The partition that the model

reports therefore cannot be used to judge whether a saving is recoverable; the closed quantity E_{nt} can. The analysis (`p1_audit_link.py`, output `results/p1_audit_link.json`) concerns one experiment of the model and is not a statement about measured soils.

Table 24: Drainage share of the change of evaporation in the mulch experiment E7 (20 combinations per soil), computed against the closed non-transpiration term $E_{nt} = ET - T$ (as in the memory analysis) and against the reported soil evaporation E_s alone. f_{E_s} is the share of the change of E_{nt} that is carried by E_s.

Soil	ψ_D against E_{nt}	ψ_D against E_s	f_{E_s}
S01	0.30	0.31	0.56
S02	0.61	0.38	0.46
S03	0.71	-0.93	-0.37
S04	0.69	-0.35	-0.40
S05	0.66	0.16	0.07
S06	0.52	0.41	0.48
S07	0.40	0.42	0.52
S08	0.55	0.36	0.39
S09	0.43	0.44	0.50
S10	0.41	0.45	0.50

4. Discussion

4.1 Memory analysis

Established findings and the operational definition of connectivity. Within a mechanistic simulator, the fraction of an evaporative change that is recoverable depends on the soil in a way that is ordered by the pore-size-distribution parameter and by saturated conductivity, and the dependence is replicated under three different perturbations (antecedent moisture, evaporation factor, mulch). A simple rate-competition law explains the memory fate with a single parameter. Two climate drivers act in opposite directions on recoverability: warming increases the recoverable share (more of the extra evaporation is met by irrigation and transpiration rather than lost water) and a wetter climate reduces it. These are the findings the data support.

Pore connectivity is given here an operational definition in terms of water fluxes. In the Mualem conductivity function, connectivity enters through the exponent λ and the tortuosity of the pore space (Ghanbarian et al., 2013; Mualem, 1976), and through n and K_s; with $\lambda = 0.5$ for all soils, the connectivity contrast among the ten soils is carried by the pore-size distribution and the saturated conductivity, and the memory fate measures its hydrological consequence directly. This is the useful reading: connectivity is expressed as a number, the share of released storage

that leaves by drainage, that can be measured in a lysimeter without imaging the pore network. The natural complement is to pair the flux diagnostics with X-ray computed-tomography network metrics on the same soils, for example following Leuther and Diamantopoulos (2024), so that the geometric and hydrological descriptions of connectivity can be related.

Interpretation of why drainage competes with evaporation. The soil's matrix acts as a reservoir with a conductance. When the profile is wetter than the field-capacity state, the unsaturated conductivity in coarse soils falls only slowly with suction, because a narrow pore-size distribution implies a rapid but late desaturation (large n), so the drainage flux stays high until the pores empty (Brady and Weil, 2016; Hillel, 1998). In fine soils, conductivity drops by orders of magnitude over the same suction range and the store is retained. When evaporation changes, the change in water remaining in the profile is shared between drainage (which responds on the timescale of K) and the irrigation or root uptake that responds on the timescale of the scheduler or crop. The parallel-sink form Eq. (8) captures the competition between those timescales in its simplest form. The fitted q^* being much larger than ET reflects that the relevant competition occurs during short, wet intervals after rain or irrigation, when conductivities are near saturation and drainage rates are of order of K_s itself, and not over the annual mean.

The two signs are consistent with the same mechanism. Warming raises evaporative demand and dries the profile, so there is less water in the high-conductivity moisture range and less drainage to be displaced; extra evaporation then draws more from irrigation and transpiration. Rainfall has the opposite effect, keeping the profile wet and drainage high. That nitrogen shows no effect is a useful negative control because N influences yield and N leaching but is not coupled to the water partition in the model.

Where ρ_R is low, in sands and loamy sands, a mulch saves much less irrigation water than its evaporation suppression suggests, because the saved water becomes deep drainage. Under those conditions more benefit may come from changing irrigation timing and depth than from reducing evaporation (Fereres and Soriano, 2007; Playán and Mateos, 2006). In contrast, in fine soils (S07, S09, S10; $\rho_R = 0.64$–0.69) reducing evaporation translates into reduced irrigation need, supporting the use of residue mulches (Balwinder-Singh et al., 2011; Vuaille et al., 2024). These conclusions are made for annual, 20-year means in a temperate climate and a one-dimensional model, and do not address field heterogeneity, preferential flow or the effects of mulch on infiltration and runoff (Beven and Germann, 2013; Jarvis, 2007). A companion software article (Paper 2) embeds a reduced form of these relationships in a decision-support tool and reports its agreement

with the simulations.

The competing rate as a physical scale. The fitted competing rate $q^* = 71.5\,\mathrm{cm\,d^{-1}}$ (bootstrap 58–105, jackknife 66–80) is two orders of magnitude above mean evapotranspiration, and this is a finding in its own right. It identifies where the competition between drainage and withdrawal is decided: in the short wet intervals after rain or irrigation, when the profile is near saturation and drainage proceeds at rates of the order of K_s, not over the seasonal mean of uptake. The law $\phi_D = K_s/(K_s+q^*)$ then has two regimes with distinct consequences. For $K_s \ll q^*$ (the fine soils, down to $1.7\,\mathrm{cm\,d^{-1}}$) the memory fate is proportional to K_s, so the uncertainty of a pedotransfer estimate of K_s passes to ϕ_D in proportion; for $K_s \gtrsim q^*$ (the coarse soils, up to $713\,\mathrm{cm\,d^{-1}}$) the fate saturates, so that further increases in conductivity change little. Both regimes are visible in the data, and the transition at $K_s = q^*$, where half of the released storage drains, is a prediction that can be located in a soil collection.

The law also compares favourably with the structural proxies that are available without conductivity data. In leave-one-soil-out validation on the eight soils with literature conductivities, the rate-competition law has an RMSE of 0.058, against 0.100 for n, 0.064 for the field-capacity saturation $S_{e,100}$ and 0.088 for the plant-available water, and 0.338 for the mean alone. On all ten soils the structural proxies give 0.097, 0.136 and 0.183 against 0.318 for the mean. The three proxies are collinear across the soil series, so their performance is similar and their combination adds little; K_s is the axis that is not a function of n, and it is the one that the law uses. Because K_s was taken from independent literature class means (Carsel and Parrish, 1988) and not from the simulator inputs, the law links an emergent property of the simulated soils to an externally measured property, so the agreement is not circular. Finally, q^* is a quantity with the units of a flux, so the law yields a testable prediction for weighing lysimeters (Section 4.1, testable predictions) that a structural index does not.

Relation to earlier work. Field and modelling studies of mulching document that evaporation falls, that the transpiration share of evapotranspiration rises, and that the response is stronger in finer-textured soils (Kang et al., 2025; Zheng et al., 2021); they report the size of the evaporation reduction and its partition, and not the share of the saved water that subsequently drains as a function of the soil's hydraulic properties, which is the quantity defined here. This analysis complements the soil-moisture memory literature (Koster and Suarez, 2001; Seneviratne et al., 2010) by quantifying not how long a moisture anomaly persists but where released storage goes. It extends the Budyko framework (Budyko, 1974; Milly, 1994), which treats annual partitioning

as a function of climate aridity and (via a parameter) catchment properties, by showing that the sub-annual competition between drainage and withdrawal, which is not captured by the single Budyko parameter, is soil-controlled. It is consistent with the finding that hydrologic connectivity constrains the partitioning of terrestrial fluxes (Good et al., 2015), and with the efficiency paradox (Grafton et al., 2018; Ward and Pulido-Velazquez, 2008), to which it adds a soil-scale mechanism: the saved water may be drainage. The need for structure-aware soil hydrology in land models (Fatichi et al., 2020; Vereecken et al., 2016, 2019) is supported by the strong dependence of ϕ_D on the pore-size distribution.

Three testable predictions follow from the results. First, the drainage share of released antecedent storage should increase with saturated conductivity along the saturating form of the rate-competition law, so that a soil with K_s equal to $q^* = 71.5\,\mathrm{cm\,d^{-1}}$ releases half of its stored water as drainage. Second, the recoverable share of an evaporation change should decrease in wetter climates and increase under warming, as found in nine of nine and ten of ten soils respectively. Third, the ranking of soils by ϕ_D should be preserved when the weather record is replaced by a different climate, because the ranking is set by hydraulic conductance and not by the weather. Each prediction can be tested with additional virtual experiments, and the first two with lysimeters in which drainage and evapotranspiration are measured on contrasting soils under imposed evaporation reduction by mulching.

In management terms, the results indicate that the saving obtained from mulching or from a lower evaporation factor is largely recoverable on fine-textured soils, where most of it appears as water that remains in the root zone, and is largely drainage on coarse soils, where it leaves the profile before the crop can use it. A mulch recommendation therefore requires a statement of the soil and of the climate, and the diagnostics defined here supply the quantities needed to make it. In land-surface and crop models the same diagnostics can be computed from a pair of runs, and they offer a compact check on whether a model's response to an evaporation change is plausible for the soil in question.

Relation of ϕ_D to the van Genuchten parameter n. The memory fate rank-orders with n (Spearman $\rho = 0.95$), as it should, since both derive from the pore-size distribution. They differ in kind. The parameter n is a static, geometric descriptor, whereas ϕ_D is an operational quantity defined on water fluxes: the fraction of released antecedent storage that leaves as drainage, which depends on the withdrawal rate imposed by climate and crop as well as on the soil. The rate-competition law adds the saturated conductivity K_s, which is not a function of n, and maps soil properties onto

a single coefficient that can be applied directly to an evaporation-saving claim. The contribution is therefore the definition, the mechanism and the transferability across experimental designs ($\rho = 0.85$ and 0.89), with a prediction accuracy that is at least that of the structural proxies and, on the eight soils with conductivity, better than any of them.

Drivers of soil hydraulic properties and the soil–plant–atmosphere continuum. The diagnostics defined here depend on soil hydraulic properties only through K_s and the pore-size parameter n, but those properties are not fixed attributes of a soil. They are set by a small number of drivers and they respond to the plants and the atmosphere that the soil exchanges water with, so that the recoverability of an evaporation saving is a property of a coupled system (Fig. 22, Table 25).

Texture and pore-size distribution. The primary control is particle-size distribution, which determines the pore-size distribution and therefore the retention curve and the conductivity function. Class-mean parameters and pedotransfer functions (Carsel and Parrish, 1988; Saxton and Rawls, 2006; Schaap et al., 2001; Weber et al., 2024) encode this control, and it is the control carried by the ten soils of this ensemble (Fig. 22a). Across them, K_s spans more than two orders of magnitude (from 1.7 to 713 cm d^{-1} in the eight soils with literature values), and the rate-competition law Eq. (8) maps this range onto a memory fate between 0.02 and 0.85.

Organic matter, aggregation and tillage. Organic matter, biological activity and aggregation create structural porosity that raises K_s and broadens the pore-size distribution relative to the texture-class value, and tillage and traffic change structure in space and in time (Brady and Weil, 2016; Hillel, 1998; Strudley et al., 2008). Because the saturated conductivity of structured soils can differ from the texture-class value by a factor of several, structural drivers move a soil along the curve of Fig. 22a. Starting from $K_s = 20$ cm d^{-1}, for which Eq. (8) gives $\phi_D = 0.22$, a threefold increase in K_s gives 0.46 and a threefold decrease gives 0.09. These values are implications of the fitted law and are not simulation results; the ensemble contains tillage and consolidation treatments (E5), but the memory fate was not computed for them in this paper.

Roots, macropores and non-equilibrium flow. Root channels, earthworm burrows and shrinkage cracks conduct water preferentially and can decouple drainage from the matrix retention curve (Beven and Germann, 1982; Jarvis, 2007). A matrix-flow model, as used here, does not represent this pathway, so the drainage shares reported are those of the matrix alone; in a soil with active macropores the drainage share of released storage would be expected to be at least as large as the matrix value, and the recoverable share correspondingly smaller. This is a direction of change

implied by the mechanism and not a measured effect. The recoverable share reported here should therefore be read as an upper bound for soils with active macropore flow.

Wetting–drying, freezing and surface condition. Cycles of wetting and drying, and of freezing and thawing, rearrange pore structure and alter retention and conductivity (Leuther and Schlüter, 2021); residue and mulch cover change the surface hydraulic boundary and the infiltration and evaporation fluxes that act on it (Leuther et al., 2026; Vuaille et al., 2024). These drivers act on the same two properties, K_s and n, and are therefore expressible in the diagnostics of this paper once their effect on the properties is known.

Coupling to plants and atmosphere. Plants link the hydraulic properties to the atmosphere in both directions. Root distribution and uptake determine how much of the stored water is withdrawn before it can drain, which is the competing process in Eq. (8); evaporative demand, set by radiation, temperature and humidity, determines the withdrawal rate that drainage competes with (Brutsaert and Parlange, 1998; Roderick and Farquhar, 2002); and the near-surface hydraulic conductivity at low suction controls how long the soil can sustain the evaporative demand at the atmospheric rate before the evaporation front recedes (Lehmann et al., 2008; Or et al., 2013; Shokri and Or, 2011). The same soil hydraulic functions therefore regulate both sides of the competition: $K(h)$ near saturation governs drainage, and $K(h)$ at intermediate suction governs the supply of water to the surface and to roots. At larger scales, soil moisture feeds back on the surface energy balance and on precipitation in regions of strong coupling (Entekhabi et al., 1996; Koster et al., 2004; Seneviratne et al., 2010), and ecohydrological theory treats the soil moisture state as the integrator of these exchanges (Rodriguez-Iturbe and Porporato, 2004). A recoverable share that depends on climate, as found here for warming and rainfall, is the soil-scale expression of this coupling: the climate changes the withdrawal rate that drainage competes with, and the resulting soil moisture state changes the evaporation and transpiration that follow.

Consequences for models. Because structure and its dynamics modify hydraulic properties over seasons, representing them in land-surface and crop models is a recognised need (Fatichi et al., 2020; Vereecken et al., 2016, 2019). The relevant outcome here is practical: a recoverability diagnostic computed from K_s and n is only as stable as those properties, so the diagnostic should be evaluated for the structural state that management produces and not only for the texture-class value.

Change over time. Hydraulic properties are not constants of a soil but state variables with their own time scales. Loosening by tillage raises the saturated conductivity and the macroporosity of the seedbed, and rainfall impact, wetting and traffic then re-consolidate and seal it over weeks to seasons (Brady and Weil, 2016; Hillel, 1998; Strudley et al., 2008); wetting–drying and freeze–thaw cycles rearrange pore structure on seasonal to annual scales (Leuther and Schlüter, 2021; Miranda-Vélez et al., 2023); and residue incorporation changes the hydraulic functions through the decomposition of the residue (Leuther et al., 2026). For a rate-competition law of the form of Eq. (8), any such change enters through $K_s(t)$, so the memory fate becomes a function of time, $\phi_D(t) = K_s(t)/[K_s(t) + q^*]$. Figure 21a shows this for an illustrative case in which a threefold loosening of K_s relaxes exponentially to the consolidated value with a rate τ_c of 0.005 or 0.02 d^{-1}, two of the consolidation rates simulated in experiment E5 (0.001 to 0.02 d^{-1}). The curves are implications of the fitted law under assumed multipliers and rates and are not simulation results. For the loam S06 ($K_s = 25\,\mathrm{cm\,d}^{-1}$, $\phi_D = 0.26$ at the consolidated state), the memory fate is 0.51 immediately after loosening; with $\tau_c = 0.02\,\mathrm{d}^{-1}$ it has fallen to 0.32 after 90 days, whereas with $\tau_c = 0.005\,\mathrm{d}^{-1}$ it is still 0.44. The recoverable share of an evaporation saving therefore depends not only on the soil but on the time elapsed since its structure was last disturbed. The same mulch is therefore expected to lose a larger share of the water it saves to drainage in a freshly loosened, fast-draining seedbed than in the same soil late in the season; this direction of effect is a prediction to be tested, not a result.

Management practices. Management acts on the same two properties through several routes. Tillage intensity and timing set the initial structural state and its decay; controlled traffic and reduced compaction keep K_s closer to the structural value; organic amendments and residue retention increase aggregation and biological activity and, over years rather than seasons, shift the pore-size distribution (Brady and Weil, 2016; Hillel, 1998); surface mulch and residue cover modify the evaporation boundary, which is the quantity whose saving is being tracked here (Leuther et al., 2026; Vuaille et al., 2024); and irrigation method and trigger determine how often the profile is brought to the wet state in which drainage competes most strongly with uptake. A strength of the recoverable-share framing is that these practices can be compared on a common footing: each is expressed as a change in K_s (or n) and read off the same curve. Figure 21b shows the consequence of a threefold increase or decrease of K_s for the eight soils with literature conductivities. The effect is strongly soil-dependent, because the law is concave on a logarithmic axis: in the loam S06, ϕ_D changes from 0.10 to 0.26 to 0.51 across the three cases, in the sandy S05

from 0.33 to 0.60 to 0.82, and in the silty clay loam S10 only from 0.008 to 0.023 to 0.066. A structural change of a given relative size matters most where the soil is intermediate, and little where the matrix is so tight that drainage is negligible whatever the management.

Soil-specific implications. Three consequences follow for practice. First, the discount that should be applied to an evaporation-saving claim is a property of the soil–management–climate combination, not of the practice alone. Second, a soil that is close to the upper end of the curve (coarse, high K_s) gains little from structural improvement in this respect, whereas a soil in the steep middle of the curve is the most sensitive to compaction and to loosening. Third, because the loss fate depends on climate (rising with warming and falling with wetter conditions in the simulations), the same management has to be assessed under the climate of the site and under projected changes, not under the long-term mean only. None of these statements has been tested against field observations of structure over time in this paper.

Evidence from the simulation ensemble. Two experiments of the Daisy ensemble bear directly on structure, time and mulching (Fig. 20); they are simulations and not field measurements. In run set E5, tillage and the rate τ_c at which the loosened soil consolidates were crossed for all ten soils, under rainfed and irrigated management. For the four finer soils with an effect (S06, S08, S09, S10) rainfed drainage rises with faster consolidation, by 1.1, 2.5, 2.1 and 1.5 mm yr^{-1} between $\tau_c = 0.001$ and $0.02\,\mathrm{d}^{-1}$, towards the no-till value (Fig. 20b,c); in S01–S05 drainage does not depend on τ_c at all, although tillage itself shifts it by -1.8 to $+5.3$ mm yr^{-1}. The time dynamics therefore matter in the simulations only for the finer soils, and the effect is small against the texture contrast (rainfed drainage from 5 to 63 mm yr^{-1} across soils). Because the dynamic parameters of Daisy change retention and conductivity together, this response cannot be attributed to K_s alone, and the multipliers of Fig. 21 remain assumptions; the ensemble indicates that, within the matrix-flow model, the structural term is second-order relative to texture. In run set E7 (strong against weak mulch, vapour-flux factor 0.3 against 0.9 at an interception capacity of 2 mm), the evaporation saved ranges from 3 mm yr^{-1} in the sand S03 to 54 mm yr^{-1} in S01, and the extra drainage from 6 to 27 mm yr^{-1} (Fig. 20a); irrigation saved is near zero in S03–S05 and 43–49 mm yr^{-1} in S01, S07, S09 and S10, which is the same texture dependence as the shares ψ_D^{E7} of Table 8. Measured $K_s(t)$ series and paired field plots remain the data needed to replace the assumptions.

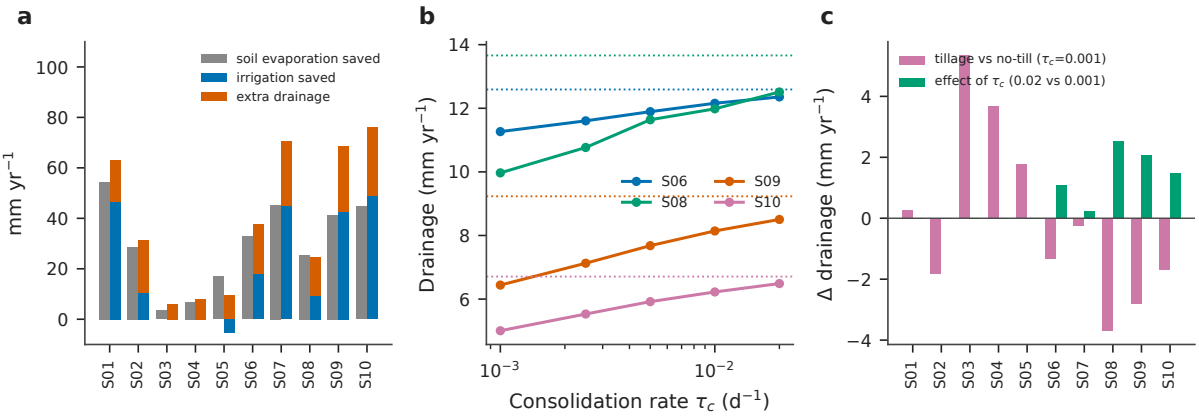


Figure 20: Simulation evidence on structure, time and mulching (Daisy ensemble, mean annual values). (a) Run set E7, strong against weak mulch (vapour-flux factor 0.3 against 0.9, capacity 2 mm): evaporation saved, irrigation saved and extra drainage by soil. (b) Run set E5, rainfed drainage of four finer soils against the consolidation rate τ_c for conventional tillage (solid) and the no-till value (dotted). (c) Effect of tillage ($\tau_c = 0.001$ against no-till) and of faster consolidation ($\tau_c = 0.02$ against 0.001) on rainfed drainage, all ten soils.

Data that would test these predictions. The predictions above are testable with three kinds of measurement: repeated infiltration or conductivity measurements across a season after tillage, which give $K_s(t)$ and its decay rate; paired drainage and evapotranspiration measurements under mulched and unmulched plots on contrasting soils, which give ψ_D directly; and structure descriptors (bulk density, aggregate stability, macroporosity) that explain why K_s departs from the texture-class value. Such data would allow the assumed multipliers and decay rates of Fig. 21 to be replaced by measured values, and would test whether the one-parameter law of Eq. (8) holds for soils whose structure changes within the season.

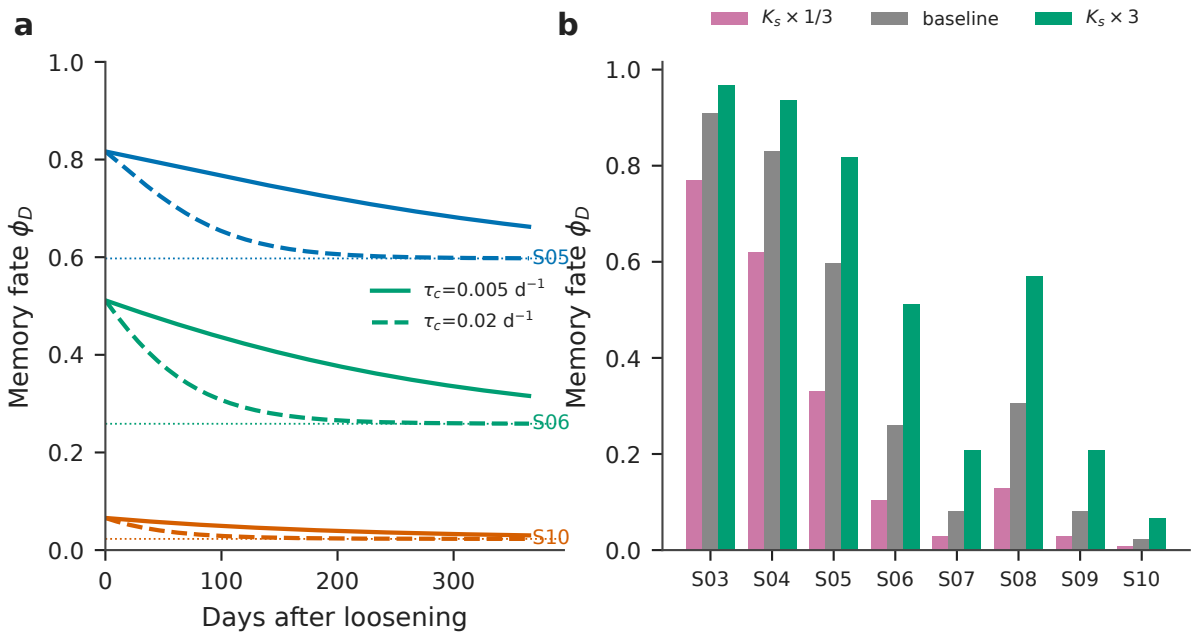


Figure 21: Implied change of the memory fate with soil structure over time and with management, from the fitted rate-competition law (Eq. (8), $q^* = 71.5\,\mathrm{cm\,d^{-1}}$). These are illustrative scenarios under assumed multipliers and rates, not simulation results. (a) A threefold loosening of K_s relaxing exponentially to the consolidated value, for the sand S05, loam S06 and silty clay loam S10; solid lines $\tau_c = 0.005\,\mathrm{d^{-1}}$, dashed lines $\tau_c = 0.02\,\mathrm{d^{-1}}$, dotted lines the consolidated baselines. (b) Memory fate for the eight soils with literature K_s at one third of, equal to and three times the baseline K_s.

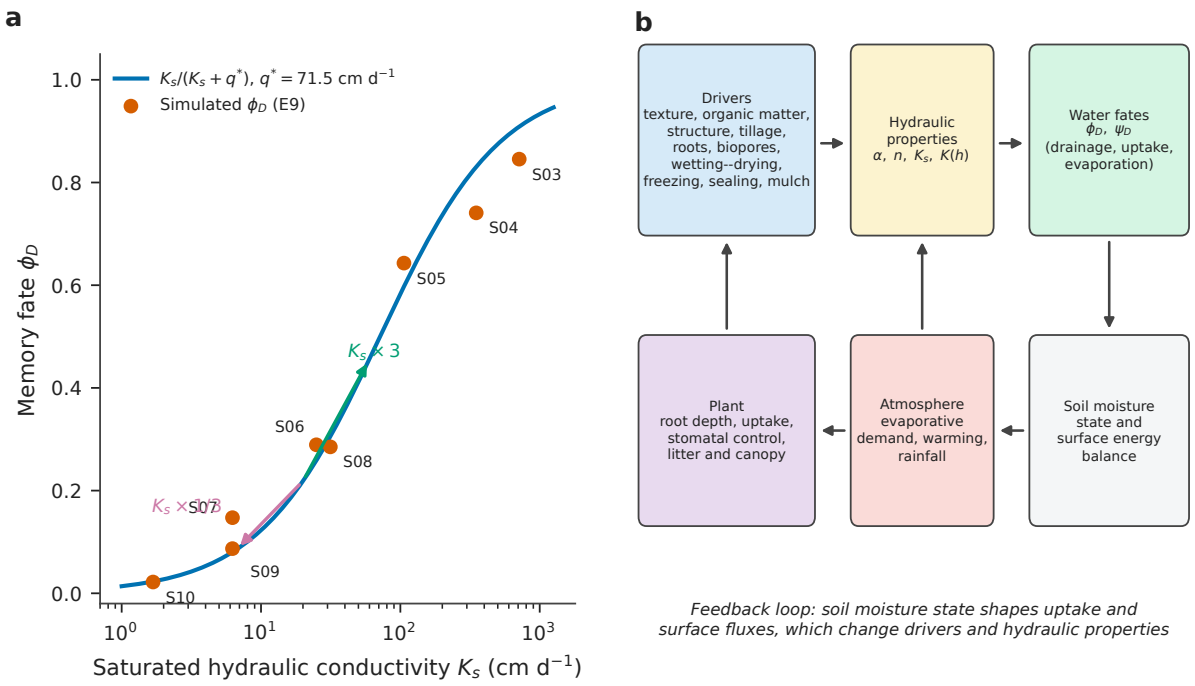


Figure 22: Drivers of soil hydraulic properties and their coupling to the soil–plant–atmosphere continuum. (a) Memory fate ϕ_D as a function of saturated conductivity K_s: simulated values for the eight soils with literature K_s (points) and the rate-competition law (line); arrows show the implied change for a threefold increase or decrease in K_s from 20 cm d^{-1} (law implication, not simulation). (b) Conceptual feedback among drivers, hydraulic properties, water fates, soil moisture state, atmosphere and plants.

Table 25: Drivers of soil hydraulic properties, their expected effect on K_s and on the memory fate through Eq. (8), and their treatment in this study.

Driver	Principal effect on hydraulic properties	Expected sign: K_s / ϕ_D	Treatment here
Texture	sets pore-size distribution, retention and conductivity	coarser: + / +	ten soils; tested
Organic matter, aggregation	structural porosity, broader pore-size distribution	+ / +	not varied
Tillage, consolidation	loosening raises, consolidation lowers conductivity	loosening: + / +	E5 in ensemble; ϕ_D not computed
Roots, macropores	preferential flow paths	+ / +	not represented (matrix flow)
Wetting–drying, freeze–thaw	rearranges structure	sign depends on soil	not represented
Surface sealing, mulch	alters surface boundary, infiltration, evaporation	sealing: − / −	mulch tested (E7, E14)
Warming, rainfall	changes the withdrawal rate that drainage competes with	— / drained share ψ_D: warming −, rain +	E11 tested

Domain of validity and the experimental programme. The conclusions are stated for the domain in which they were established, and each boundary of that domain defines an experiment.

1. *Soil series.* Ten soils span coarse to fine textures and more than two orders of magnitude in K_s. Soil-level statistics are reported with permutation tests and bootstrap intervals; the ordering of ϕ_D with n and K_s is firm ($\rho = 0.95$ and 0.97), whereas the E16 association with n (bootstrap -0.06 to 1.00) is indicative, and a larger soil library sampled from pedotransfer posteriors extends it.
2. *Process representation.* The mechanisms are those of matrix flow in Daisy. Macropore bypass (Beven and Germann, 1982; Jarvis, 2007) is absent, so the recoverable share is an upper bound for soils with active macropore flow, and replication in dual-permeability and Richards-based models such as HYDRUS, DSSAT, APSIM or AquaCrop (Jones et al., 2003; Keating et al., 2003; Šimůnek et al., 2008, 2016; Steduto et al., 2009) is the direct test of its structural generality.
3. *Connectivity.* Defined through the Mualem function and the flux response (Section 4.1.1).
4. *Conductivity source.* K_s is taken from independent literature class means for eight soils, which keeps the law non-circular; the remaining two soils are compared through n.
5. *Initial-state design.* In E9, h_0 is the initial suction, so ϕ_D is the response to antecedent state, and its link to scheduling is established through the independent E16 and E7 experiments.
6. *Time scale and climate.* Twenty-year means in one temperate climate; transfer to other climates is the second testable prediction below.
7. *Completeness.* 10 041 of 10 190 runs (98.5 %) completed; the 149 failures include 13 in S03 within E9 and a large fraction of S02 within E16 (37 of 64 combinations valid), so the S02 loss fates are read as indicative.
8. *Evidence status.* The conclusions are model-based and are formulated as quantitative predictions for lysimeter experiments; no field measurements enter the analysis.

Table 26 translates the results into a qualitative guide. The recoverable fraction $\rho_R = 1 - \psi_D$ is quoted from E16 for the central climate; because Fig. 13 shows that ψ_D varies by an order of magnitude with rainfall and warming in a single soil, the table is a ranking aid and not a prediction.

Table 26: Qualitative reading of the diagnostics for management. ρ_R from E16 (Table 8); "mulch" column from E7.

Soil group	Memory ϕ_D and ρ_R	Evaporation-saving practice (e.g. mulch)	Where the saved water goes
Sand, loamy sand, sandy loam (S03–S05)	high ϕ_D (0.64–0.85); $\rho_R \leq 0.42$	small irrigation benefit; $\psi_I^{E7} \approx 0$	mainly deep drainage
Loam, sandy clay loam, coarse sand (S06, S08, S02)	intermediate (≈ 0.26–0.29)	moderate; $\psi_D^{E7} = 0.52$–0.61	split between drainage and irrigation
Silt loam, coarse loam, clay loam, silty clay loam (S07, S01, S09, S10)	low (0.02–0.15); $\rho_R \approx 0.64$–0.71	largest irrigation benefit; $\psi_I^{E7} = 0.58$–0.63	retained in root zone; irrigation and transpiration

Land-surface schemes typically parameterise drainage with a single free-drainage lower boundary and a texture-based pedotransfer function (Vereecken et al., 2019; Weber et al., 2024). The strong dependence of ϕ_D on K_s and n suggests that, for evaporation-change questions, the uncertainty of the pedotransfer function enters the answer almost linearly through K_s when $K_s \ll q^*$. Because pedotransfer functions have large uncertainty in K_s (often an order of magnitude), the recoverable fraction of fine soils should be treated as uncertain by a comparable relative amount, while for sandy soils the saturating form of Eq. (8) compresses the uncertainty. The absence of soil structure in most Earth system models (Fatichi et al., 2020) means that the memory fate may be misrepresented precisely where structure is most important (fine soils, tillage).

(i) Extend the soil library by sampling parameters from pedotransfer-function posterior distributions and estimate the uncertainty of q^*. (ii) Repeat the E9, E16 and E7 designs in a dual-permeability model and in HYDRUS-1D (Šimůnek et al., 2008). (iii) Test the predictions of the Testable predictions subsection in lysimeters with controlled evaporation treatments and structure imaging. (iv) Examine climate beyond Taastrup, in particular Mediterranean climates where irrigation dominates the balance. (v) Link ρ_R to economic and groundwater-recharge accounting (Grafton et al., 2018; Ward and Pulido-Velazquez, 2008).

(i) In weighing lysimeters, the drainage response to a standard extra-evaporation treatment (shade or mulch) should scale as $K_s/(K_s + q^*)$ with q^* of order 10^1–10^2 cm d^{-1}. (ii) For two soils of equal *TAW* but different n, the one with the larger n should show a smaller recoverable fraction. (iii) Under warming, the recoverable fraction should rise in all soils where irrigation is

responsive. (iv) The recoverable fraction should be insensitive to nitrogen supply.

4.2 Accounting audit

Contributions of a validated discovery protocol and a quantified accounting gap. The paper contributes a discovery protocol in which every stage is tested against a known answer, and a first application that exposes an unclosed term in a widely used process model. The pipeline is blind to variable roles; it recovers the water balance as the dominant invariant, with a permutation null (smallest singular value 5.9 against 33.3 for permuted data), and this positive control is what licenses the interpretation of every other result. Applied to the evapotranspiration partition, the same pipeline quantifies a term that the reported components do not carry, $R \approx 164\,\mathrm{mm\,yr^{-1}}$ or about 30 % of ET, and a storage non-closure in two experiments (median 16 %). A compact three-term law then predicts ET across withheld experiments and withheld soils (LOEO $R^2 = 0.898$, LOSO 0.904) as accurately as gradient boosting (0.879 and 0.894). Quantifying a closure gap has a long record as a productive step in flux science: the energy-balance closure problem at eddy-covariance sites, in which turbulent fluxes typically fall 10–30 % short of available energy, was characterised across a network and then drove advances in measurement and in the interpretation of flux data (Foken, 2008; Wilson et al., 2002). The unattributed term R has the same function for model-output accounting: it converts an implicit assumption (that reported components are the total) into a measured quantity that can be traced to a pathway. Stating the recovered invariant for what it is, the known mass balance returned by a blind method, is the credential that makes the new findings credible in a field where discovery claims are often overstated (Nearing et al., 2021; Rudin, 2019).

Evapotranspiration partitioning is a central topic, from global studies that infer transpiration fractions (Jasechko et al., 2013; Schlesinger and Jasechko, 2014) to field studies of mulch effects (Balwinder-Singh et al., 2011). If a model's reported ET components omit a third of the flux, then a partition fraction computed as $T/(T + E_s + E_c)$ is larger than T/ET by a factor $1/(1 - 0.30) \approx 1.4$ on average, an error comparable with the differences between management treatments that are routinely reported. The size of R also helps to explain why the closure-derived and component-based estimates of soil evaporation differ. For users of Daisy and similar models, the practical recommendation is to compute ET as the model's own total and to report R alongside the components, and to document, in the model output specification, exactly which processes enter the total (Holbak et al., 2026).

A map of predictability across the flux budget. The ET law works because ET is the large, smooth, sub-linear function of supply that the water-balance literature has described for decades, and because the model's ET is energy- and storage-limited (Budyko, 1974; Milly, 1994; Zhang et al., 2001). The drainage fraction is the small residual of two large numbers, it is strongly nonlinear in the wetness of the profile, and it depends on thresholds (field capacity, trigger) that a log-linear library cannot represent (Rodriguez-Iturbe and Porporato, 2004); gradient boosting captures some of this and still only reaches $R^2 \approx 0.5$ in LOEO. This locates the information boundary: the content that drainage needs lies beyond the annual-flux library (e.g. intra-annual timing, event intensity, antecedent state). The memory analysis shows that antecedent state is indeed decisive for drainage in coarse soils.

The three-term law is competitive with gradient boosting under LOEO and LOSO, which supports the view that accuracy does not always require opacity (Rudin, 2019). The law does not extend to the regimes in which the physics it omits is first-order (initial-state experiments, dynamic hydraulic properties), and the gradient-boosting benchmark fails there too. This agreement is itself informative: it shows that the boundary is set by the information in the library and not by the model class, which tells a modeller which targets can be emulated by compact laws and which need timing and state variables.

Equation discovery in soil hydrology has so far targeted the governing partial differential equation, notably the Richards equation recovered from soil-moisture time series by group-sparse regression (Song et al., 2022, 2025), and deep-learning studies have imposed water-balance closure as a constraint on forecasts (Li et al., 2024). The present paper is complementary and differs in its object and its emphasis: it works with annual flux budgets, tests whether a pipeline recovers a known balance before interpreting any other result, audits the closure of the partition that supplies its targets, and validates by withheld regimes. The present approach shares with SINDy-style discovery (Brunton et al., 2016) the use of sparse regression over a library, but differs in two respects: it targets steady annual relations rather than dynamics, and it places the validation burden on leave-one-regime-out testing (Roberts et al., 2017). It shares with physics-informed learning (Karniadakis et al., 2021; Raissi et al., 2019) the emphasis on conservation, but imposes it as a *diagnostic* (closure residual) rather than as a constraint (Beucler et al., 2021); applying it as a constraint would have hidden R, which is the finding most relevant to model users. Related work on learning conserved quantities (Iten et al., 2020; Liu and Tegmark, 2021) would be a natural extension to daily dynamics.

Emulators are used to make process-model knowledge available in fast decision tools (Asher et al., 2015; Razavi et al., 2012). The results here indicate which emulator targets are safe: ET and transpiration are predictable from compact laws; drainage and soil evaporation are not, and should be emulated with care or by closure.

A software-level audit would resolve the identity of R directly. Four questions, in order of cost: (i) Which state variables are summed into the output called "evapotranspiration" in the log file used here, and which into the three reported components? (ii) Is evaporation from surface ponds, mulch and litter reported in a separate column that was not exported in the ensemble? (iii) Does the canopy-evaporation column represent interception only, excluding evaporation of water retained on residue? (iv) Is the storage term in the model's own water balance consistent with the exported depth and boundary definitions (for example, is the 1 m drainage column the flux at the bottom of the root zone or the soil column)? If the answer to (ii) is yes, the unattributed term would be explained and the diagnostic here would have served its purpose as an audit.

Random k-fold cross-validation of the same models would have produced much higher scores, because runs inside an experiment are close neighbours in factor space. To illustrate the effect on the present data: a model that predicts $\ln ET$ from $\ln W$ alone has LOEO R^2 of 0.80 on the full data and 0.60 when E8 and E9 are removed. Leave-one-experiment-out removes entire designs, so the model must extrapolate to unseen combinations of factor ranges, which is exactly what a "law" is supposed to do (Roberts et al., 2017). The leave-one-soil-out scores answer a second question: would the law hold for a soil not used to fit it? That the two scores agree for ET (0.898 and 0.904) is evidence that the law is not driven by a single experiment or soil, but it is not evidence of validity in other climates. A nested evaluation, in which the forward selection was repeated inside each training set of 16 experiments and scored only on the held-out experiment, gives a pooled $R^2 = 0.895$, against 0.898 for the single-level score, and selects the same three terms in 15 of 17 outer folds (the two-term set $\ln W$, n in the other two); the optimism from selecting on the evaluation score is therefore small for this law.

The sub-linear exponent $\partial \ln ET / \partial \ln W = 0.646$ can be compared with the elasticity of Budyko-type curves. For a Budyko curve $ET/P = F(\Phi)$ with aridity index $\Phi = PET/P$, the elasticity of ET to P lies between 0 (energy limit) and 1 (water limit) and is about 0.5–0.8 at intermediate aridity (Budyko, 1974; Milly, 1994; Zhang et al., 2001). The fitted value is within that range. Unlike the Budyko approach, which treats catchment properties via a single parameter, the present law includes a soil term (n): the fitted coefficient implies that a unit increase in n

reduces ET by 26 % ($1 - e^{-0.299}$) at fixed W, which can be read as the dependence of the effective Budyko parameter on soil hydraulic uniformity. The law is not claimed to be a Budyko curve; the mapping would require PET as a variable, and PET is constant within the weather series used.

A sequential protocol follows from these results and is offered as a general template for discovery studies in hydrology. The first element is a positive control: the pipeline is applied to a system whose governing balance is known, and it must recover that balance with a null-referenced uncertainty, as the permutation test and bootstrap do here for the water balance (Efron and Tibshirani, 1993; Good, 2005). The second is a closure audit of every partition used as a target, since a decomposition that omits a term will propagate that omission into every law fitted to it, as the unattributed term R propagates into the transpiration and soil-evaporation fractions. The third is regime-wise validation, in which whole experiments, soils or climates are withheld, because random folds inside a designed ensemble measure interpolation between neighbouring factor combinations and not transfer to a new regime (Roberts et al., 2017). The fourth is a comparison with a flexible benchmark, reported with the cases in which the benchmark is better, so that the interpretability premium can be read directly (Friedman, 2001; Rudin, 2019). The fifth is a stated domain of validity for each law, expressed in the experiments and soils for which the leave-one-out score exceeds a stated threshold.

The unattributed term R also defines a concrete verification programme for the process model. Three comparisons would identify it. First, an ablation in which individual surface and canopy evaporation pathways are switched off in turn would show which pathway removes R from the balance. Second, a tracer-style accounting that records every term added to the evaporative flux in the model log, with its state variable of origin, would establish whether R is a reporting convention or a physical pathway. Third, a comparison across structurally different models (Richards-based, capacity-based and radiation-driven) on the same designed ensemble would show whether a non-closing partition is a feature of one model family or a property of how evapotranspiration is partitioned in general (Jones et al., 2003; Keating et al., 2003; Šimůnek et al., 2008; Steduto et al., 2009). The experimental data that would constrain the physical interpretation are lysimeter and eddy-covariance records in which evaporation of intercepted and ponded water is measured separately from transpiration (Good et al., 2015; Jasechko et al., 2013; Schlesinger and Jasechko, 2014).

Beyond hydrology, the same sequence of controls is relevant to other settings in which a regression of a conserved quantity on its drivers is read as a discovered relation. Sparse regression

over a library is attractive because the result is an equation that can be examined, dimensionally checked and compared with theory (Brunton et al., 2016; Champion et al., 2019; Udrescu and Tegmark, 2020). Its reliability, however, depends on the design of the data. In an ensemble built by crossing factors, several drivers are correlated by construction, and the selection path of a forward algorithm reflects that correlation; the leave-one-experiment-out score then serves two roles, as selection criterion and as validation, which is why the LOSO scores, which are less affected by selection, are reported alongside it. The practical consequence is that the coefficients of a discovered law should be reported with the validation regime in which they were obtained, and that a law should be described as discovered only for the regimes in which it is validated.

Finally, the results bear on the design of the emulators and decision tools that draw on process-model ensembles. A tool that reports evapotranspiration components should state whether they sum to the total; one that reports drainage should state whether it is simulated or derived by closure, and the evidence here shows that closure-derived drainage is accurate in sign and rank order but under-estimates large drainage. Emulators built on such ensembles should also be tested against held-out simulations (Asher et al., 2015; Razavi et al., 2012; Wilkinson et al., 2016).

Domain of validity of the discovered laws. Each law is reported with the regimes in which it was validated, and each boundary defines the next test.

1. *Process-model domain.* The results describe Daisy output. R is a property of the model's accounting; its counterpart in nature is measured in lysimeters and eddy-covariance records that separate intercepted and ponded water from transpiration.
2. *Failed runs.* The 149 excluded runs (87 % in S02) form a pattern that identifies them as solver failures (Section 2.2); re-running with tighter tolerances is the confirming step, and S02 is under-represented at large doses and extreme rain scales.
3. *Model and weather.* One model and one climate; the laws are conditional on both, and the leave-one-regime-out protocol is designed to be repeated on further models and climates.
4. *Selection.* Forward selection used the LOEO score; the nested evaluation, with selection repeated inside each training set, gives $R^2 = 0.895$ against 0.898 for the ET law, so the selection optimism is small. LOSO values are less affected by selection.
5. *Dependence of runs.* Coefficient standard errors assume independent runs; cross-fold variation is the reported guide to uncertainty.
6. *Soil covariates.* Ten soils; n carries several correlated soil properties.
7. *Identification of R.* Its surface-layer origin is inferred from its driver dependence, and the

four-question audit of the model code in this section settles it.

8. *Aggregation.* Twenty-year means average over seasonality and event structure; the memory analysis addresses antecedent state.
9. *Benchmark reproducibility.* The gradient-boosting benchmark is run with a fixed random seed. Across unseeded repeats the LOSO score varied by up to 0.013, so differences below about 0.02 between a law and the benchmark are not interpreted.

(1) Begin with a positive control with a known answer. (2) Use a permutation null for any decomposition-based invariant. (3) Validate across *regimes* (experiments, soils, climates), not only across random folds. (4) Audit partitions for closure before using them as targets. (5) Compare with a flexible benchmark and report where the benchmark wins. (6) State a domain of validity for each law. (7) Publish failures.

5. What the two analyses say together

The audit identifies which parts of the output of a process model can carry a conclusion, and the memory analysis shows what a conclusion built on them is worth. Three statements follow, each limited to the simulated system. First, an evaporation saving has to be judged against a closed quantity: $E_{nt} = ET - T$ closes by construction, whereas the reported components E_s and E_c leave out the unattributed term and give a different, in two soils opposite, answer (Section 3.2). Second, the soils in which drainage competes most with evaporation are the soils in which the partition is least settled, so the soil-specific discount of a saving cannot be replaced by a single factor. Third, the experiments that start away from equilibrium (E8, E9) are those in which storage does not close (16 %), and the memory fate was therefore estimated from the slope of drainage on storage release over many starting states and not from a single balance. None of this is evidence about measured soils: a lysimeter with controlled evaporation, measured drainage and a weighed mass balance would test the recoverable share and the unattributed term together.

5.1 Reproducing and extending the analysis with the software and its manual

The ensemble of this study is not only archived (Halilu, 2026a): every one of its 17 experiments is a prepared experiment of the Daisy workbench of SWAFL Desktop 1.1.0, documented in its technical manual (Halilu, 2026b) (*Experiments and scenarios*, p. 592; *The factors and responses of the Experiments tab*, p. 877; *Results and analysis*, p. 660). A reader can load an experiment,

change the soil, the weather or a level, and run it with Daisy on a laptop; the number of runs set by the reader is a ceiling, kept by a balanced (Latin-hypercube) choice when the combinations are more. Each run is kept file by file next to Daisy, its summary gives the range of every result and the effect of every input on every result, and every quantity of every log is drawn as a figure through the year, by month, by year or cumulatively. The quantities this paper rests on (soil evaporation, drainage below 1 m, the storage change, the non-transpiration term) are logs of every run, so the closure test (Eq. (12) and Table 17) and the fate of a released storage can be repeated on any soil a reader enters.

Five factors of the workbench scale the van Genuchten–Mualem parameters of every horizon of a run (K_{sat}, α, $n-1$, θ_s, and a shift of l), the parameters that this paper finds to govern hydraulic memory. One run of the loam with Daisy 7.1.14 illustrates the direction (Fig. 23): with the parameters of a faster-draining soil ($K_{\mathrm{sat}} \times 4$, $\alpha \times 2$, $(n-1) \times 1.3$, $\theta_s \times 1.1$, $l+1$), cumulative matrix percolation over two seasons rose from 494 to 575 mm ($+81$ mm), actual evapotranspiration fell from 1050 to 981 mm (-69 mm) and runoff vanished (16 mm before): water that a slowly draining soil returns to the atmosphere is drained by a fast one, which is the rate competition of Section 2.1 seen in a single pair of runs. It is one pair, not a test of the law; the factors make the test of the law on new soils a matter of one experiment of the workbench.

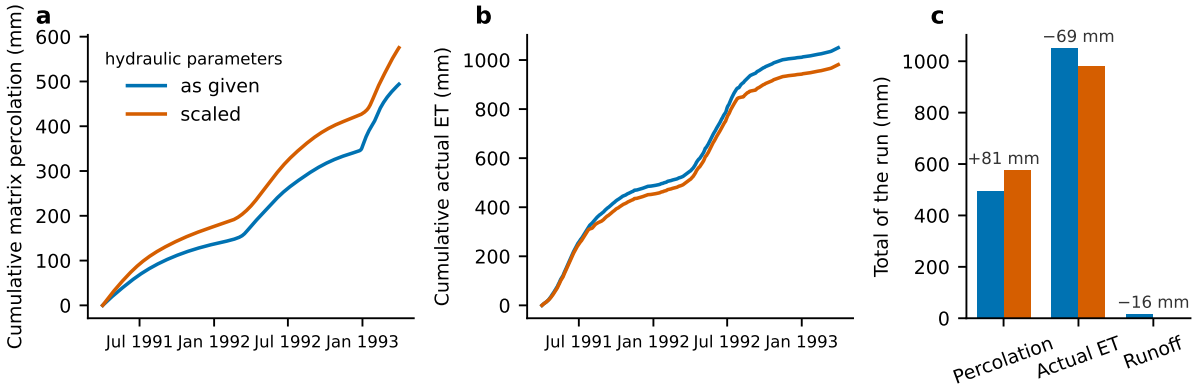
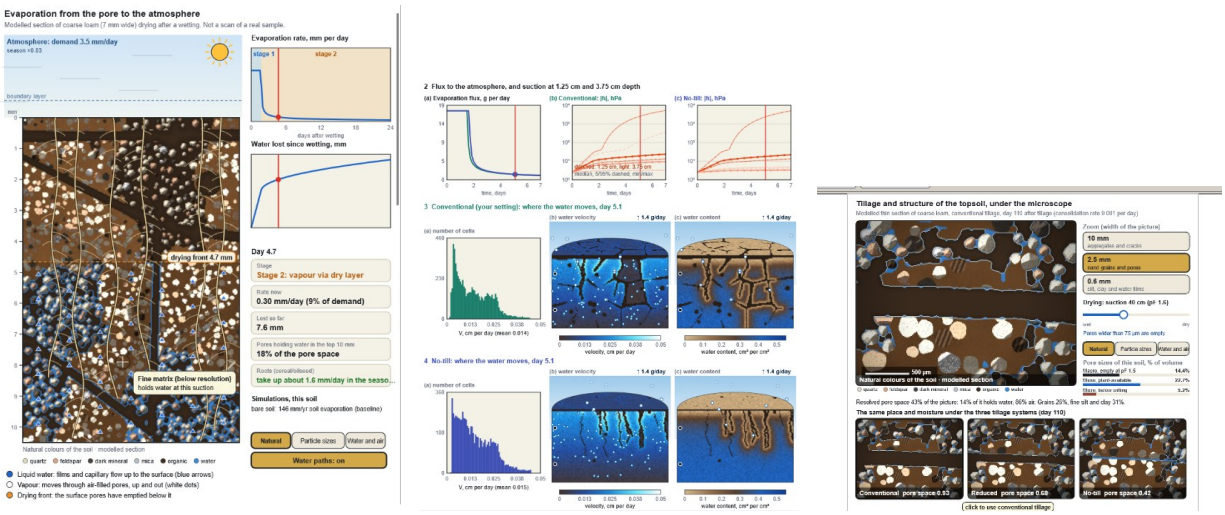


Figure 23: One Daisy 7.1.14 run of the loam written by SWAFL 1.1.0 with the van Genuchten–Mualem parameters of every horizon as given (blue) and scaled toward a faster-draining soil (orange: $K_{\mathrm{sat}} \times 4$, $\alpha \times 2$, $(n-1) \times 1.3$, $\theta_s \times 1.1$, $l+1$). (a) Cumulative matrix percolation, (b) cumulative actual evapotranspiration, (c) totals of the two seasons with the difference. Daily logs of Daisy, unchanged; script `reproducibility/fig_vg_check.py`.



(a) Evaporation from the pore to the atmosphere: stage 1 and 2, drying front.

(b) Flux and suction at 1.25 and 3.75 cm; where the water moves under two tillage systems.

(c) Pore space of the topsoil under three tillage systems, at the suction chosen.

Figure 24: The processes behind hydraulic memory as the user sees them in SWAFL Desktop 1.1.0 (real screenshots): evaporation drying the surface pores (a), the flux to the atmosphere against drainage in the cells of the profile (b), and the pore space that sets n and K_s (c). Manual pp. 66–110.

Novelty in one view. (1) *Hydraulic memory* is defined, measured and predicted: the share of a released storage that drains spans 0.02 to 0.85 across ten soils and is predicted for held-out soils by a one-parameter rate-competition law (leave-one-soil-out RMSE 0.058). (2) *A blind discovery protocol* (linear-invariant search with a permutation null, positive and negative controls) recovers the water balance of a process model without being told it and exposes an unattributed evaporative term of about 30 % of ET. (3) *The two results change each other*: judged against the reported soil evaporation alone, the drainage share of an evaporation change is misjudged in most soils and changes sign in two. (4) *Everything is repeatable and extensible* in an offline application whose manual documents every experiment, factor and output (Halilu, 2026b), so a reader can rerun, extend and falsify the claims on a soil of their own.

6. Conclusions

Hydraulic memory, defined as the drainage share of released antecedent storage, ranges from 0.02 to 0.85 across ten simulated soils, follows the pore-size parameter ($\rho = 0.95$) and saturated conductivity ($\rho = 0.97$), and is captured out of sample by a one-parameter rate-competition law (LOSO RMSE 0.058). It transfers to independent evaporation-factor and mulch experiments ($\rho = 0.85, 0.89$). The recoverable share of an evaporation change is soil-specific and climate-modified: lower in coarse soils (0.20 in loamy sand; none in sand), and shifted by -0.16 with warming and

$+0.42$ with wetter climate in the drainage share. The accounting under these results was audited: a blind linear-invariant search recovers the water balance with every coefficient within 5 % of the exact value and flags an unattributed evapotranspiration term averaging 164 mm yr^{-1} (30 % of ET) and a storage non-closure of 16 % in the two initial-moisture experiments; a three-term law predicts evapotranspiration as well as gradient boosting in withheld experiments ($R^2 = 0.898$) and soils ($R^2 = 0.904$), and drainage resists both. Judged against the reported soil evaporation alone, the drainage share of an evaporation change would differ by more than 0.1 in six of ten soils and change sign in two. Together these give a transferable, mechanism-based recoverability diagnostic for ten soils and a protocol for deciding which outputs of a process model can carry a conclusion, with quantitative predictions that experiments controlling pore-network structure, antecedent state and evaporation can test.

Companion papers and release 1.1.0

The simulation ensemble analysed here (17 experiments, 10 190 runs, with the run time of every run, mean 148 s, in the comprehensive summary of the repository) is also the basis of the decision layers of SWAFL Desktop 1.1.0. A fourth paper uses the per-experiment outputs of the same data folder to derive a closed form, from the van Genuchten–Mualem parameters, for the largest suction a crop can impose at a sensor (the reach of a suction trigger, beyond which a trigger never fires) and to map the fate of each irrigation millimetre across soils, doses, seasons and methods (Paper 4). Paper 2 describes the engines built on it, Paper 3 the whole suite, including the graphical Daisy workbench, the bridge that connects Daisy to the decision layers, deterministic phenotyping, the field sketch, Sentinel-2 indices and the crop library. The results of both analyses of this paper were regenerated after the changes of release 1.1.0 and are unchanged.

Data and code availability

Data. The complete Daisy ensemble outputs (all 10 190 runs, raw and derived) are archived on Zenodo, doi: 10.5281/zenodo.23043626. The raw run table used here, and a linked list of every data file used, are in the Data folder of the project repository (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data).

Full simulation outputs. Every value of every log of every run (the annual water and nitrogen balances, the harvests and the crop state, year by year), one compressed table per experiment (`E00_Baselines` to `E16_Edge_Cases`), the one-row-per-run summary of all 10 190 runs (`00_summary_all_simulations`), the per-experiment summary (`00_SUMMARY_ALL_EXPERIMENTS`) and the script that made them (`summarize_results3.py`) are in the folder `Data/Comprehensive_Summary` of the repository (`https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data/Comprehensive_Summary`); their 20-year means reproduce the run table within 2 %.

The data of every table and figure. One Excel workbook, `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, holds the data used in all four papers: the run tables, the per-experiment summary, the derived results behind every table and figure, and a map that names, for each table and figure of each paper, the script that makes it, its input and its output. The map is also in `Data/Manuscript_Data/README.md`, and the workbook is rebuilt by `Data/Manuscript_Data/make_workbook.py`.

Code. The analysis and figure scripts are in the Code folder (`https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Manuscripts`). The folder `Manuscripts/reproducibility` is a self-contained package: pinned software versions (Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, statsmodels 0.15.0, Matplotlib 3.11.2, Pillow 12.3.0), fixed random seeds in every stochastic step, SHA-256 checksums, and one command (`run_all.sh`) that regenerates all results and figures from the ensemble and then compares every regenerated value with the archived result and with the headline numbers printed in the manuscript. Each table and figure names the script that produced it (Appendix B). Daisy itself was not re-run for this package, no field measurements are used, and small intermediate arrays used only by figures are archived rather than regenerated; these limits are listed in `README_REPRODUCE.md`.

Author contributions

N.H.: conceptualisation, simulation, analysis, software, visualisation, writing.

Funding

No funding was received for this work.

Competing interests

The author declares no competing interests.

Acknowledgements

The Daisy model is developed and maintained at the University of Copenhagen. The author thanks the maintainers of the open-source libraries cited.

References

John Aitchison. *The Statistical Analysis of Compositional Data*. Chapman and Hall, London, 1986.

M. J. Asher, B. F. W. Croke, A. J. Jakeman, and L. J. M. Peeters. A review of surrogate models and their application to groundwater modeling. *Water Resources Research*, 51:5957–5973, 2015. doi: 10.1002/2015WR016967.

Balwinder-Singh, Philip L. Eberbach, Elizabeth Humphreys, and Sat Sharan Kukal. The effect of rice straw mulch on evapotranspiration, transpiration and soil evaporation of irrigated wheat in punjab, india. *Agricultural Water Management*, 98:1847–1855, 2011. doi: 10.1016/j.agwat.2011.05.007.

Neil D. Bennett, Barry F. W. Croke, Giorgio Guariso, Joseph H. A. Guillaume, Serena H. Hamilton, Anthony J. Jakeman, et al. Characterising performance of environmental models. *Environmental Modelling and Software*, 40:1–20, 2013. doi: 10.1016/j.envsoft.2012.09.011.

Alexis Berg and Justin Sheffield. Climate change and drought: the soil moisture perspective. *Current Climate Change Reports*, 4:180–191, 2018. doi: 10.1007/s40641-018-0095-0.

Tom Beucler, Michael Pritchard, Stephan Rasp, Jordan Ott, Pierre Baldi, and Pierre Gentine. Enforcing analytic constraints in neural networks emulating physical systems. *Physical Review Letters*, 126:098302, 2021. doi: 10.1103/PhysRevLett.126.098302.

Keith Beven and Peter Germann. Macropores and water flow in soils. *Water Resources Research*, 18:1311–1325, 1982. doi: 10.1029/WR018i005p01311.

Keith Beven and Peter Germann. Macropores and water flow in soils revisited. *Water Resources Research*, 49:3071–3092, 2013. doi: 10.1002/wrcr.20156.

Martin J. Blunt. *Multiphase Flow in Permeable Media: A Pore-Scale Perspective*. Cambridge University Press, Cambridge, 2017. doi: 10.1017/9781316145098.

Nyle C. Brady and Ray R. Weil. *The Nature and Properties of Soils*. Pearson, Columbus, OH, 15 edition, 2016.

Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113:3932–3937, 2016. doi: 10.1073/pnas.1517384113.

Wilfried Brutsaert and Marc B. Parlange. Hydrologic cycle explains the evaporation paradox. *Nature*, 396:30, 1998. doi: 10.1038/23845.

Edgar Buckingham. On physically similar systems; illustrations of the use of dimensional equations. *Physical Review*, 4:345–376, 1914. doi: 10.1103/PhysRev.4.345.

Mikhail I. Budyko. *Climate and Life*. Academic Press, New York, 1974.

Robert F. Carsel and Rudolph S. Parrish. Developing joint probability distributions of soil water retention characteristics. *Water Resources Research*, 24:755–769, 1988. doi: 10.1029/WR024i005p00755.

Kathleen Champion, Bethany Lusch, J. Nathan Kutz, and Steven L. Brunton. Data-driven discovery of coordinates and governing equations. *Proceedings of the National Academy of Sciences*, 116:22445–22451, 2019. doi: 10.1073/pnas.1906995116.

Aiguo Dai. Increasing drought under global warming in observations and models. *Nature Climate Change*, 3:52–58, 2013. doi: 10.1038/nclimate1633.

Suat Datta and Saleh Taghvaeian. Soil water sensors for irrigation scheduling in the united states: a systematic review of literature. *Agricultural Water Management*, 278:108148, 2023. doi: 10.1016/j.agwat.2023.108148.

Brian M. de Silva, Kathleen Champion, Markus Quade, Jean-Christophe Loiseau, J. Nathan Kutz, and Steven L. Brunton. PySINDy: a Python package for the sparse identification of nonlinear dynamical systems from data. *Journal of Open Source Software*, 5:2104, 2020. doi: 10.21105/joss.02104.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, New York, 1993.

Juan J. Egozcue, Vera Pawlowsky-Glahn, Glòria Mateu-Figueras, and Carles Barceló-Vidal. Isometric logratio transformations for compositional data analysis. *Mathematical Geology*, 35: 279–300, 2003. doi: 10.1023/A:1023818214614.

Dara Entekhabi, Ignacio Rodriguez-Iturbe, and Fabio Castelli. Mutual interaction of soil moisture state and atmospheric processes. *Journal of Hydrology*, 184:3–17, 1996. doi: 10.1016/0022-1694(95)02965-6.

Simone Fatichi, Dani Or, Robert Walko, Harry Vereecken, Michael H. Young, Teamrat A. Ghezzehei, et al. Soil structure is an important omission in Earth System Models. *Nature Communications*, 11:522, 2020. doi: 10.1038/s41467-020-14411-z.

Elias Fereres and María Auxiliadora Soriano. Deficit irrigation for reducing agricultural water use. *Journal of Experimental Botany*, 58:147–159, 2007. doi: 10.1093/jxb/erl165.

Thomas Foken. The energy balance closure problem: an overview. *Ecological Applications*, 18: 1351–1367, 2008. doi: 10.1890/06-0922.1.

Jerome H. Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29:1189–1232, 2001. doi: 10.1214/aos/1013203451.

Laura García, Lorena Parra, Jose M. Jimenez, Jaime Lloret, and Pascal Lorenz. IoT-based smart irrigation systems: an overview on the recent trends on sensors and IoT systems for irrigation in precision agriculture. *Sensors*, 20:1042, 2020. doi: 10.3390/s20041042.

Sam Geerts and Dirk Raes. Deficit irrigation as an on-farm strategy to maximize crop water productivity in dry areas. *Agricultural Water Management*, 96:1275–1284, 2009. doi: 10.1016/j.agwat.2009.04.009.

Behzad Ghanbarian, Allen G. Hunt, Robert P. Ewing, and Muhammad Sahimi. Tortuosity in porous media: a critical review. *Soil Science Society of America Journal*, 77:1461–1477, 2013. doi: 10.2136/sssaj2012.0435.

Phillip I. Good. *Permutation, Parametric and Bootstrap Tests of Hypotheses*. Springer, New York, 3 edition, 2005.

Stephen P. Good, David Noone, and Gabriel Bowen. Hydrologic connectivity constrains partitioning of global terrestrial water fluxes. *Science*, 349:175–177, 2015. doi: 10.1126/science.aaa5931.

R. Quentin Grafton, John Williams, Chris J. Perry, François Molle, Claudia Ringler, Pasquale Steduto, et al. The paradox of irrigation efficiency. *Science*, 361:748–750, 2018. doi: 10.1126/science.aat9314.

Naziru Halilu. Coupled water fluxes and pore scale dynamics under joint precision irrigation and evaporation control mulching: ensemble of 10 190 Daisy simulations. Zenodo dataset, 2026a.

Naziru Halilu. *SWAFL Desktop 1.1.0: Technical Manual (1155 pages)*, 2026b. Distributed with the software and at https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Software/v1.1.0/Manual.

Søren Hansen, Hans E. Jensen, Niels E. Nielsen, and Henrik Svendsen. Simulation of nitrogen dynamics and biomass production in winter wheat using the danish simulation model DAISY. *Fertilizer Research*, 27:245–259, 1991. doi: 10.1007/BF01051131.

Søren Hansen, Per Abrahamsen, Christian T. Petersen, and Merete Styczen. Daisy: model use, calibration, and validation. *Transactions of the ASABE*, 55:1317–1335, 2012. doi: 10.13031/2013.42244.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, New York, 2 edition, 2009.

Daniel Hillel. *Environmental Soil Physics*. Academic Press, San Diego, 1998.

Maja Holbak, Per Abrahamsen, Celine R. Grønning, Tobias Klöffel, Magdalena V. Schwartzkopff, Merete E. Styczen, Simon F. Svane, and Efstathios Diamantopoulos. The Daisy model: a tool for simulating agricultural systems — recent (2012–2025) developments, calibrations and validations. *Agricultural Systems*, 238:104846, 2026. doi: 10.1016/j.agsy.2026.104846.

John D. Hunter. Matplotlib: a 2D graphics environment. *Computing in Science and Engineering*, 9:90–95, 2007. doi: 10.1109/MCSE.2007.55.

Christopher Hutton, Thorsten Wagener, Jim Freer, Dawei Han, Chris Duffy, and Berit Arheimer. Most computational hydrology is not reproducible, so is it really science? *Water Resources Research*, 52:7548–7555, 2016. doi: 10.1002/2016WR019285.

Raban Iten, Tony Metger, Henrik Wilming, Lídia del Rio, and Renato Renner. Discovering physical concepts with neural networks. *Physical Review Letters*, 124:010508, 2020. doi: 10.1103/PhysRevLett.124.010508.

Nicholas J. Jarvis. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *European Journal of Soil Science*, 58:523–546, 2007. doi: 10.1111/j.1365-2389.2007.00915.x.

Scott Jasechko, Zachary D. Sharp, John J. Gibson, S. Jean Birks, Yi Yi, and Peter J. Fawcett. Terrestrial water fluxes dominated by transpiration. *Nature*, 496:347–350, 2013. doi: 10.1038/nature11983.

James W. Jones, Gerrit Hoogenboom, Cheryl H. Porter, Kenneth J. Boote, William D. Batchelor, L. A. Hunt, et al. The DSSAT cropping system model. *European Journal of Agronomy*, 18: 235–265, 2003. doi: 10.1016/S1161-0301(02)00107-7.

Wenrong Kang, Yongyong Zhang, Wenzhi Zhao, and Shaoxiong Wu. The effect of gravel mulch on soil evaporation and resistance: experimental findings and modeling. *Journal of Soil Science and Plant Nutrition*, 25(1):1135–1148, 2025. doi: 10.1007/s42729-024-02190-1.

George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021. doi: 10.1038/s42254-021-00314-5.

Brian A. Keating, Peter S. Carberry, Graeme L. Hammer, Merv E. Probert, Michael J. Robertson, Dean Holzworth, et al. An overview of APSIM, a model designed for farming systems simulation. *European Journal of Agronomy*, 18:267–288, 2003. doi: 10.1016/S1161-0301(02)00108-9.

James W. Kirchner. Getting the right answers for the right reasons: linking measurements, analyses, and models to advance the science of hydrology. *Water Resources Research*, 42:W03S04, 2006. doi: 10.1029/2005WR004362.

Randal D. Koster and Max J. Suarez. Soil moisture memory in climate models. *Journal of Hydrometeorology*, 2:558–570, 2001.

Randal D. Koster, Paul A. Dirmeyer, Zhichang Guo, Gordon Bonan, Edmond Chan, Peter Cox, C. T. Gordon, Shinjiro Kanae, Eva Kowalczyk, David Lawrence, et al. Regions of strong coupling between soil moisture and precipitation. *Science*, 305:1138–1140, 2004. doi: 10.1126/science.1100217.

Frederik Kratzert, Daniel Klotz, Claire Brenner, Karsten Schulz, and Mathew Herrnegger. Rainfall–runoff modelling using long short-term memory (LSTM) networks. *Hydrology and Earth System Sciences*, 22:6005–6022, 2018. doi: 10.5194/hess-22-6005-2018.

Frederik Kratzert, Daniel Klotz, Guy Shalev, Günter Klambauer, Sepp Hochreiter, and Grey Nearing. Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrology and Earth System Sciences*, 23:5089–5110, 2019. doi: 10.5194/hess-23-5089-2019.

Peter Lehmann, Shmuel Assouline, and Dani Or. Characteristic lengths affecting evaporative drying of porous media. *Physical Review E*, 77:056309, 2008. doi: 10.1103/PhysRevE.77.056309.

Frederic Leuther and Efstathios Diamantopoulos. Simulating bare soil evaporation for undisturbed soil cores: using HYDRUS 3d simulation on X-ray μCT determined soil macrostructures. *Vadose Zone Journal*, 23:e20339, 2024. doi: 10.1002/vzj2.20339.

Frederic Leuther and Steffen Schlüter. Impact of freeze-thaw cycles on soil structure and soil hydraulic properties. *SOIL*, 7:179–191, 2021. doi: 10.5194/soil-7-179-2021.

Frederic Leuther, Annika Langanki, Eva Lehndorff, and Efstathios Diamantopoulos. Fresh and degraded maize shoot and root residues temporarily change soil hydraulic properties. *EGUsphere*, 2026:1–21, 2026. doi: 10.5194/egusphere-2026-1063.

L. Li, Y. Dai, Z. Wei, W. Shangguan, Y. Zhang, N. Wei, and Q. Li. Enforcing water balance in multitask deep learning models for hydrological forecasting. *Journal of Hydrometeorology*, 25 (1):89–103, 2024. doi: 10.1175/JHM-D-23-0073.1.

Ziming Liu and Max Tegmark. Machine learning conservation laws from trajectories. *Physical Review Letters*, 126:180604, 2021. doi: 10.1103/PhysRevLett.126.180604.

Edward E. Miller and Raymond D. Miller. Physical theory for capillary flow phenomena. *Journal of Applied Physics*, 27:324–332, 1956. doi: 10.1063/1.1722370.

Paul C. D. Milly. Climate, soil water storage, and the average annual water balance. *Water Resources Research*, 30:2143–2156, 1994. doi: 10.1029/94WR00586.

Juan F. Miranda-Vélez, Frederic Leuther, John M. Köhne, Lars J. Munkholm, and Iris Vogeler. Effects of freeze-thaw cycles on soil structure under different tillage and plant cover management practices. *Soil and Tillage Research*, 225:105540, 2023.

Daniel N. Moriasi, Jeffrey G. Arnold, Michael W. Van Liew, Ronald L. Bingner, R. Daren Harmel, and Tamie L. Veith. Model evaluation guidelines for systematic quantification of accuracy in watershed simulations. *Transactions of the ASABE*, 50:885–900, 2007. doi: 10.13031/2013.23153.

Yechezkel Mualem. A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, 12:513–522, 1976. doi: 10.1029/WR012i003p00513.

Grey S. Nearing, Frederik Kratzert, Alden Keefe Sampson, Craig S. Pelissier, Daniel Klotz, Jonathan M. Frame, Cristina Prieto, and Hoshin V. Gupta. What role does hydrological science play in the age of machine learning? *Water Resources Research*, 57:e2020WR028091, 2021. doi: 10.1029/2020WR028091.

Dani Or, Peter Lehmann, Ebrahim Shahraeeni, and Nima Shokri. Advances in soil evaporation physics—a review. *Vadose Zone Journal*, 12(4), 2013. doi: 10.2136/vzj2012.0163.

Naomi Oreskes, Kristin Shrader-Frechette, and Kenneth Belitz. Verification, validation, and confirmation of numerical models in the earth sciences. *Science*, 263:641–646, 1994. doi: 10.1126/science.263.5147.641.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Luís S. Pereira, Ian Cordery, and Iacovos Iacovides. Improved indicators of water use performance and productivity for sustainable water conservation and saving. *Agricultural Water Management*, 108:39–51, 2012. doi: 10.1016/j.agwat.2011.08.022.

Chris Perry, Pasquale Steduto, and Fawzi Karajeh. Does improved irrigation technology save water? a review of the evidence. *FAO Report, Food and Agriculture Organization of the United Nations*, 2017. URL https://www.fao.org/3/I7090EN/i7090en.pdf.

Enrique Playán and Luciano Mateos. Modernization and optimization of irrigation systems to increase water productivity. *Agricultural Water Management*, 80:100–116, 2006. doi: 10.1016/j.agwat.2005.07.007.

Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.

Saman Razavi, Bryan A. Tolson, and Donald H. Burn. Review of surrogate modeling in water resources. *Water Resources Research*, 48:W07401, 2012. doi: 10.1029/2011WR011527.

Jens Christian Refsgaard and Hans Jørgen Henriksen. Modelling guidelines—terminology and guiding principles. *Advances in Water Resources*, 27:71–82, 2004. doi: 10.1016/j.advwatres.2003.08.006.

Markus Reichstein, Gustau Camps-Valls, Bjorn Stevens, Martin Jung, Joachim Denzler, Nuno Carvalhais, and Prabhat. Deep learning and process understanding for data-driven Earth system science. *Nature*, 566:195–204, 2019. doi: 10.1038/s41586-019-0912-1.

Lorenzo A. Richards. Capillary conduction of liquids through porous mediums. *Physics*, 1:318–333, 1931. doi: 10.1063/1.1745010.

David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40:913–929, 2017. doi: 10.1111/ecog.02881.

Michael L. Roderick and Graham D. Farquhar. The cause of decreased pan evaporation over the past 50 years. *Science*, 298:1410–1411, 2002. doi: 10.1126/science.1075390.

Ignacio Rodriguez-Iturbe and Amilcare Porporato. *Ecohydrology of Water-Controlled Ecosystems: Soil Moisture and Plant Dynamics*. Cambridge University Press, Cambridge, 2004. doi: 10.1017/CBO9780511535741.

Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1:206–215, 2019. doi: 10.1038/s42256-019-0048-x.

Keith E. Saxton and Walter J. Rawls. Soil water characteristic estimates by texture and organic matter for hydrologic solutions. *Soil Science Society of America Journal*, 70:1569–1578, 2006. doi: 10.2136/sssaj2005.0117.

Marcel G. Schaap, Feike J. Leij, and Martinus Th. van Genuchten. Rosetta: a computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. *Journal of Hydrology*, 251:163–176, 2001. doi: 10.1016/S0022-1694(01)00466-8.

William H. Schlesinger and Scott Jasechko. Transpiration in the global water cycle. *Agricultural and Forest Meteorology*, 189–190:115–117, 2014. doi: 10.1016/j.agrformet.2014.01.011.

Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324:81–85, 2009. doi: 10.1126/science.1165893.

Sonia I. Seneviratne, Thierry Corti, Edouard L. Davin, Martin Hirschi, Eric B. Jaeger, Irene Lehner, Boris Orlowsky, and Adriaan J. Teuling. Investigating soil moisture–climate interactions in a changing climate: a review. *Earth-Science Reviews*, 99:125–161, 2010. doi: 10.1016/j.earscirev.2010.02.004.

Chaopeng Shen. A transdisciplinary review of deep learning research and its relevance for water resources scientists. *Water Resources Research*, 54:8558–8593, 2018. doi: 10.1029/2018WR022643.

Steven Sherwood and Qiang Fu. A drier future? *Science*, 343:737–739, 2014. doi: 10.1126/science.1247620.

Nima Shokri and Dani Or. What determines drying rates at the onset of diffusion controlled stage-2 evaporation from porous media? *Water Resources Research*, 47:W09513, 2011. doi: 10.1029/2010WR010284.

Jiří Šimůnek, Martinus Th. van Genuchten, and Miroslav Šejna. Development and applications of the HYDRUS and STANMOD software packages and related codes. *Vadose Zone Journal*, 7:587–600, 2008. doi: 10.2136/vzj2007.0077.

Jiří Šimůnek, Martinus Th. van Genuchten, and Miroslav Šejna. Recent developments and applications of the HYDRUS computer software packages. *Vadose Zone Journal*, 15(7), 2016. doi: 10.2136/vzj2016.04.0033.

Wenxiang Song, Liangsheng Shi, Lijun Wang, Yanling Wang, and Xiaolong Hu. Data-driven discovery of soil moisture flow governing equation: a sparse regression framework. *Water Resources Research*, 58(8), 2022. doi: 10.1029/2022WR031926.

Wenxiang Song, Liangsheng Shi, Leilei He, et al. Deep learning identification of the governing equation for water flow in heterogeneous soils from data. *Water Resources Research*, 61(3), 2025. doi: 10.1029/2024WR037786.

Pasquale Steduto, Theodore C. Hsiao, Dirk Raes, and Elias Fereres. AquaCrop—the FAO crop model to simulate yield response to water: I. concepts and underlying principles. *Agronomy Journal*, 101:426–437, 2009. doi: 10.2134/agronj2008.0139s.

Mark W. Strudley, Timothy R. Green, and James C. Ascough. Tillage effects on soil hydraulic properties in space and time: state of the science. *Soil and Tillage Research*, 99:4–48, 2008. doi: 10.1016/j.still.2008.01.007.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58:267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Silviu-Marian Udrescu and Max Tegmark. AI Feynman: a physics-inspired method for symbolic regression. *Science Advances*, 6:eaay2631, 2020. doi: 10.1126/sciadv.aay2631.

Martinus Th. van Genuchten. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44:892–898, 1980. doi: 10.2136/sssaj1980.03615995004400050002x.

Harry Vereecken, Andrea Schnepf, Jan W. Hopmans, Mathieu Javaux, Dani Or, Tiina Roose, Jan Vanderborght, et al. Modeling soil processes: review, key challenges, and new perspectives. *Vadose Zone Journal*, 15(5), 2016. doi: 10.2136/vzj2015.09.0131.

Harry Vereecken, Lutz Weihermüller, Shmuel Assouline, Jiří Šimůnek, Anne Verhoef, Michael Herbst, et al. Infiltration from the pedon to global grid scales: an overview and outlook for land surface modeling. *Vadose Zone Journal*, 18:180191, 2019. doi: 10.2136/vzj2018.10.0191.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.

Jessica Vuaille, Per Abrahamsen, Signe M. Jensen, Efstathios Diamantopoulos, Tobias S. Wacker, and Christian T. Petersen. Modelling pesticide degradation and leaching in conservation agriculture: effect of no-till and mulching. *Science of the Total Environment*, 929:172559, 2024. doi: 10.1016/j.scitotenv.2024.172559.

Frank A. Ward and Manuel Pulido-Velazquez. Water conservation in irrigation can increase water use. *Proceedings of the National Academy of Sciences*, 105:18215–18220, 2008. doi: 10.1073/pnas.0805554105.

Tobias K. D. Weber, Lutz Weihermüller, Attila Nemes, Michel Bechtold, Aurélie Degré, Efstathios Diamantopoulos, Simone Fatichi, Vilim Filipović, Surya Gupta, Tobias L. Hohenbrink, Daniel R. Hirmas, Conrad Jackisch, Quirijn de Jong van Lier, John Koestel, Peter Lehmann, Toby R. Marthews, Budiman Minasny, Holger Pagel, Martine van der Ploeg, Shahab A. Shojaeezadeh, Sigrid F. Svane, Brigitta Szabó, Harry Vereecken, Anne Verhoef, Michael Young, Yijian Zeng, Yonggen Zhang, and Sara Bonetti. Hydro-pedotransfer functions: a roadmap for future development. *Hydrology and Earth System Sciences*, 28:3391–3433, 2024. doi: 10.5194/hess-28-3391-2024.

Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, et al. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016. doi: 10.1038/sdata.2016.18.

Kell Wilson, Allen Goldstein, Eva Falge, Marc Aubinet, Dennis Baldocchi, Paul Berbigier, Christian Bernhofer, Reinhart Ceulemans, Han Dolman, Chris Field, Achim Grelle, Andreas Ibrom, Beverly E. Law, Andrew Kowalski, Tilden Meyers, John Moncrieff, Russell Monson, Walter Oechel, John Tenhunen, Riccardo Valentini, and Shashi Verma. Energy balance closure at FLUXNET sites. *Agricultural and Forest Meteorology*, 113:223–243, 2002. doi: 10.1016/S0168-1923(02)00109-0.

Svante Wold, Kim Esbensen, and Paul Geladi. Principal component analysis. *Chemometrics and Intelligent Laboratory Systems*, 2:37–52, 1987. doi: 10.1016/0169-7439(87)80084-9.

Lu Zhang, W. R. Dawes, and G. R. Walker. Response of mean annual evapotranspiration to vegetation changes at catchment scale. *Water Resources Research*, 37:701–708, 2001. doi: 10.1029/2000WR900325.

Jing Zheng, Junliang Fan, Fucang Zhang, and Qianlai Zhuang. Evapotranspiration partitioning and water productivity of rainfed maize under contrasting mulching conditions in northwest china. *Agricultural Water Management*, 243:106473, 2021. doi: 10.1016/j.agwat.2020.106473.

A. Supplementary analyses

Replacing K_s in Eq. (8) by $K(h^*)$ of Eq. (2) with a fitted h^* yields $h^* = 6.7\,\mathrm{cm}$ and $q = 15.4\,\mathrm{cm\,d^{-1}}$ for the eight soils; in-sample RMSE is 0.027 and leave-one-soil-out RMSE 0.040 (S03: observed 0.846, in-sample 0.835, LOO 0.822). The variant has one more fitted parameter, uses the same eight soils, and is mildly better than the K_s form; the choice between them cannot be resolved with eight points. The simpler form is therefore presented in the main text.

The 64 combinations per soil are written `C#_P#_T#_h#_D#_N#_Ep#`. Valid combinations: 64 in nine soils and 37 in S02, so ψ_D for S02 is estimated from fewer pairs; paired differences use only complete pairs.

Mass-balance closure. Median $|\Delta S|/W$ is 0.9 % over all valid runs; excluding E8 and E9 it is 0.65 % (95th percentile 2.3 %); for E8/E9 it is 16 %, as expected for experiments that deliberately start away from equilibrium. Fate shares in Eq. (5) sum to 1 to within rounding.

Fig. 25 shows the recoverable share $1-\psi_D$ for E16 and E7. Because E16 and E7 differ in how evaporation is changed, the two estimates differ in the order of ± 0.2 except in the sandy soils.

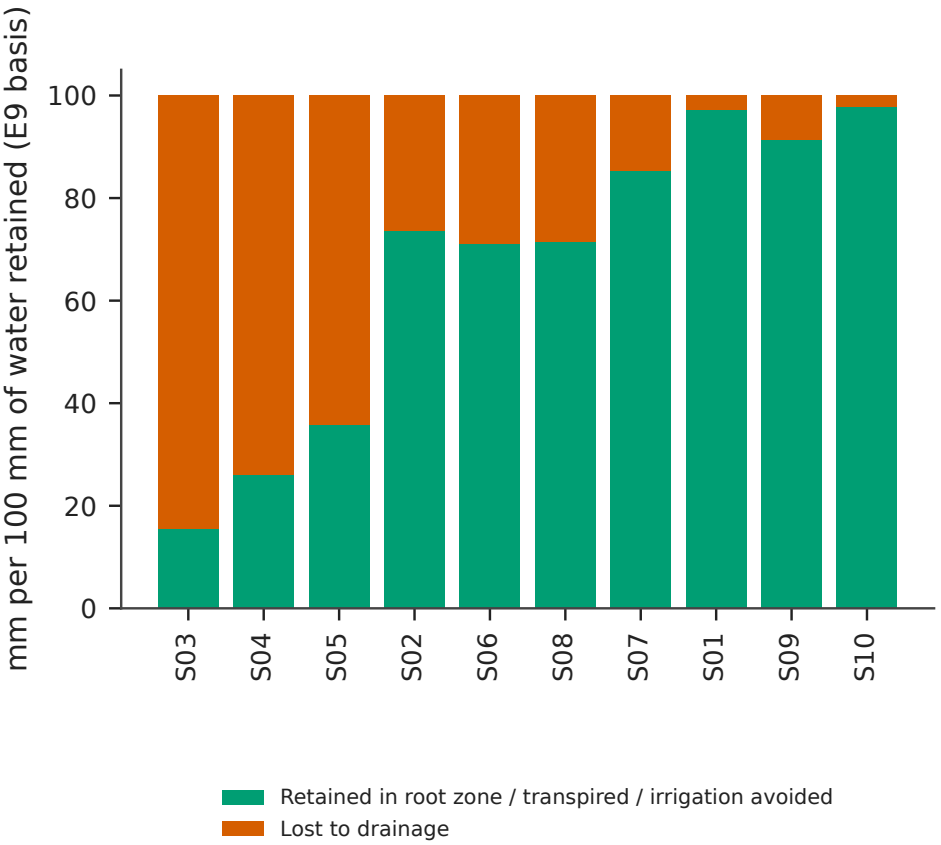


Figure 25: Recoverable fraction $1 - \psi_D$ by soil for the E16 and E7 contrasts.

Table 27: E9 reference values at $P = 1$. storage release $\bar{R}$ (mm yr^{-1}), drainage D and transpiration T for wet ($h_0 = 30$ or 100 cm) and dry ($h_0 = 1500$ or 3000 cm) starts. Negative D denotes net capillary rise across the 1 m boundary; "–" marks a failed run (S03 at 3000 cm).

Soil	Texture	$\bar{R}_{30}$	$\bar{R}_{100}$	$\bar{R}_{1500}$	$\bar{R}_{3000}$	D_{100}	D_{3000}	T_{100}	T_{3000}
S01	Coarse loam	236.2	69.7	-13.1	-23.1	-2	-1	80	24
S02	Coarse sand	214.8	106.9	2.8	-6.1	23	26	88	29
S03	Sand	3.2	-24.1	-28.2	–	66	–	26	–
S04	Loamy sand	21.3	-27.5	-40.7	-41.0	47	40	25	20
S05	Sandy loam	72.6	-20.8	-62.1	-63.3	18	2	31	15
S06	Loam	197.3	84.6	-47.2	-51.8	9	-0	116	13
S07	Silt loam	258.2	177.2	-31.0	-41.8	10	0	190	16
S08	Sandy clay loam	117.6	5.3	-51.7	-55.4	7	-0	50	15
S09	Clay loam	192.9	139.9	-27.5	-38.7	4	0	167	21
S10	Silty clay loam	189.5	158.2	-14.8	-29.6	0	0	186	26

Table 28: Closure of the loss fates. ΔET_{nt}: mean paired change in non-transpirational evapotranspiration for $E_p = 0.3 \to 1.2$ (mm yr^{-1}); ψ_D, ψ_I, ψ_T: shares; Σ: their sum (the remainder is storage and runoff); span E7: range of ET_{nt} across the mulch lattice (mm yr^{-1}).

Soil	ΔET_{nt}	ψ_D	ψ_I	ψ_T	Σ	span E7	ψ_D^{E7}	ψ_I^{E7}
S01	218	0.295	0.619	0.111	1.025	167	0.298	0.583
S02	144	0.193	0.658	0.157	1.008	86	0.607	0.339
S03	100	1.210	-0.384	0.178	1.004	59	0.706	0.000
S04	115	0.799	0.219	-0.007	1.011	58	0.687	-0.023
S05	141	0.582	0.415	0.015	1.012	74	0.663	-0.027
S06	191	0.469	0.433	0.107	1.009	122	0.521	0.379
S07	218	0.362	0.528	0.117	1.007	156	0.396	0.605
S08	152	0.533	0.386	0.091	1.010	111	0.549	0.332
S09	217	0.312	0.645	0.048	1.005	152	0.426	0.584
S10	222	0.310	0.838	-0.017	1.131	160	0.414	0.629

B. Reproducibility map of the memory analysis

Table 29: Result-to-script map (scripts in `Manuscripts/reproducibility/analysis/`, outputs in the work directory `out/`).

Result	Script / output
Ensemble table and run counts	`make_data.py` → `data/daisy_ensemble.csv.gz`
Memory and loss fates (Tables 8)	`p1_core.py` → `p1_core.json`, `p1_table.csv`
Rank statistics (Table 10)	`p1_stats.py` → `p1_table2.csv`; `p1_figs_d.py` → `p1_rank.json`
Rate-competition law: q^*, R^2, bootstrap and leave-one-out intervals, LOSO errors (Fig. 9, Table 9)	`p1_rate_law.py` → `p1_rate_law.json`
Earlier two-parameter variant; logistic-in-n LOO RMSE	`p1_mech.py` → `p1_mech.json`
Modifiers (Table 11)	`p1_figs_c.py` → `p1_modifiers.json`
Figures	`p1_figs_a--e.py`, `p1_fig12--14*.py`, `p1_fig1_new.py`, `figstyle.py`
All steps, checks and limits	`run_all.sh`, `compare_results.py`, `audit_numbers.py`, `README_REPRODUCE.md`

C. Supplementary information

The analysis was exploratory and was not pre-registered. The main analytical decisions were: the four controls (permutation null, bootstrap, LOSO/LOEO, gradient-boosting benchmark); the forward-selection stopping threshold (0.003); log-targets; the exclusion of E8 and E9 as a sensitivity analysis for the invariant search and for closure-derived drainage; and the treatment of R as a remainder. The candidate library was assembled by the author with knowledge of the ensemble and of the hydraulic-memory results (for example, the choice to include n, $\ln\alpha$ and $\ln W/TAW$),

which increases researcher degrees of freedom; readers should treat LOEO and LOSO scores as indicative rather than as confirmatory error rates.

The seven-flux search also returns weights on transpiration, soil evaporation and canopy evaporation in its smallest direction (rain 0.896, irrigation 0.834, ET -1, transpiration 0.367, soil evaporation 0.121, canopy evaporation -0.122, drainage -0.933), which is not a clean invariant: because $T + E_s + E_c + R = ET$ and R is not a column, the seven-column matrix has an additional near-dependence that mixes components. This is why the four-flux system is the clean positive control.

Table 30: Columns of the ensemble used in this paper.

Column	Unit	Definition
`experiment`, `treatment`, `run_id`	–	experiment code, treatment string encoding factor levels, unique run identifier
`soil`	–	S01–S10
`rain_mm`, `irrigation_mm`	mm yr^{-1}	20-year mean annual rainfall and irrigation
`ET_mm`	mm yr^{-1}	model evapotranspiration (total)
`transpiration_mm`, `soil_evap_mm`, `canopy_evap_mm`	mm yr^{-1}	reported components of ET
`drainage_1m_mm`	mm yr^{-1}	flux across the 1 m boundary (negative = net rise)
`water_stress_d`	d yr^{-1}	days of water stress
`yield_Mg_ha`	Mg ha^{-1}	grain/biomass yield
`N_fertiliser_kg_ha`, `N_leached_kg_ha`	kg ha^{-1}	nitrogen applied and leached
`IE_kg_m3`, `DPF`, `NLF`	–	irrigation efficiency, drainage fraction D/W, N-leaching fraction (definitional; excluded as targets)
`P`, `dT`, `Ep`, `trig`, `z`, `amt_mm`, `vff`, `cap`, `Nrate`	various	factor levels (rain scale, warming, evaporation factor, trigger, sensor depth, dose, bare fraction, mulch capacity, N rate)

DPF$= D/W$ is exact by construction and its "prediction" is therefore the same task as predicting D; NLF and IE are ratios of other columns. They are not used as independent evidence in any result.

D. Reproducibility map of the accounting audit

Table 31: Result-to-script map.

Result	Script / output
Ensemble table and run counts	`make_data.py` → `data/daisy_ensemble.csv.gz`
SVD, permutation null, closure (Eq. (17), Fig. 14)	`p2_a.py`, `p2_laws.py` → `p2_a.json`, `p2_laws.json`
Unattributed term R (Eq. (18), Fig. 15)	`p2_R.py` → `p2_R.json`
clr-PCA (Fig. 16, Table 15)	`p2_a.py`, `p2_figs.py`, `p2_fig4.py` → `p2_pca.json`, `p2_pca_sens.json`
Sparse laws (Table 21, Eq. (19))	`p2_et.py`, `p2_dpf.py`, `p2_figs2.py`, `p2_nested.py` → `p2_et.json`, `p2_dpf.json`, `p2_etlaw.json`, `p2_nested.json`; exclusion sensitivity `p2_s02_excl.py` → `p2_s02_excl.json`
Cluster bootstrap, law without E8 and E9, descriptive statistics (Table 5) and R by experiment (Table 14)	`p2_robust.py` → `p2_robust.json`
Compensation (Fig. 19)	`p2_laws2.py` → `p2_laws2.json`
Figures	`p2_figs.py`, `p2_figs2.py`, `p2_fig4.py`, `p2_fig6.py`, `p2_fig7.py`, `p2_figfail.py`, `p2_fig1_new.py`
All steps, checks and limits	`run_all.sh`, `compare_results.py`, `audit_numbers.py`, `README_REPRODUCE.md`